

Biochemistry

retake review notes

Foundations

Biochemical foundations · Water, acids, bases and buffers

Proteins

Amino acids · Protein structure and folding · Hemoglobin and globular proteins · Collagen, elastin and fibrous proteins

Enzymes and energy

Enzymes, kinetics and regulation · Bioenergetics and oxidative phosphorylation

carbohydrates

Carbohydrate structure and stereochemistry · Glycolysis · Pyruvate dehydrogenase and the TCA cycle · Gluconeogenesis · Glycogen metabolism · Fructose, galactose and lactose · Pentose phosphate pathway and NADPH · Glycoconjugates

Lipids

Dietary lipids and absorption · Fatty acids, triacylglycerols and ketones · Complex lipids and eicosanoids · Cholesterol, lipoproteins and steroids

Nitrogen metabolism

Amino acids: disposal of nitrogen · Amino acid degradation and synthesis · Heme, bilirubin and amine products · Nucleotide metabolism

Metabolic integration

Insulin and glucagon · Fed-fast integration · Diabetes mellitus · Obesity and energy balance

Nutrition and vitamins

Nutrition · Vitamins

Molecular biology

DNA structure, replication and repair · RNA, transcription and processing · Translation · Gene regulation · Biotechnology

Laboratory

Laboratory principles and tests

coral headings, numbered key points, things to learn

blue clinical links and practice-bank topics

grey box a past paper asked this, with the corrected reasoning

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Biochemical foundations

Teacher slides and laboratory material; teacher slides cited under each idea.

What biochemistry explains in medicine

Biochemistry applies chemical principles to biological processes at the molecular and cellular levels. It connects a molecular change to altered cell function, a clinical phenotype, a diagnostic marker, and sometimes a targeted treatment. A useful medical answer therefore moves from molecule to mechanism to patient rather than stopping at a definition.

1. Biochemistry studies the chemistry of living systems.
2. Molecular abnormalities can explain disease mechanisms and laboratory findings.
3. Mechanistic understanding supports diagnosis, prevention, and rational treatment.

→ clinic: A pathogenic enzyme variant can change a metabolite concentration and create a recognizable disease pattern.

→ clinic: Biomarkers are clinically useful because they report an underlying molecular or cellular process.

INTRODUCTION.ppt, slides 3–6 (definition, scope, and medical relevance). · biochemistry-scope.md, Primary course files: INTRODUCTION.ppt.

Elements and major biomolecule classes

Most biological matter is built chiefly from carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur. Carbohydrates, lipids, proteins, and nucleic acids have distinct recurring chemical features and major roles, but they interact in cells rather than operating as isolated lists. Proteins perform much of the cell's catalytic and structural work, whereas nucleic acids store and transmit hereditary information.

1. CHNOPS denotes carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur.
2. Carbohydrates commonly provide fuel and participate in structure and recognition.
3. Lipids store energy, form membranes, insulate, and generate signaling molecules.
4. Proteins act as enzymes, receptors, transporters, antibodies, and structural elements.
5. Nucleic acids store and express genetic information.

→ clinic: Different biomolecule classes require different laboratory measurements and therapeutic approaches.

INTRODUCTION.ppt, slides 9–13 (CHNOPS, cells, biomolecule classes, and central dogma).

Central dogma and genotype-to-phenotype reasoning

The usual flow of sequence information is DNA to RNA to protein. A DNA variant can alter an RNA sequence or its expression and thereby change the amount or amino-acid sequence of a protein. The resulting change in protein structure or function is the mechanistic bridge from genotype to phenotype.

1. Transcription transfers information from DNA to RNA.
2. Translation uses messenger RNA to specify a polypeptide sequence.
3. Mutations can affect protein sequence, abundance, localization, or regulation.

→ clinic: Missense variants can replace one amino acid and change enzyme or hemoglobin function.

→ **clinic:** Some disease variants affect RNA processing or expression without changing a coding sequence.

INTRODUCTION.ppt, slide 13 (central dogma and information flow).

Anabolism, catabolism, and ATP coupling

Catabolic pathways break molecules down and often capture released energy, whereas anabolic pathways build molecules and require energy and reducing power. ATP couples favorable energy-releasing reactions to otherwise unfavorable cellular work through group transfer and linked reactions. ATP is an immediate energy currency, not the body's only energy store.

1. Catabolism is degradative and commonly energy yielding.
2. Anabolism is synthetic and commonly energy requiring.
3. ATP contains adenine, ribose, and three phosphate groups.
4. ATP hydrolysis can be coupled to biosynthesis, transport, and mechanical work.

→ **clinic:** Hypoxia impairs ATP generation and rapidly compromises ion pumps and cell viability.

INTRODUCTION.ppt, slides 16–19 (anabolism, catabolism, and ATP).

Functional groups determine biochemical behavior

Functional groups give biomolecules predictable acid-base behavior, polarity, reactivity, and interaction patterns. Hydroxyl groups are polar; carboxyl and amino groups participate in proton transfer; carbonyl groups define aldehydes and ketones; phosphate groups carry charge and transfer energy; and sulfhydryls can form disulfides. Recognizing the group is more useful than memorizing an isolated molecule because the same chemistry reappears in proteins, carbohydrates, lipids, and nucleotides.

1. Hydroxyl groups support polarity and hydrogen bonding.
2. Carboxyl groups can donate protons; amino groups can accept protons.
3. Carbonyl groups occur in aldehydes and ketones.
4. Phosphate groups contribute negative charge and participate in energy and signal transfer.
5. Two oxidized cysteine sulfhydryls can form a disulfide bond.

→ **clinic:** Ionization of functional groups affects drug solubility, membrane passage, and protein binding.

INTRODUCTION.ppt, slides 20–22 (major biological functional groups).

covalent and noncovalent forces in biomolecules

Covalent bonds share electrons and build stable molecular backbones, while ionic interactions, hydrogen bonds, and hydrophobic effects organize molecules and are often reversible. Unequal electron sharing makes a bond polar and creates partial charges that can interact with water. Nonpolar groups cluster in water because this minimizes disruption of water-water hydrogen bonding, a major driver of membranes and protein folding.

1. Ionic interactions are attractions between opposite charges.
2. Polar covalent bonds share electrons unequally; nonpolar covalent bonds share them more evenly.
3. Hydrogen bonds are individually weak but collectively important.
4. Disulfide bonds are covalent links between cysteine residues.
5. The hydrophobic effect drives nonpolar groups away from water.

→ **clinic:** Heat, extreme pH, and detergents can disrupt noncovalent forces and denature proteins.

→ clinic: A mutation that replaces a charged residue with a hydrophobic one can promote abnormal protein aggregation.

INTRODUCTION.ppt, slides 24-31 (ionic/covalent bonds, polarity, disulfides, hydrogen bonds, and hydrophobic interactions).

past paper asked

Which substance is produced in every dehydration synthesis reaction?

Water is the best answer. Condensation joins monomers while releasing water.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q30

past paper asked

Formic acid is also called

Methanoic acid is the best answer. The IUPAC name for one-carbon formic acid is methanoic acid.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q60

past paper asked

The general formula of amines is

R-NH₂ is the best answer. A primary amine has the general form R-NH₂.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q1

past paper asked

If you replace the oxygen of the hydroxyl group with sulfur in R-OH, which functional group will you get?

Thiol is the best answer. Replacing O in an alcohol with S gives a thiol, R-SH.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q14

Test yourself

1. Which set contains the six elements emphasized as CHNOPS?

- | | |
|---|---|
| A. Calcium, hydrogen, neon, oxygen, potassium, and sodium | B. Carbon, helium, nitrogen, osmium, phosphorus, and selenium |
| C. Carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur | D. Copper, hydrogen, nickel, oxygen, potassium, and sulfur |

2. What makes a covalent bond polar?

- | | |
|---|--|
| A. Complete loss of both electrons | B. Absence of nuclei |
| C. Unequal electron sharing caused by an electronegativity difference | D. Equal attraction by identical atoms |

Answers: 1 C · 2 C

water, acids, bases and buffers

Teacher slides and laboratory material; teacher slides cited under each idea.

calculation toolkit

Relationship	Use / trap
$\text{pH} = -\log_{10}[\text{H}^+]$	One pH unit is a tenfold change in hydrogen-ion activity (concentration approximation in dilute solutions).
$\text{pH} = \text{pK}_a + \log_{10}([\text{A}^-]/[\text{HA}])$	At equal conjugate base and weak acid, $\text{pH} = \text{pK}_a$. Use concentrations after any strong acid/base neutralization.
Buffering range $\approx \text{pK}_a \pm 1$	A useful rule of thumb, not an abrupt cutoff. Higher total concentration increases buffer capacity at the same ratio.
0.070 M lactate / 0.080 M lactic acid; pK_a 3.74	$3.74 + \log_{10}(0.875) = 3.68$. Concentration ratio is base/acid, not acid/base.
pI for an amino acid	Average the two pK_a values that bracket the electrically neutral species. Do not blindly average all pK_a values.

Body-water compartments and ionic composition

Total body water is distributed mainly between intracellular fluid and extracellular fluid, with roughly two thirds intracellular. Extracellular fluid includes plasma and interstitial fluid, while transcellular fluids are specialized small compartments such as cerebrospinal fluid. Sodium and chloride predominate extracellularly, whereas potassium, magnesium, and phosphate are relatively enriched intracellularly.

1. About two thirds of total body water is intracellular.
2. ECF comprises plasma, interstitial fluid, and small transcellular compartments.
3. Sodium is the major ECF cation; potassium is the major ICF cation.
4. Solute distribution and membrane permeability govern water movement.

→ **clinic:** Fluid therapy changes ECF first and then redistributes according to tonicity and permeability.

→ **clinic:** Serum sodium is primarily a marker of water balance relative to exchangeable body solute.

Water and Buffer 1404.ppt; slides 4-5 (body water and compartments). · Marks water/buffer reference, local reference pages 2-4 (compartments, electrolytes, and osmolality).

water polarity, hydrogen bonding, and solubility

Water is polar because oxygen attracts shared electrons more strongly than hydrogen and the molecule has a bent geometry. Its partial charges permit hydrogen bonding and hydration shells around ions and polar solutes. Amphipathic molecules contain both hydrophilic and hydrophobic regions and therefore organize into structures such as membranes and micelles.

1. Water is a dipole with partial negative charge on oxygen and partial positive charge on hydrogen.

2. Hydrogen bonding explains much of water's solvent behavior.
3. Hydrophilic solutes interact favorably with water; hydrophobic solutes do not.
4. Amphipathic molecules contain both polar and nonpolar regions.

→ clinic: Amphipathic bile salts emulsify dietary lipids.

→ clinic: Drug polarity influences distribution between aqueous compartments and membranes.

Water and Buffer 1404.ppt, slides 4 and 6–8 (water polarity, hydrogen bonds, and solubility classes).

pH, pOH, and logarithmic concentration reasoning

pH is the negative base-10 logarithm of hydrogen-ion concentration, so each pH unit represents a tenfold change in hydrogen-ion concentration. At 25°C, pH plus pOH equals 14, which allows conversion between hydrogen and hydroxide concentrations. Because physiology operates over a narrow pH range, small numerical pH shifts represent substantial chemical changes.

1. $\text{pH} = -\log_{10}[\text{H}^+]$.
2. At 25°C, $\text{pH} + \text{pOH} = 14$.
3. A fall of one pH unit means a tenfold increase in $[\text{H}^+]$.
4. Acidemia and alkalemia refer to measured blood pH, not the underlying process.

→ clinic: Arterial pH below the reference range is acidemia; above it is alkalemia.

→ clinic: A change from pH 7.4 to 7.1 is chemically much larger than the decimal difference suggests.

Water and Buffer 1404.ppt, slides 9–15 (pH, pOH, clinical pH, acidosis, and alkalosis).

Acids, bases, conjugate pairs, K_a , and pK_a

A Brønsted acid donates a proton and a Brønsted base accepts one; loss of a proton converts an acid into its conjugate base. Strong acids dissociate more completely than weak acids in water. A larger K_a and therefore a lower pK_a indicate a stronger acid and make quantitative prediction of ionization possible.

1. Acid and conjugate base differ by one proton.
2. K_a measures the extent of acid dissociation.
3. $\text{pK}_a = -\log_{10} K_a$.
4. Lower pK_a means stronger acid.
5. Weak does not mean dilute; it describes incomplete dissociation.

→ clinic: The pK_a of a drug helps predict its charge state and membrane permeability at a given pH.

Water and Buffer 1404.ppt, slides 14 and 17–20 (acid-base definitions, conjugate pairs, K_a , and pK_a).

Henderson-Hasselbalch equation and ionization ratios

The Henderson-Hasselbalch equation is $\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$. It converts the difference between pH and pK_a into the ratio of conjugate base to weak acid: one unit above pK_a means 10:1, and one unit below means 1:10. At pH equal to pK_a , acid and conjugate base are equal.

1. $\text{pH} - \text{pK}_a = \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$.
2. $\text{pH} = \text{pK}_a$ gives $[\text{A}^-] = [\text{HA}]$.
3. Each unit of $\text{pH} - \text{pK}_a$ changes the ratio tenfold.

4. Above pK_a the deprotonated form predominates; below pK_a the protonated form predominates.

→ clinic: Ion trapping of weak acids and bases follows the same pH- pK_a logic.

→ clinic: The bicarbonate equation applies this relationship to blood acid-base physiology.

Water and Buffer 1404.ppt, slides 24–28 and teacher MCQs 37–38. · Lippincott 6e, Chapter 1 acid-base section, local PDF pages 18–21.

Titration curves and how buffers resist pH change

A buffer contains a weak acid and its conjugate base, allowing added H^+ to be consumed by base and added OH^- to be consumed by acid. Its flattest titration-curve region lies near pK_a , with greatest effectiveness roughly within pK_a plus or minus one pH unit. Buffer capacity depends on the total concentration of the pair, not just their ratio.

1. At half-equivalence, $pH = pK_a$ and buffering is maximal.
2. The useful buffer range is approximately $pK_a \pm 1$.
3. The conjugate base consumes added H^+ .
4. Higher total buffer concentration gives greater capacity at the same pH and ratio.

→ clinic: A patient's ability to buffer an acid load depends on available buffer stores as well as ventilation and renal response.

Water and Buffer 1404.ppt, slides 22–28 (titration, buffer region, range, and capacity).

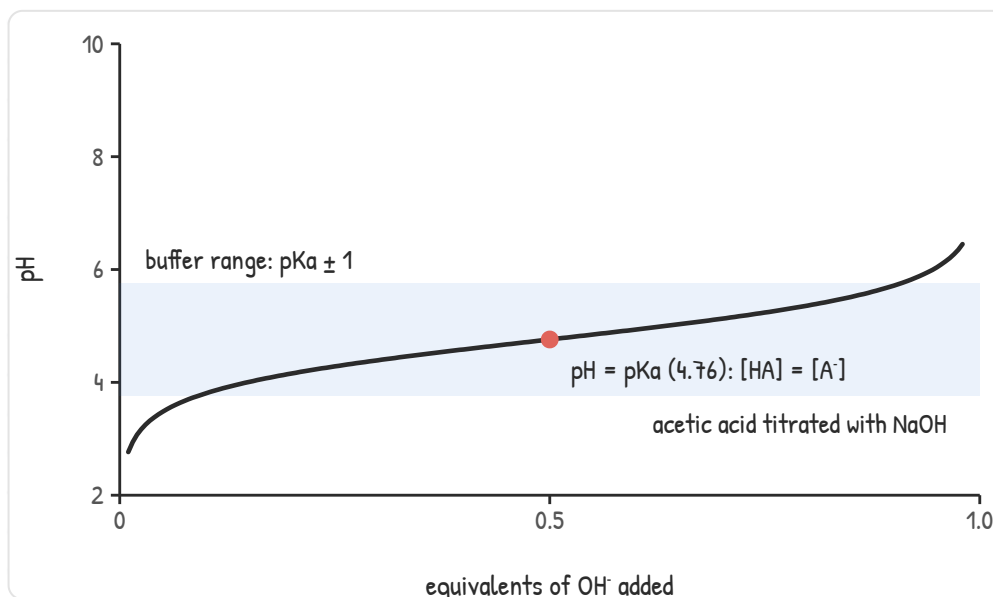


fig. 1 Weak-acid titration curve, computed from Henderson-Hasselbalch for acetic acid (pK_a 4.76). The flat middle is where the buffer resists pH change.

Bicarbonate, phosphate, proteins, and hemoglobin in acid-base control

The major physiological buffers are bicarbonate, phosphate, plasma proteins, and hemoglobin. Bicarbonate is an open system because the lungs regulate CO_2 rapidly and the kidneys regulate bicarbonate and net acid more slowly. Protein side chains, especially in hemoglobin, accept or donate protons; compensation limits a pH disturbance but does not remove its primary cause.

1. Bicarbonate couples blood chemistry to pulmonary CO_2 elimination.

2. Phosphate is important intracellularly and in urine.
3. Ionizable protein side chains buffer H^+ .
4. Deoxyhemoglobin binds H^+ more readily than oxyhemoglobin.
5. Respiratory compensation for metabolic acidosis lowers $PaCO_2$.

→ clinic: Kussmaul breathing in ketoacidosis lowers CO_2 and partially raises pH.

→ clinic: Carbonic anhydrase and the chloride shift support CO_2 transport by red cells.

Water and Buffer 1404.ppt, slides 29-35 (physiological buffers). · Marks water/buffer reference, local reference pages 6-8 (bicarbonate, hemoglobin, chloride shift, and urinary buffering).

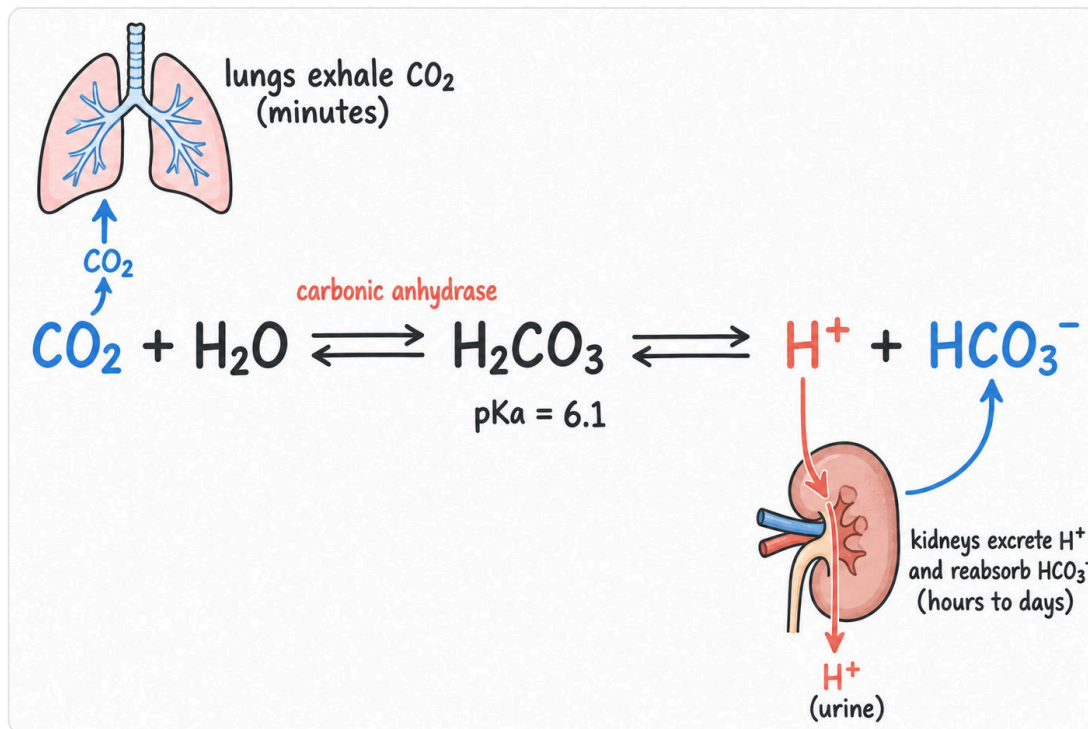


fig. 2 The bicarbonate buffer couples rapid pulmonary CO_2 removal to slower renal H^+ excretion and bicarbonate reabsorption.

past paper asked

A buffer solution is 0.080 M in lactic acid ($pK_a=3.74$) and 0.070 M in sodium lactate. The pH of the solution is

3.68 is the best answer. Henderson-Hasselbalch gives $3.74 + \log(0.070/0.080) \approx 3.68$.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q31

past paper asked

The chemical reactions in the small intestine take place in a basic (alkaline) environment. Which number indicates this basic pH?

8.5 is the best answer. pH 8.5 is clearly alkaline; values below 7 are acidic.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q34

past paper asked

For octanoic acid, $pK_a = 4.89$. Which pH range would be the best buffering region for this acid?

3.89–5.89 is the best answer. A weak acid buffers effectively within about one pH unit of its pK_a .

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q40

past paper asked

Hydroxyapatite has a very small dissolution equilibrium constant (approximately 10^{-26} in the source). Which description is correct under physiological conditions?

A very small dissolution equilibrium constant favors solid hydroxyapatite. Acid promotes dissolution; added calcium/phosphate can suppress it by the common-ion effect. Source correction: hydroxyapatite is the inorganic mineral component of bone, not the organic matrix.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q51

past paper asked

As the pK_a of an acid increase, the acid will be:

Weaker is the best answer. A larger pK_a means a smaller K_a and therefore a weaker acid.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q27

past paper asked

Which of the following is not an example of a weak acid?

Sulfuric acid is the best answer. Sulfuric acid is treated as a strong acid, whereas lactic, pyruvic, and carbonic acids are weak acids.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q54

past paper asked

The strength of an organic acid depends on:

Electronegativity is the best answer. Electron-withdrawing electronegativity stabilizes the conjugate base and increases organic-acid strength.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q59

past paper asked

Polyprotic acids such as H_3PO_4 can act as acid-base buffers:

At pH values around any of their pK_a 's is the best answer. Each ionizable proton provides a separate buffer region centered near its own pK_a .

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q21

past paper asked

A buffer solution contains propionic acid and its conjugate base; the pKa of propionic acid is 4.88. At what pH would the solution buffer best?

5.0 is the best answer. A buffer works best near its pKa; pH 5.0 is closest to pKa 4.88.

Cell. & Molecules · September 2021 · Biochemistry; Biochemistry · original Q10

Test yourself

1. Which ions are most characteristic of extracellular fluid?

- A. Potassium and magnesium
- B. Phosphate and protein
- C. Sodium and chloride
- D. Iron and copper

2. At the midpoint of a weak-acid titration, which relation is true?

- A. $\text{pH} = 0$
- B. $[\text{A}^-]$ is zero
- C. pH must equal 7
- D. $\text{pH} = \text{pKa}$ and $[\text{A}^-] = [\text{HA}]$

3. How is pH defined?

- A. The hydrogen-ion concentration itself
- B. The positive logarithm of hydroxide concentration
- C. The negative base-10 logarithm of hydrogen-ion concentration
- D. The ratio of sodium to potassium

Answers: 1 C · 2 D · 3 C

Amino acids

Lippincott 6e, Chapter 1; teacher slides cited under each idea.

common amino-acid structure and side-chain classification

A standard alpha-amino acid has an alpha carbon attached to an amino group, a carboxyl group, a hydrogen, and a distinctive side chain. The side chain determines most residue-specific chemistry and supports the practical classification into nonpolar, uncharged polar, acidic, and basic groups. At physiological pH, the alpha amino and carboxyl groups are usually protonated and deprotonated respectively.

1. The R group distinguishes one amino acid from another.
2. Nonpolar side chains favor hydrophobic interactions.
3. Uncharged polar side chains can hydrogen bond.
4. Acidic side chains are usually negative and basic side chains often positive at physiological pH.
5. Proline has a secondary amino group; glycine has two alpha-carbon hydrogens.

→ **clinic:** A single side-chain substitution can alter folding, solubility, charge, or catalytic activity.

Amino acids and Proteins.Part1.pdf, pages 7-15 (structure and side-chain classes). · Lippincott 6e, Chapter 1, local extraction items 11 and 135.

Nonpolar residues and the hydrophobic effect

Nonpolar side chains avoid favorable contact with water and therefore cluster in the interior of soluble globular proteins. In membrane proteins the orientation can reverse, with hydrophobic residues exposed to lipid. This hydrophobic effect is a major force driving folding and can become pathological when a mutation creates an exposed hydrophobic patch.

1. Nonpolar residues usually do not form ionic or hydrogen bonds through their side chains.
2. They pack inside soluble proteins but can face membrane lipids.
3. Hydrophobic clustering stabilizes a protein's three-dimensional structure.

→ **clinic:** The beta-globin Glu-to-Val substitution in HbS creates a hydrophobic interaction that promotes deoxyhemoglobin polymerization.

Amino acids and Proteins.Part1.pdf, pages 10-15 (amino-acid side-chain classes). · Lippincott 6e, Chapter 1, local extraction items 20 and 23.

Polar residues, phosphorylation, and disulfide bonds

Serine, threonine, and tyrosine contain hydroxyl groups that form hydrogen bonds and can be phosphorylated. Cysteine contains a sulfhydryl group; oxidation of two cysteines forms a covalent disulfide bond and the paired product is cystine. These modifications can alter protein activity, localization, or structural stability.

1. Serine, threonine, and tyrosine are common phosphorylation sites.
2. Asparagine and glutamine have polar amide side chains.
3. Two cysteine thiols can oxidize to a disulfide.
4. Disulfides are especially common in extracellular and secreted proteins.

→ **clinic:** Kinases regulate many enzymes and receptors through reversible phosphorylation.

→ **clinic:** The oxidizing extracellular environment favors disulfide stabilization.

Amino acids and Proteins.Part1.pdf, pages 12-13 and 31-33. · Lippincott 6e, Chapter 1, local extraction item 35.

Acidic and basic side chains, especially histidine

Aspartate and glutamate carry negatively charged carboxylate side chains at physiological pH, whereas lysine and arginine are usually positively charged. Histidine's imidazole has a pKa near the physiological range, so local protein environments can switch it between protonated and neutral states. That makes histidine particularly useful for acid-base catalysis and buffering.

1. Aspartate and glutamate are acidic and usually negative.
2. Lysine and arginine are basic and usually positive.
3. Histidine can donate or accept protons near physiological pH.
4. Local electrostatic environment can shift a side-chain pKa.

→ **clinic:** Histidines contribute to hemoglobin buffering and the Bohr effect.

→ **clinic:** Catalytic histidine often acts as a proton shuttle in enzyme active sites.

Amino acids and Proteins.Part1.pdf, pages 14-15 (acidic/basic residues and histidine).

Amino-acid stereochemistry and the glycine exception

Except for glycine, standard amino acids have four different substituents on the alpha carbon and are chiral. Human ribosomal proteins are synthesized from L-amino acids. Glycine is achiral because its side chain is hydrogen, leaving two hydrogens attached to the alpha carbon.

1. Chiral amino acids occur as D and L enantiomers.
2. Human proteins use L-amino acids.
3. Glycine is achiral.
4. D-amino acids occur in some bacterial cell walls and antibiotics.

→ **clinic:** Stereochemistry affects enzyme recognition and drug design.

Amino acids and Proteins.Part1.pdf, page 20 (glycine and L-amino acids). · Lippincott 6e, Chapter 1 stereochemistry section.

Amino-acid ionization, titration, and isoelectric point

Amino acids are amphoteric and can carry positive, zero, or negative net charge as pH changes. The isoelectric point is the pH at which the average net charge is zero; below pI a molecule is more protonated and positive, while above pI it is more deprotonated and negative. Buffering is greatest near each ionizable group's pKa, not specifically at pI unless a pKa coincides with it.

1. At pI, average net charge is zero.
2. pH below pI gives a net positive charge.
3. pH above pI gives a net negative charge.
4. Titration curves reveal pKa values and buffering regions.
5. Ionizable side chains add extra titration steps.

→ **clinic:** Electrophoresis separates proteins according to net charge at the buffer pH.

→ **clinic:** Solubility is often lowest near a protein's pI.

Aromatic amino acids and analytical recognition

Phenylalanine, tyrosine, and tryptophan contain aromatic rings, with tryptophan contributing most strongly to protein absorbance near 280 nm and tyrosine contributing substantially. Tyrosine also contains a phenolic hydroxyl that can hydrogen bond, be phosphorylated, and ionize at alkaline pH. Aromatic-ring nitration underlies the xanthoproteic protein test.

1. Tryptophan is the strongest common contributor to A280.
2. Tyrosine also absorbs near 280 nm; phenylalanine contributes less.
3. Tyrosine is aromatic yet has a polar hydroxyl group.
4. The xanthoproteic reaction detects aromatic residues by a yellow-to-orange color.

→ **clinic:** A280 is commonly used to estimate protein concentration, but nucleic acids and other chromophores can interfere.

→ **clinic:** Tyrosine phosphorylation is a central signaling mechanism.

Amino acids and Proteins.Part1.pdf, page 21 (aromatic amino acids and absorbance). · Confirmed TUMS practical curriculum: xanthoproteic test.

past paper asked

Which of the following statements about solutions of amino acids at physiologic pH is true?

All amino acids contain both positive and negative charges is the best answer. At physiological pH, free amino acids generally contain both NH_3^+ and COO^- groups as zwitterions.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q30

past paper asked

Choose the incorrect statement about amino acids.

D-amino acids occur, for example, in bacterial peptidoglycan and as D-serine in humans. Most ribosomally incorporated amino acids are L, with achiral glycine. The vague source phrase "tyrosine is a modified amino acid" is clarified here.

Biochemistry 1 · Finals 2022; Biochemistry · original Q19

past paper asked

An amino acid not found in proteins is ...

β -Alanine is the best answer. β -Alanine is not one of the standard α -amino acids incorporated into proteins.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q36

Test yourself

1. Which atoms are attached to the alpha carbon of a standard alpha-amino acid?
 - A. Two phosphate groups and a base
 - B. An amino group, carboxyl group, hydrogen, and side chain
 - C. Three fatty acids and glycerol
 - D. A sugar, base, and phosphate
2. An amino acid is placed in a solution whose pH is below its pI. What is its predominant net charge?
 - A. Negative
 - B. Always exactly zero
 - C. It has no ionizable groups
 - D. Positive
3. What type of covalent cross-link can form between two cysteine residues in proteins, and what is the resulting linkage called?
 - A. Under oxidizing conditions, the two cysteines can form an internal disulfide (covalent) bond.
 - B. Glycine is the exception because its alpha-carbon has two hydrogen substituents.
 - C. Proline's side chain and alpha-amino nitrogen form a rigid five-membered ring structure.
 - D. Two cysteine -SH groups can be oxidized to form a covalent disulfide bond (-S-S-).

Answers: 1 B · 2 D · 3 D

Protein structure and folding

Lippincott 6e, Chapter 2; teacher slides cited under each idea.

Primary structure encodes folding and function

Primary structure is the linear amino-acid sequence of a polypeptide, conventionally read from N terminus to C terminus. Sequence determines the pattern of side-chain interactions available during folding and therefore strongly constrains final structure and function. A sequence-changing mutation can cause loss of function, instability, mislocalization, or toxic aggregation.

1. Primary structure is amino-acid sequence.
2. Sequences are written N terminus to C terminus.
3. The sequence contains the information needed for native folding, although cells often require chaperones.
4. A changed sequence can produce a changed phenotype.

→ **clinic:** Missense variants can cause sickle cell disease or destabilize collagen.

→ **clinic:** Sequence comparison helps interpret genetic disease variants.

Amino acids and Proteins.Part1.pdf, pages 31-33 and 43-49. · Lippincott 6e, Chapter 2, local extraction items 5-6.

Peptide-bond formation, resonance, and planarity

A peptide bond is an amide linkage formed between an alpha-carboxyl group and an alpha-amino group. Resonance gives the C-N bond partial double-bond character, making the peptide unit rigid and planar and restricting rotation. Rotation around bonds adjacent to the alpha carbon still permits organized folding, and peptide bonds are usually trans.

1. Peptide formation is a condensation reaction.
2. The peptide C-N bond has partial double-bond character.
3. Restricted rotation makes the peptide unit planar.
4. Most peptide bonds adopt the trans configuration.
5. Denaturation usually does not hydrolyze peptide bonds.

→ **clinic:** Proteases hydrolyze selected peptide bonds during digestion, signaling, and zymogen activation.

Amino acids and Proteins.Part1.pdf, pages 31-33 (peptide bond). · Lippincott 6e, Chapter 2, local PDF pages 30-54.

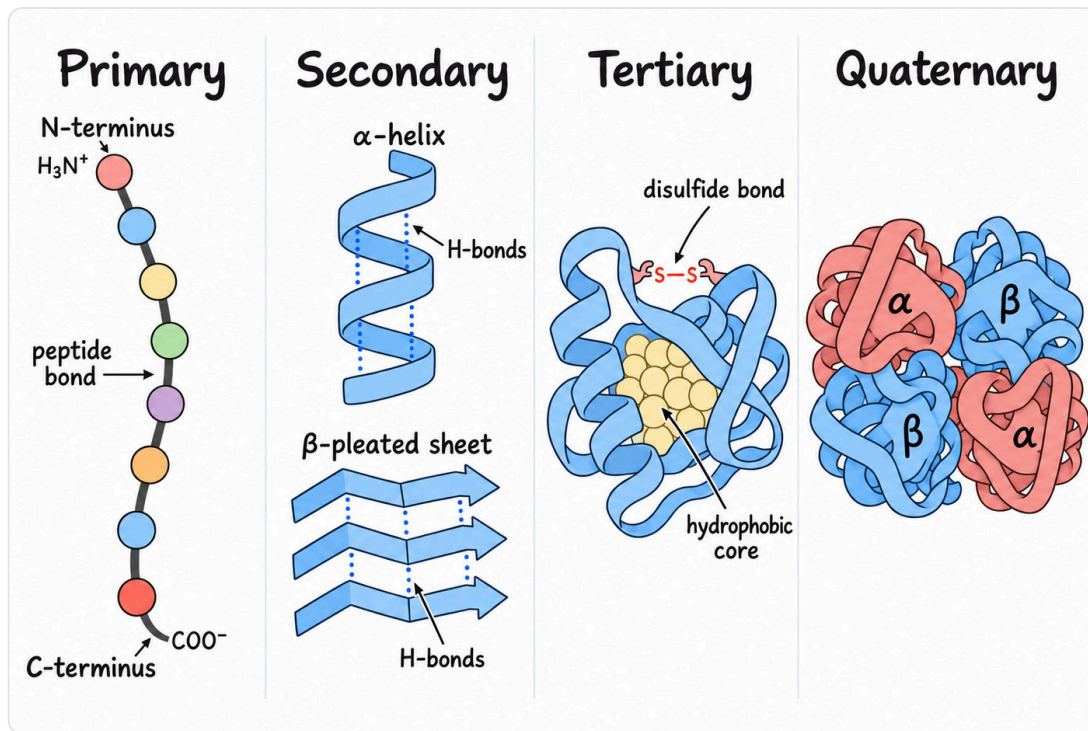


fig. 3 Protein structure progresses from amino-acid sequence through local secondary structure and a folded chain to assemblies of multiple subunits.

Alpha helices, beta sheets, and beta turns

Secondary structure is local, repetitive backbone folding stabilized mainly by hydrogen bonds between peptide carbonyl and amide groups. In an alpha helix, hydrogen bonds form within one chain and the side chains project outward; beta sheets contain extended strands hydrogen-bonded in parallel or antiparallel arrangements. Beta turns reverse chain direction and often connect nearby antiparallel strands.

1. An alpha helix has about 3.6 residues per turn and i-to-i+4 backbone hydrogen bonds.
2. Proline often disrupts an alpha helix.
3. Beta sheets are stabilized by backbone hydrogen bonds between extended strands.
4. Beta turns permit compact folding and often contain glycine or proline.

→ **clinic:** Amyloid fibrils are enriched in ordered beta-sheet structure.

Amino acids and Proteins.Part1.pdf, pages 36-40 (alpha helix, beta sheet, and beta turn).

Tertiary structure, motifs, domains, and side-chain forces

Tertiary structure is the overall three-dimensional arrangement of one polypeptide, including how secondary elements assemble into motifs and domains. A domain is a compact structural and often functional unit that can fold somewhat independently. Hydrophobic interactions, hydrogen bonds, ionic interactions, and disulfide bonds collectively stabilize the native conformation.

1. Motifs are recurring combinations of secondary structures.
2. Domains are modular structural and functional units.
3. Hydrophobic burial is a major folding force in aqueous solution.
4. Ionic and hydrogen bonds are sensitive to pH and solvent.
5. Disulfides provide covalent stabilization, especially extracellularly.

→ **clinic:** A domain-specific mutation can selectively abolish binding or catalysis while leaving other domains intact.

Amino acids and Proteins.Part1.pdf, page 43 (protein domains). · Lippincott 6e, Chapter 2, Local extraction item 91.

Quaternary structure requires multiple subunits

Quaternary structure describes the arrangement and interaction of two or more polypeptide subunits. A one-chain protein can have tertiary structure but has no quaternary structure. Subunits are held mainly by noncovalent forces, although occasional interchain disulfides can contribute, and subunit communication permits cooperativity and allostery.

1. One polypeptide has no quaternary structure.
2. Two or more interacting polypeptides create quaternary structure.
3. Hydrophobic, ionic, and hydrogen-bond interactions usually dominate subunit association.
4. Hemoglobin is an $\alpha_2\beta_2$ cooperative tetramer.

→ **clinic:** A variant at a subunit interface can alter cooperativity without directly changing an active or ligand-binding site.

Amino acids and Proteins.Part1.pdf, page 49 (quaternary structure). · Lippincott 6e, Chapter 2, Local extraction item 160.

Protein folding and molecular chaperones

Amino-acid sequence drives folding toward a low-energy native conformation, with the hydrophobic effect as a major early force. Molecular chaperones bind exposed hydrophobic regions, prevent inappropriate aggregation, and often use ATP to give a protein another opportunity to fold correctly. Chaperones assist folding but do not dictate a different final sequence-encoded structure.

1. Folding is ordered rather than a random search of all conformations.
2. Hydrophobic collapse helps organize early folding intermediates.
3. Hsp70 protects exposed hydrophobic regions of nascent chains.
4. Chaperonins provide a protected folding chamber.
5. Many chaperone cycles require ATP.

→ **clinic:** Cell stress increases heat-shock proteins.

→ **clinic:** Failure of proteostasis contributes to neurodegeneration and some loss-of-function diseases.

Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 2. · Lippincott 6e, Chapter 2, Local extraction items 101, 110, and 115.

Denaturation disrupts higher structure, not primary sequence

Denaturation unfolds and disorganizes secondary and tertiary structure without ordinarily hydrolyzing peptide bonds, so primary structure remains intact. Heat, extreme pH, organic solvents, detergents, urea, and heavy metals can disrupt the interactions maintaining the native state. Some proteins refold after the denaturant is removed, but aggregation makes denaturation commonly irreversible in practice.

1. Denaturation causes loss of native conformation and usually function.
2. Peptide bonds are generally preserved.
3. Hydrophobic exposure can cause precipitation and aggregation.
4. Reversibility depends on sequence, conditions, and avoidance of aggregation.

→ clinic: Fever and severe acidosis can impair enzyme function, although ordinary fever is not equivalent to wholesale protein denaturation.

→ clinic: Heat inactivates many enzymes used in laboratory samples.

Amino acids and Proteins.Part1.pdf, page 48 (protein denaturation). · Lippincott 6e, Chapter 2, local extraction item 107.

Protein misfolding, amyloid, and prion propagation

Misfolded proteins are normally refolded or targeted for degradation, but failed quality control can allow toxic aggregates to accumulate. Amyloid is an ordered, insoluble, beta-sheet-rich fibrillar assembly rather than a random denatured precipitate. In prion disease, a pathogenic PrP conformation templates conversion of normal host PrP into the same abnormal conformation.

1. Amyloid fibrils are ordered and beta-sheet rich.
2. Alzheimer plaques contain A beta fragments produced from APP.
3. PrPSc promotes conformational conversion of PrPC.
4. Sequence mutation is not required for every misfolding disease.

→ clinic: Amyloid disorders include Alzheimer disease and systemic amyloidoses.

→ clinic: Prion diseases are transmissible spongiform encephalopathies and resist ordinary proteolysis.

Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 2. · Lippincott 6e, Chapter 2, local extraction items 125, 148, and 172.

past paper asked

A tripeptide has?

3 amino acids and 2 peptide bonds is the best answer. Three amino acids are linked by two peptide bonds.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q29

past paper asked

Peptide bond is a?

Covalent bond is the best answer. A peptide bond is a covalent amide bond.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q39

past paper asked

The oxygen carrier of muscle is the globular protein myoglobin. Which one of the following amino acids is highly likely to be localized within the interior of the molecule?

Valine is the best answer. Valine is nonpolar and is therefore favored in the hydrophobic interior of a globular protein.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q13

past paper asked

A coiled peptide chain held in place by hydrogen bonding between peptide bonds in the same chain (not between two chains) is?

α -helix is the best answer. An α -helix is stabilized by intrachain backbone hydrogen bonds.

Cell. & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q68

past paper asked

The secondary structure of protein is held together by which type of bonding?

Hydrogen bonding is the best answer. Backbone hydrogen bonds stabilize α -helices and β -sheets.

Cell. & Molecules · September 2021 · Biochemistry; Biochemistry · original Q34

Test yourself

1. Which statement about a peptide bond is correct?

- | | |
|--|---|
| A. It is a phosphodiester bond | B. It is freely rotating like a single carbon-carbon bond |
| C. It is an amide linkage between an alpha-carboxyl and an alpha-amino group | D. It is disrupted by every noncovalent denaturant |

2. List the four main types of interactions that stabilize the tertiary structure of globular proteins.

- | | |
|---|---|
| A. The four are disulfide bonds, hydrophobic interactions, hydrogen bonds, and ionic interactions. | B. A polypeptide is an unbranched chain formed by linking many amino acids through peptide bonds. |
| C. The primary structure of a protein is the linear sequence of amino acids in the polypeptide chain. | D. A monomeric protein consists of a single polypeptide chain and therefore lacks quaternary structure. |

3. Which structural level usually remains intact during reversible protein denaturation?

- | | |
|-----------------------|-------------------------|
| A. Primary structure | B. Secondary structure |
| C. Tertiary structure | D. Quaternary structure |

Answers: 1 C · 2 A · 3 A

Hemoglobin and globular proteins

Lippincott 6e, Chapter 3; teacher slides cited under each idea.

Heme chemistry and HbA architecture

Heme consists of protoporphyrin IX coordinated to ferrous iron, Fe^{2+} , which can bind oxygen reversibly. Major adult hemoglobin HbA is an $\alpha_2\beta_2$ tetramer with one heme per subunit and therefore four oxygen-binding sites. Oxidation of iron to Fe^{3+} produces methemoglobin, which cannot bind oxygen normally.

1. Functional heme oxygen binding requires Fe^{2+} .
2. HbA contains two alpha and two beta chains.
3. Each globin subunit contains one heme.
4. Hemoglobin subunits associate mainly through noncovalent forces.

→ **clinic:** Oxidant drugs can convert Fe^{2+} to Fe^{3+} and cause methemoglobinemia.

exam-scope.md, August 25 active biochemistry scope: proteins and hemoglobin; complete Lippincott Chapters 1–7 scaffold. · Lippincott 6e, Chapter 3, local extraction items 8 and 28.

Myoglobin stores oxygen; hemoglobin transports it

Myoglobin is a monomeric muscle hemeprotein that stores and facilitates intracellular oxygen movement, whereas hemoglobin is a tetrameric red-cell transporter. Myoglobin binds one oxygen and has a hyperbolic, high-affinity curve without cooperativity. Hemoglobin binds up to four oxygen molecules and uses subunit interactions to load in lungs and unload in tissues.

1. Myoglobin is monomeric and found in cardiac and skeletal muscle.
2. Hemoglobin is tetrameric and found in erythrocytes.
3. Myoglobin has a hyperbolic oxygen-binding curve.
4. Hemoglobin has a sigmoidal oxygen-binding curve.

→ **clinic:** Myoglobin rises early after muscle injury but is less cardiac-specific than troponin.

Lippincott 6e, Chapter 3, local extraction items 7, 12–16, and 41–53. · exam-scope.md, complete confirmed Lippincott Chapters 1–7 scaffold.

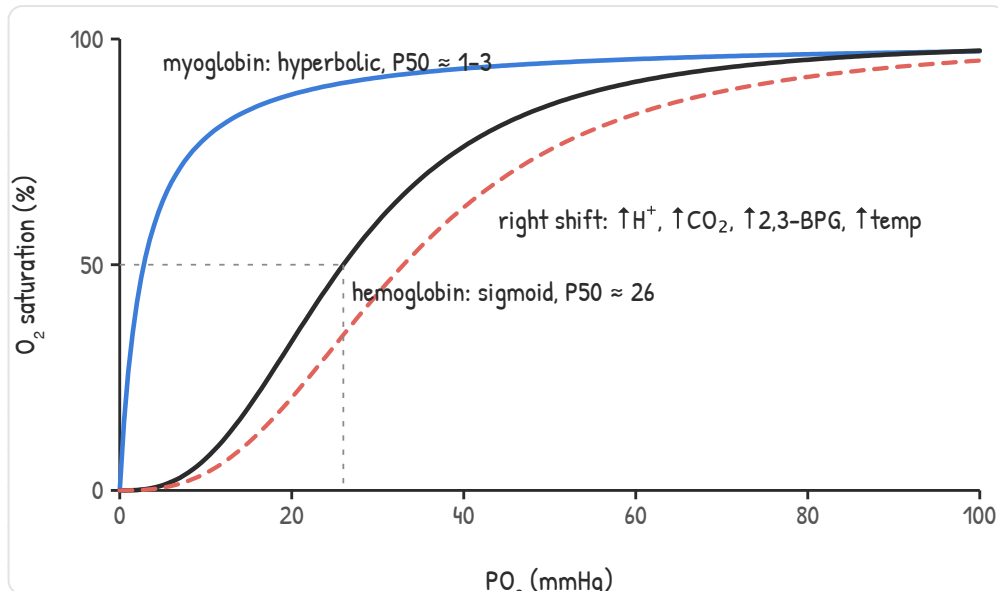


fig. 4 Oxygen-binding curves computed with the Hill equation (Hb $n = 2.7$). A right shift raises P₅₀ and unloads more O₂ in tissues.

cooperativity, T/R states, and P₅₀

Deoxyhemoglobin favors the low-affinity T state, whereas oxygen binding shifts subunits toward the high-affinity R state. Binding at one heme increases affinity at the remaining hemes, producing the sigmoidal curve and efficient tissue unloading. P₅₀ is the oxygen partial pressure at half-saturation; a higher P₅₀ means lower affinity and a right shift.

1. T state is tense, deoxygenated, and lower affinity.
2. R state is relaxed, oxygenated, and higher affinity.
3. Positive cooperativity creates a sigmoidal curve.
4. Higher P₅₀ means lower oxygen affinity.

→ clinic: Right shifts improve unloading but can impair loading if extreme; left shifts do the reverse.

Lippincott 6e, Chapter 3, local extraction items 39-40, 49, and 53-64.

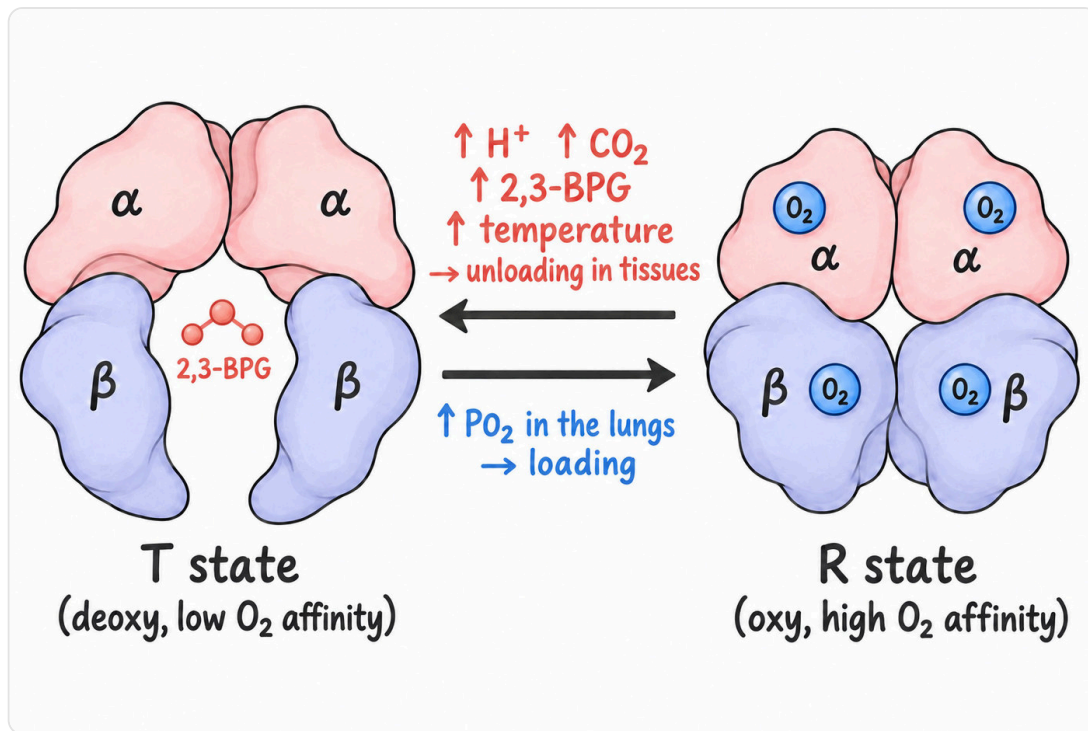


fig. 5 Hemoglobin shifts between the low-affinity T state favored by tissue conditions and the high-affinity R state favored by oxygen loading in the lungs.

Bohr effect: active tissues promote oxygen unloading

In metabolically active tissues, increased H^+ and CO_2 stabilize deoxyhemoglobin's T state and lower oxygen affinity, shifting the curve right. In the lungs, CO_2 removal and higher pH favor oxygen loading. CO_2 also forms carbamates at globin N termini, while deoxyhemoglobin helps buffer protons.

1. Lower pH decreases hemoglobin oxygen affinity.
2. Higher CO_2 favors oxygen unloading.
3. The effect is reversed in the lungs.
4. Deoxyhemoglobin binds H^+ more strongly than oxyhemoglobin.

→ clinic: Exercise increases local CO_2 , H^+ , and temperature, promoting oxygen delivery.

→ clinic: Acidosis can worsen HbS deoxygenation and sickling.

Water and Buffer 1404.ppt, slides 33-35 (hemoglobin buffering and CO_2 transport). · Lippincott 6e, Chapter 3, local extraction item 65.

2,3-BPG stabilizes deoxyhemoglobin

2,3-BPG is an anionic red-cell metabolite that binds the positively charged central cavity of deoxyhemoglobin. It stabilizes the T state, lowers oxygen affinity, shifts the curve right, and thereby enhances tissue unloading. Its concentration rises in chronic hypoxia and anemia, whereas stored blood loses 2,3-BPG and initially unloads oxygen less effectively.

1. One 2,3-BPG binds per deoxyhemoglobin tetramer.
2. It binds chiefly between the beta chains.
3. It lowers oxygen affinity and raises P_{50} .
4. Hypoxia and chronic anemia increase its concentration.

→ clinic: High-altitude adaptation includes increased 2,3-BPG.

→ **clinic:** Freshly transfused stored red cells may temporarily have reduced unloading capacity.

Lippincott 6e, Chapter 3, local extraction items 77-91.

Fetal hemoglobin and placental oxygen transfer

Fetal hemoglobin is $\alpha_2\gamma_2$. Gamma chains bind 2,3-BPG less strongly than adult beta chains, so HbF has higher oxygen affinity than HbA in the same red-cell environment. This left shift enables fetal blood to acquire oxygen from maternal blood across the placenta.

1. HbF composition is $\alpha_2\gamma_2$.
2. HbF binds 2,3-BPG weakly.
3. Weaker BPG binding increases oxygen affinity.
4. HbF predominates before birth and falls after birth.

→ **clinic:** Increasing HbF with hydroxyurea reduces HbS polymerization and sickle crises.

Lippincott 6e, Chapter 3, local extraction items 108 and 116-119.

carbon monoxide poisoning versus methemoglobinemia

Carbon monoxide occupies heme sites and stabilizes the R state, reducing oxygen capacity while making remaining sites hold oxygen more tightly. Methemoglobin instead contains Fe^{3+} , which cannot bind oxygen normally and can produce brown blood with cyanosis. Both cause tissue hypoxia, but pulse oximetry and treatment considerations differ.

1. CO reduces available sites and left-shifts remaining oxygen binding.
2. Methemoglobin contains Fe^{3+} rather than Fe^{2+} .
3. Chocolate-brown blood suggests methemoglobinemia.
4. Oxidant drugs such as nitrates or benzocaine can cause acquired methemoglobinemia.

→ **clinic:** CO poisoning is treated with high-concentration oxygen and sometimes hyperbaric oxygen.

→ **clinic:** Methylene blue can reduce methemoglobin in appropriate patients but requires clinical caution.

Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 3. · Lippincott 6e, Chapter 3, local extraction items 97-104 and 167-174.

HbA1c records cumulative glucose exposure

HbA1c forms by slow nonenzymatic glycation of HbA, especially at the N-terminal valines of beta chains. Its abundance reflects average glucose exposure over the circulating red cell population rather than a momentary glucose value. Altered red-cell lifespan, transfusion, or some hemoglobin variants can make HbA1c misleading.

1. Glycation is nonenzymatic and differs from enzyme-directed glycosylation.
2. The extent of HbA1c formation rises with ambient glucose concentration and time.
3. HbA1c reflects roughly the preceding two to three months with greater weighting of recent weeks.
4. Conditions changing erythrocyte lifespan can bias the result.

→ **clinic:** HbA1c is used to diagnose and monitor diabetes when the assay is reliable.

→ **clinic:** Recent blood loss or hemolysis can produce an unexpectedly low HbA1c.

Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 3. · Lippincott 6e, Chapter 3, local extraction items 122-128 and 204.

HbS: a primary-structure change causes polymerization

HbS contains a beta-chain Glu6Val substitution that creates a hydrophobic patch. When deoxygenated, HbS molecules polymerize into rigid fibers that distort red cells, impair deformability, and occlude small vessels. Low oxygen, acidosis, dehydration, and other conditions favoring deoxyhemoglobin increase sickling, whereas HbF reduces polymerization.

1. The mutation is beta6 glutamate to valine.
2. Deoxy-HbS, not oxy-HbS, polymerizes most readily.
3. Polymerization causes hemolysis and vaso-occlusion.
4. Hydroxyurea helps by increasing HbF.

→ **clinic:** Vaso-occlusion causes painful crises, infarction, and organ damage.

→ **clinic:** Newborn screening enables prophylaxis before severe infection occurs.

Lippincott 6e, Chapter 1 local extraction item 24 and Chapter 3 local extraction items 151-158.

Thalassemia is impaired globin-chain production

Thalassemias result from decreased or absent synthesis of alpha- or beta-globin chains rather than a single abnormal hemoglobin structure alone. The unbalanced normally synthesized partner chains precipitate, damage erythroid precursors, and cause ineffective erythropoiesis and hemolysis. Alpha-thalassemia is commonly deletional, while beta-thalassemia commonly reflects point mutations affecting expression or RNA processing.

1. Thalassemia is primarily a quantitative globin defect.
2. Beta-thalassemia leaves excess alpha chains that precipitate.
3. Severity depends on how many genes are affected and whether production is reduced or absent.
4. Beta-thalassemia often raises HbA2 and HbF.

→ **clinic:** Severe disease produces anemia, marrow expansion, transfusion dependence, and iron overload.

→ **clinic:** Microcytosis out of proportion to anemia can suggest thalassemia trait.

Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 3. · Lippincott 6e, Chapter 3, local extraction items 175-193.

past paper asked

Which of the following mutations would produce a minor thalassemia?

Deletion of one β -globin locus is the best answer. Loss of one β -globin locus can produce β -thalassemia trait/minor.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q32

past paper asked

The ability of hemoglobin to serve as an effective transporter of oxygen and carbon dioxide between lungs and tissues is explained by which of the following properties?

In contrast to myoglobin, hemoglobin exhibits greater changes in secondary and tertiary structure after oxygen binding is the best answer. Oxygen binding produces cooperative conformational changes in hemoglobin, unlike monomeric myoglobin.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q83

past paper asked

Which of the following would produce Methemoglobinemia?

Converting Ferrous iron (Fe^{2+}) to Ferric (Fe^{3+}) is the best answer. Oxidation of heme iron from Fe^{2+} to Fe^{3+} forms methemoglobin, which cannot bind oxygen normally.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q47

past paper asked

Blood is drawn from a child with severe anemia and the hemoglobin protein is degraded for the analysis of peptide and amino acids. Of the results below, which change in hemoglobin primary structure is most likely to correlate with the clinical phenotype of anemia?

Leu-Glu-Ile to Leu-Val-Ile is the best answer. Replacing glutamate with valine reproduces the key nonconservative change of sickle hemoglobin.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q57

past paper asked

All of the following statements correctly describes transport of O_2 by hemoglobin except

Increased CO_2 concentrations increase O_2 affinity is the best answer. Higher CO_2 lowers hemoglobin oxygen affinity through the Bohr effect rather than increasing it.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q51

past paper asked

In sickle cell anemia, the basis of the malfunction of the hemoglobin molecule is ...

Substitution of a single amino acid is the best answer. Sickle hemoglobin results from a single β -globin substitution, Glu6Val, changing primary structure.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q20

Test yourself

1. What specific amino acid substitution occurs in HbS and at which position of the beta chain?
 - A. It occurs in approximately 1 out of every 500 newborn African American infants.
 - B. Glutamate at position 6 of the beta chain is replaced by valine (Glu6Val).
 - C. Myoglobin is a globular hemeprotein present in heart and skeletal muscle.
 - D. Myoglobin has a higher oxygen affinity than hemoglobin at all pO₂ values.
2. Why does fetal hemoglobin have a higher oxygen affinity than adult HbA?
 - A. Its heme contains Fe³⁺ instead of Fe²⁺
 - B. Its gamma chains bind 2,3-BPG less strongly than beta chains
 - C. It is a monomer like myoglobin
 - D. It cannot undergo an R-state transition
3. A rise in hemoglobin P50 indicates which change?
 - A. Increased oxygen affinity and a left shift
 - B. Reduced oxygen affinity and a right-shifted dissociation curve
 - C. Loss of all cooperative binding
 - D. Conversion of hemoglobin to methemoglobin only

Answers: 1 B · 2 B · 3 B

collagen, elastin and fibrous proteins

Lippincott 6e, Chapter 4; teacher slides cited under each idea.

collagen triple helix, Gly-X-Y, and major types

Collagen is a triple helix of three alpha chains with a repeating Gly-X-Y sequence, commonly containing proline and hydroxyproline. Glycine must occur every third residue because only its small hydrogen side chain fits at the helix center. Types I, II, and III form fibrils, whereas type IV forms a sheet-like basement-membrane network.

1. Collagen is the most abundant protein in the body.
2. Gly-X-Y repeats permit triple-helix packing.
3. Type I predominates in bone, skin, tendon, and scar.
4. Type II predominates in cartilage and vitreous body.
5. Type III forms reticular fibers; type IV forms basement-membrane networks.

→ **clinic:** Tissue distribution helps predict the phenotype of a collagen defect.

exam-scope.md, complete confirmed Lippincott Chapters 1-7 scaffold. · Lippincott 6e, Chapter 4, local PDF pages 87-104.

Intracellular collagen synthesis and vitamin C

Prepro-alpha chains enter rough ER, lose their signal sequence, and undergo selected proline and lysine hydroxylation plus hydroxylysine glycosylation. Hydroxylases require oxygen, Fe²⁺, and vitamin C; terminal propeptides align three chains and prevent premature fibril formation. The chains assemble from the C terminus into a procollagen triple helix before Golgi processing and secretion.

1. Hydroxylation occurs in rough ER after residues enter the chain.
2. Vitamin C keeps iron in the reduced state needed by hydroxylases.
3. Hydroxyproline stabilizes the triple helix through hydrogen bonding.
4. Propeptides align chains and keep procollagen soluble inside the cell.

→ **clinic:** Vitamin C deficiency causes weak collagen, bleeding, and poor wound healing.

Lippincott 6e, Chapter 4, local PDF pages 87-104 and extraction items 31-39.

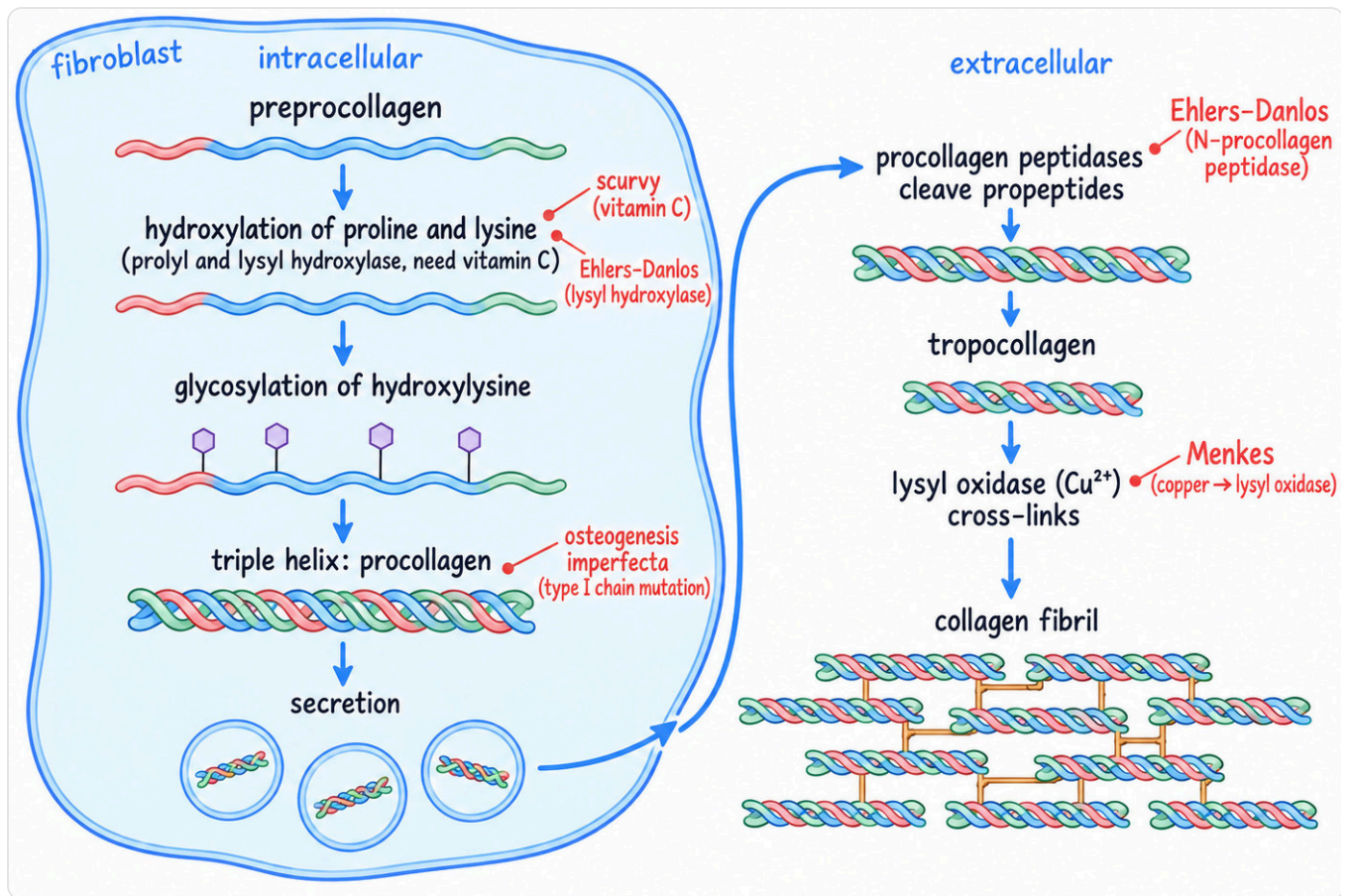


fig. 6 Collagen is modified and assembled as procollagen inside fibroblasts, then cleaved and cross-linked extracellularly into fibrils.

Extracellular procollagen cleavage and copper-dependent cross-linking

After secretion, N- and C-procollagen peptidases remove terminal propeptides to produce tropocollagen. Tropocollagen molecules align into staggered fibrils, and copper-dependent lysyl oxidase creates reactive aldehydes from lysine or hydroxylysine. Covalent intermolecular cross-links then provide tensile strength.

1. Procollagen cleavage occurs extracellularly.
2. Tropocollagen self-assembles into staggered fibrils.
3. Lysyl oxidase requires copper.
4. Cross-link density increases tensile strength with maturation.

→ **clinic:** Copper deficiency or lysyl-oxidase inhibition weakens collagen cross-linking.

→ **clinic:** Defective procollagen peptidases can cause Ehlers-Danlos phenotypes.

Lippincott 6e, Chapter 4, local PDF pages 87-104 and extraction items 40-47.

collagen degradation and matrix remodeling

Collagen is long lived but not permanent; collagenases initiate cleavage of the otherwise resistant triple helix. These enzymes belong to the matrix metalloproteinase family and produce fragments that other proteases further degrade. Synthesis and degradation must remain balanced during development, wound healing, and tissue maintenance.

1. Collagenases are matrix metalloproteinases.

2. Initial collagenase cleavage enables further proteolysis.
3. MMP activity is regulated by activation and tissue inhibitors.
4. Excess breakdown weakens matrix; excess deposition causes fibrosis.

→ clinic: MMP activity contributes to arthritis, tumor invasion, and vascular remodeling.

→ clinic: Chronic injury can shift balance toward fibrotic collagen deposition.

Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 4. · Lippincott 6e, Chapter 4, local extraction items 52-53.

Scurvy, osteogenesis imperfecta, and Ehlers-Danlos syndrome

Scurvy is acquired defective hydroxylation from vitamin C deficiency, causing capillary fragility and poor wound healing.

Osteogenesis imperfecta usually involves type I collagen quantity or structure and produces fractures, blue sclerae, and sometimes hearing or dental abnormalities. Ehlers-Danlos syndromes arise from varied collagen or processing defects and commonly feature skin hyperextensibility and joint hypermobility; vascular type III collagen defects risk arterial rupture.

1. Scurvy impairs proline and lysine hydroxylation.
2. Osteogenesis imperfecta commonly involves type I collagen.
3. Glycine substitution can disrupt the entire triple helix.
4. Classic EDS often involves type V; vascular EDS involves type III.
5. Dominant-negative chains can damage a multichain collagen molecule.

→ clinic: Bleeding plus dietary deficiency suggests scurvy; congenital recurrent fractures suggest OI.

→ clinic: Marked hypermobility with fragile hyperextensible skin suggests EDS.

Lippincott 6e, Chapter 4, local PDF pages 87-104 and extraction item 58.

Elastin structure permits stretch and recoil

Collagen resists stretch, whereas elastin permits tissues to extend and recoil. Tropoelastin is rich in small nonpolar residues and lysine and is deposited on a fibrillin microfibril scaffold. Lysyl oxidase contributes to desmosine cross-links, creating an extensible, amorphous network important in lungs, large arteries, and elastic ligaments.

1. Elastin provides elasticity rather than high tensile strength.
2. Tropoelastin is the soluble precursor.
3. Fibrillin microfibrils guide elastic-fiber assembly.
4. Desmosine cross-links connect tropoelastin chains.
5. Elastic recoil is crucial in alveoli and large arteries.

→ clinic: Loss of alveolar elastin reduces recoil and promotes expiratory airway collapse.

Lippincott 6e, Chapter 4, local extraction items 68-74.

Fibrillin defects and unopposed elastase injury

Fibrillin-1 forms the microfibril scaffold for elastic fibers, and its mutation causes Marfan syndrome with skeletal, ocular, and cardiovascular manifestations. Alpha1-antitrypsin normally restrains neutrophil elastase in the lung; deficiency permits destruction of alveolar elastic tissue and panacinar emphysema. The common severe Z variant also polymerizes in hepatocyte ER, producing both low circulating inhibitor and liver injury.

1. Fibrillin-1 mutation causes Marfan syndrome.

2. Alpha1-antitrypsin inhibits neutrophil elastase.
3. Unopposed elastase destroys alveolar walls and recoil.
4. Misfolded Z-variant AAT can accumulate in hepatocytes.

→ clinic: Marfan syndrome raises risk of aortic root disease and lens dislocation.

→ clinic: AAT deficiency causes early panacinar emphysema, worsened by smoking, and may cause cirrhosis.

Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 4. · Lippincott 6e, Chapter 4, local extraction items 74 and 78-85.

past paper asked

All of the following statements regarding fibrous proteins are true except

Hemoglobin is the best example is the best answer. Hemoglobin is globular, whereas collagen and keratin are fibrous structural proteins.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q33

past paper asked

Which of the substances below is primarily found in tendons?

Collagen is the best answer. Protein function follows primary sequence and higher-order folding interactions.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q49

past paper asked

During synthesis of mature collagen fiber, which one of the following steps would occur within the fibroblast?

Formation of a triple helix is the best answer. Protein function follows primary sequence and higher-order folding interactions.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q43

past paper asked

Deficiency of lysyl hydroxylase causes syndrome characterized by hypermobile joints and skin abnormalities.

Ehlers-Danlos is the best answer. Protein function follows primary sequence and higher-order folding interactions.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q80

past paper asked

A child with tall stature, loose joints, and detached retinas is found to have a mutation in type II collagen. Recall that collagen consists of a repeating tripeptide motif where the first amino acid of each tripeptide is the same. Which of the following amino acids is the recurring amino acid most likely to be altered in mutations that distort collagen molecules?

Glycine is the best answer. Protein function follows primary sequence and higher-order folding interactions.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q11

past paper asked

The highest concentration of cystine can be found in

Keratin is the best answer. Keratin is rich in cystine disulfide bonds, which strengthen hair and nails.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q27

past paper asked

Which one of the following results in Ehlers-Danlos syndrome?

Deficiency of lysyl hydroxylase is the best answer. Lysyl hydroxylase defects impair collagen maturation and can cause hyperextensible skin and hypermobile joints.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q32

past paper asked

Glycine and proline are the most abundant amino acids in the structure of

Collagen is the best answer. Collagen is rich in glycine and proline/hydroxyproline, supporting its triple helix.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q44

Test yourself

1. What class of enzymes degrades collagen, and to which larger enzyme family do they belong?

- | | |
|---|---|
| A. In the cornea, collagen fibers are stacked in an orderly manner to transmit light with minimum scattering. | B. Some hydroxylysine residues in collagen are glycosylated with glucose or with a disaccharide glucosyl-galactose. |
| C. Type IV collagen assembles into sheet-like or mesh structures that form a major component of basement membranes. | D. Collagen is degraded by collagenases, which belong to the larger family of matrix metalloproteinases (MMPs). |

2. Hydroxylation of selected proline and lysine residues during collagen synthesis requires which vitamin?

- | | |
|--------------|----------------|
| A. Vitamin K | B. Vitamin B12 |
| C. Vitamin A | D. Vitamin C |

3. Where do proline and lysine hydroxylation of collagen chains occur?

- | | |
|--|--|
| A. Inside lysosomes after collagen degradation | B. Only in the nucleus |
| C. In the rough endoplasmic reticulum before secretion | D. In the extracellular matrix after fibril assembly |

Answers: 1 D · 2 D · 3 C

Enzymes, kinetics and regulation

Lippincott 6e, Chapter 5; teacher slides cited under each idea.

Enzyme kinetics at a glance

Pattern	K_m	v_{max} / distinguishing point
Michaelis-Menten	$[S]$ at $v = v_{max}/2$	$v = v_{max}[S]/(K_m + [S])$; at $[S] = 3K_m$, $v = 0.75 v_{max}$.
More enzyme	Unchanged	v_{max} rises proportionally when other conditions are fixed.
Competitive inhibition	Increased apparent K_m	v_{max} unchanged; high substrate can overcome reversible competition.
Pure noncompetitive inhibition	Unchanged	v_{max} decreased. Mixed inhibition need not leave K_m unchanged.
Uncompetitive inhibition	Decreased apparent K_m	v_{max} decreased in the same ratio; inhibitor binds ES.
Irreversible inactivation	No universal K_m prediction	Active enzyme falls; removing free inhibitor does not restore already-inactivated enzyme.
Catalytic performance	K_m alone is insufficient	$k_{cat} = v_{max}/[E]_{total}$; k_{cat}/K_m is the specificity constant at low $[S]$.

Active sites create specificity and induced fit

An enzyme active site is a three-dimensional pocket formed by residues that may be far apart in primary sequence. Complementary shape, charge, polarity, and dynamics give substrate and reaction specificity. Substrate binding can induce a conformational change that positions catalytic groups and stabilizes the transition state.

1. The active site is created by tertiary structure.
2. Specificity can be absolute, group, bond, or stereochemical.
3. Induced fit is conformational adjustment after binding.
4. Catalytic residues need not be adjacent in sequence.

→ clinic: Many drugs work by occupying an enzyme active site or an allosteric pocket.

→ clinic: Stereospecificity explains why drug enantiomers can differ in effect.

exam-scope.md, August 25 active biochemistry scope: enzymes and regulation; complete Lippincott Chapters 1-7 scaffold. · Lippincott 6e, Chapter 5, local extraction items 12, 14, and 18.

Enzymes lower activation energy without changing equilibrium

Enzymes accelerate reactions by stabilizing the transition state and providing a lower-activation-energy pathway. They do not change the overall free-energy difference between reactants and products and do not move the equilibrium position.

They accelerate both forward and reverse reactions, allowing equilibrium to be reached faster.

1. Activation energy is the barrier to reaching the transition state.
2. Enzymes lower activation energy.
3. Enzymes do not change reaction ΔG .

4. Enzymes do not change the equilibrium constant.
5. Transition-state stabilization is central to catalysis.

→ clinic: Transition-state analogs can be potent enzyme inhibitors and useful drugs.

Lippincott 6e, Chapter 5, local extraction items 29–36.

Apoenzyme, holoenzyme, cofactor, and coenzyme

An apoenzyme is the protein portion without a required nonprotein component, whereas the complete active form is the holoenzyme. Metal ions can act as cofactors; organic helpers are coenzymes, many derived from vitamins. A cosubstrate binds transiently and leaves chemically changed, while a prosthetic group remains tightly bound and is regenerated on the enzyme.

1. Apoenzyme plus required nonprotein component equals holoenzyme.
2. Metal ions such as Zn^{2+} or Fe^{2+} can be cofactors.
3. NAD^+ is a vitamin-derived coenzyme and cosubstrate.
4. Tightly bound FAD can function as a prosthetic group.

→ clinic: Vitamin deficiencies can reduce the activity of all enzymes requiring the corresponding coenzyme.

→ clinic: Metal chelation can inhibit metalloenzymes.

Vitamin 2024.pdf, pages 4–12 (cofactors, coenzymes, and prosthetic groups). · Lippincott 6e, Chapter 5, local extraction items 20–25 and 114.

Temperature and pH alter enzyme rate

Raising temperature initially increases molecular collisions and reaction rate, but excessive heat unfolds enzymes and reduces activity. pH changes the protonation of catalytic and substrate-binding residues, and extreme pH can disrupt the protein's stabilizing interactions. Each enzyme therefore has an activity profile suited to its physiological compartment.

1. Rate rises with temperature until structural instability dominates.
2. High heat decreases activity through denaturation.
3. pH controls active-site residue ionization.
4. Extreme pH can disrupt tertiary structure.
5. Pepsin and cytosolic enzymes have different pH optima.

→ clinic: Specimen handling temperature and pH can change measured enzyme activity.

→ clinic: Acidic lysosomes support hydrolases that would be less active in neutral cytosol.

Lippincott 6e, Chapter 5, local extraction items 43–48.

Michaelis-Menten saturation, K_m , and V_{max}

For a simple Michaelis-Menten enzyme, velocity rises hyperbolically with substrate concentration and approaches V_{max} as active sites become saturated. K_m is the substrate concentration at half V_{max} and is often used as an inverse indicator of apparent substrate affinity. V_{max} depends on enzyme concentration, whereas K_m for a given enzyme-substrate system does not.

1. $v = V_{max}[S]/(K_m + [S])$.
2. At $[S] = K_m$, $v = V_{max}/2$.
3. Lower K_m commonly means higher apparent affinity.

4. At saturation, adding substrate does not raise velocity further.

5. Increasing enzyme concentration increases V_{max} .

→ clinic: K_m and V_{max} help distinguish inhibitor mechanisms and compare enzyme variants.

Enzyme Kinetics and regulation- Session2 2024.pptx, slides 7 and 9-10.

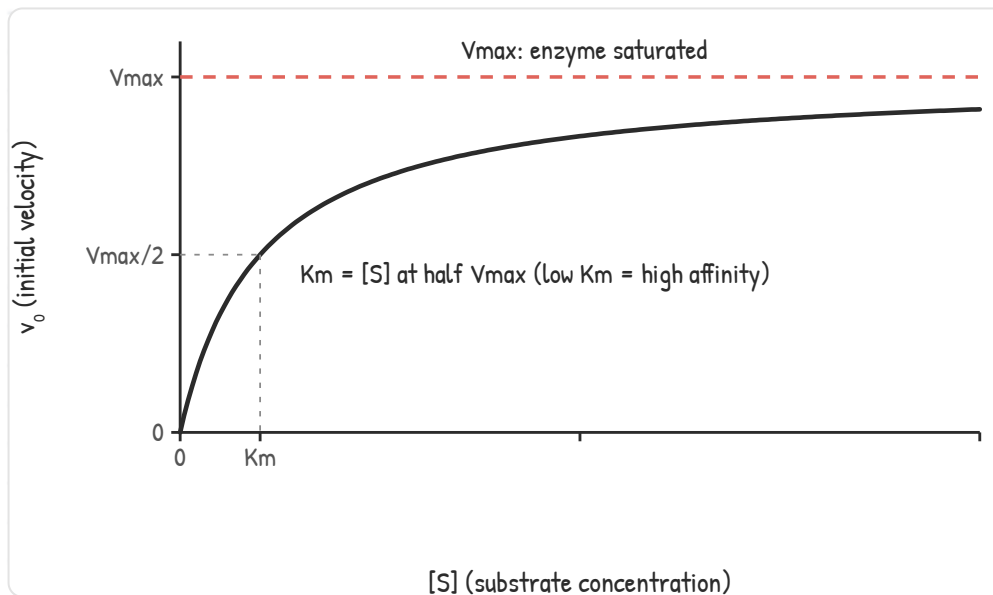


Fig. 7 Michaelis-Menten curve $v_0 = V_{max}[S]/(K_m + [S])$, computed with $K_m = 2$ and $V_{max} = 1$.

Lineweaver-Burk plot interpretation

The Lineweaver-Burk plot linearizes Michaelis-Menten kinetics by plotting $1/v$ against $1/[S]$. Its y-intercept is $1/V_{max}$, x-intercept is $-1/K_m$, and slope is K_m/V_{max} . Although modern fitting generally uses the original nonlinear equation, the plot remains a common exam tool for recognizing parameter changes.

1. y-intercept = $1/V_{max}$.
2. x-intercept = $-1/K_m$.
3. slope = K_m/V_{max} .
4. Competitive and noncompetitive inhibitors produce different intersection patterns.

Enzyme Kinetics and regulation- Session2 2024.pptx, slides 13-14. · Lippincott 6e, Chapter 5, local extraction items 62-65.

competitive inhibition raises apparent K_m

A competitive inhibitor and substrate compete for the enzyme's active site. It increases apparent K_m because more substrate is needed to reach half V_{max} , but V_{max} is unchanged because sufficiently high substrate can outcompete reversible inhibitor. On a Lineweaver-Burk plot, lines share the y-intercept.

1. Competitive inhibitor binds the active site.
2. Apparent K_m increases.
3. V_{max} is unchanged.
4. Increasing substrate can overcome reversible competitive inhibition.

→ clinic: Fomepizole or ethanol competitively limits toxic alcohol metabolism by alcohol dehydrogenase.

→ clinic: Statins competitively inhibit HMG-CoA reductase.

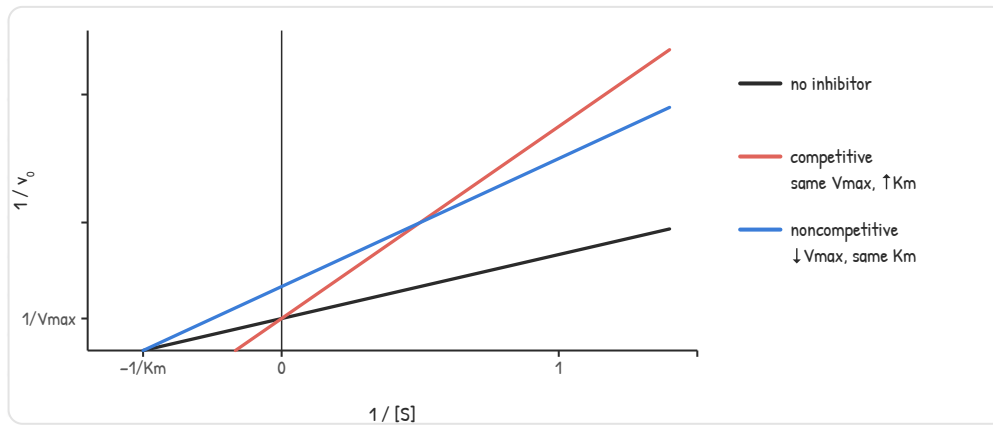


fig. 8 Lineweaver-Burk lines computed from $1/v_0 = (K_m/V_{max})(1/[S]) + 1/V_{max}$: competitive inhibitors share the y-intercept, noncompetitive inhibitors share the x-intercept.

Noncompetitive versus irreversible inhibition

A pure noncompetitive inhibitor binds enzyme and enzyme-substrate complex outside the active site, lowering V_{max} without changing K_m . An irreversible inhibitor permanently inactivates enzyme, commonly through covalent modification, so activity returns only when new enzyme is synthesized. Neither mechanism is overcome simply by increasing substrate concentration.

1. Pure noncompetitive inhibition decreases V_{max} and leaves K_m unchanged.
2. Irreversible inhibition permanently reduces active enzyme concentration.
3. Increasing substrate does not restore the original V_{max} .
4. Real mixed inhibition can change both K_m and V_{max} .

→ clinic: Aspirin irreversibly acetylates cyclooxygenase.

→ clinic: Organophosphates irreversibly inhibit acetylcholinesterase unless treated before aging.

Enzyme Kinetics and regulation- Session2 2024.pptx, slides 19-24. · Lippincott 6e, Chapter 5, local extraction item 82.

Allosteric enzymes control pathway flux

Allosteric effectors bind regulatory sites distinct from the active site and shift an enzyme between activity states. These enzymes are often multisubunit, show sigmoidal rather than simple Michaelis-Menten kinetics, and catalyze rate-limiting or committed steps. Feedback inhibition occurs when a pathway end product inhibits an earlier regulatory enzyme, preventing wasteful overproduction.

1. Allosteric sites are distinct from catalytic sites.
2. Allosteric enzymes often show cooperative sigmoidal kinetics.
3. Regulatory enzymes commonly occur at irreversible pathway-control steps.
4. End-product feedback conserves energy and precursors.

→ clinic: Allosteric drugs can modulate activity without competing with substrate.

Enzyme Kinetics and regulation- Session2 2024.pptx, slides 35-38.

Fast and slow modes of enzyme regulation

Reversible phosphorylation rapidly changes existing enzyme activity and commonly targets serine, threonine, or tyrosine. Proteolytic cleavage activates zymogens irreversibly, which is useful when enzymes must remain off until they reach a specific site. Induction and repression alter enzyme abundance through synthesis or degradation and therefore act more slowly than allosteric or covalent control.

1. Kinases add phosphate; phosphatases remove it.
2. Phosphorylation can activate one enzyme and inhibit another.
3. Zymogen activation is proteolytic and irreversible for that molecule.
4. Induction increases enzyme synthesis over hours to days.
5. Multiple regulation modes can act on the same pathway.

→ **clinic:** Digestive proteases and coagulation factors are synthesized as zymogens.

→ **clinic:** Inappropriate protease activation can cause pancreatitis or uncontrolled coagulation.

Enzyme Kinetics and regulation- Session2 2024.pptx, slides 39, 41, and 43.

Plasma enzymes, isoenzymes, and tissue injury

Many intracellular enzymes appear in plasma at low steady-state concentrations and rise when cells are injured. Isoenzymes catalyze the same reaction but differ in structure and tissue distribution, helping identify a source of damage. Biomarker interpretation requires tissue specificity and time course; for myocardial infarction, cardiac troponin has replaced CK-MB as the preferred marker in most settings.

1. Plasma enzyme elevation can indicate cell injury rather than increased pathway activity in plasma.
2. Isoenzymes differ in sequence, charge, kinetics, or tissue distribution.
3. ALT is relatively useful for hepatocellular injury.
4. Cardiac troponin is the preferred myocardial-injury biomarker.
5. A result must be interpreted with symptoms, timing, and reference intervals.

→ **clinic:** Serial troponin measurements help detect an evolving myocardial injury pattern.

→ **clinic:** An enzyme pattern can support a tissue source but rarely establishes a diagnosis alone.

Exam scope update 2026-08-20: complete confirmed Lippincott scaffold includes Chapter 5. · Lippincott 6e, Chapter 5, local extraction items 96-110.

Phosphorylation changes enzyme conformation and activity

Protein kinases transfer a phosphate from ATP to specific amino-acid side chains, most often serine, threonine, or tyrosine. The new negative charges change intramolecular interactions and can shift an enzyme into a more active or less active conformation. Protein phosphatases remove the phosphate and reverse this regulatory modification. Thus phosphorylation activates glycogen phosphorylase but inhibits glycogen synthase, without changing the equilibrium constant of either reaction.

1. Reversible phosphorylation changes the activity of an existing protein.
2. A phosphate group can alter charge interactions and protein conformation.
3. Phosphorylation can activate or inhibit, depending on the enzyme.
4. Protein kinases add phosphate; protein phosphatases remove it.

creatinine kinase, alkaline phosphatase, and urinary amylase

Enzyme measurements are interpreted using both tissue distribution and removal from the circulation. Skeletal muscle contains predominantly CK-MM, and muscle necrosis in rhabdomyolysis releases large amounts of creatine kinase into blood. The two principal sources of serum alkaline phosphatase are the hepatobiliary system and bone, so an elevated value alone does not identify which tissue is responsible. Amylase is a relatively small enzyme that can undergo glomerular filtration and appear in urine. When amylase forms a large complex with an immunoglobulin, its urinary clearance falls despite increased serum activity.

1. Creatine kinase is a sensitive biochemical marker of skeletal-muscle injury.
2. CK-MM predominates in skeletal muscle.
3. Serum alkaline phosphatase commonly originates from liver/biliary tissue or bone.
4. Amylase can be detected in urine after renal filtration.
5. Macroamylase is poorly filtered because the complex is large.

→ clinic: Rhabdomyolysis elevates CK and may cause myoglobin-associated acute kidney injury.

→ clinic: Cholestasis and increased osteoblastic activity can both elevate ALP.

Lippincott 6e, Chapter 5, pp. 65–67 (enzyme leakage, tissue specificity, and CK isoenzymes); the ALP and urinary-amylase details are supplemented below. · Mayo Clinic Laboratories, Creatine Kinase (CK), Serum, Clinical Information and Interpretation: <https://www.mayocliniclabs.com/test-catalog/Overview/8336> · Mayo Clinic Laboratories, Alkaline Phosphatase, Serum, Interpretation: <https://www.mayocliniclabs.com/test-catalog/Overview/800001> · Mayo Clinic Laboratories, Amylase, Total, Serum, Interpretation and Cautions: <https://www.mayocliniclabs.com/test-catalog/Overview/8352>

past paper asked

Which of the following statements about Michaelis–Menten constant is correct?

A low K_m indicates a high affinity of an enzyme for its substrate. is the best answer. Michaelis–Menten parameters relate substrate concentration to reaction velocity. K_m is independent of total enzyme concentration in the Michaelis–Menten model. Low K_m is often used as an affinity shorthand, but K_m is not universally a dissociation constant. The original wording permits both A and the customary textbook interpretation of B.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q14

past paper asked

When $K_m = S$, reaction velocity (v) would be equal to.....

$1/2 V_{max}$ is the best answer. Michaelis–Menten parameters relate substrate concentration to reaction velocity.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q43

past paper asked

Which of the following can act as a feedback inhibitor of an enzyme?

End product is the best answer. An end product commonly inhibits an early committed enzyme in its own biosynthetic pathway.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q53

past paper asked

Zymogen or proenzyme is a?

Enzyme precursor is the best answer. A zymogen is an inactive enzyme precursor activated by a later step such as proteolysis.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q65

past paper asked

The effect of reversible competitive inhibitor can be canceled out by?

Increasing the substrate concentration is the best answer. High substrate can outcompete a reversible competitive inhibitor for the active site.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q44

past paper asked

Which enzyme class catalyzes the reaction shown?

Lactate dehydrogenase couples lactate oxidation to NAD⁺ reduction. It is an oxidoreductase.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q22

past paper asked

When $S=3K_m$, the V is equal to..... ?

$3/4 V_{max}$ is the best answer. Michaelis-Menten parameters relate substrate concentration to reaction velocity.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q35

past paper asked

Ibuprofen is a reversible inhibitor of the COX-1 enzyme which is involved in Arachidonic Acid metabolism. Blocking of enzyme action by blocking its active site is called as?

Competitive inhibition is the best answer. Ibuprofen reversibly competes at the cyclooxygenase active site; aspirin inhibits COX irreversibly by acetylation.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q49

past paper asked

The catalytic efficiency of two enzymes acting on a substrate is commonly compared using which parameter?

Editorial repair: the source offered K_m alone. k_{cat}/K_m , not K_m alone, is the specificity constant used for low-substrate catalytic performance.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q8

past paper asked

Which one of the following is a reversible inhibitor of the COX-1 enzyme that is involved in prostaglandin synthesis?
Ibuprofen is the best answer. Ibuprofen reversibly competes at the cyclooxygenase active site; aspirin inhibits COX irreversibly by acetylation.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q53

past paper asked

A non-competitive inhibitor of an enzyme ...

Decreases V_{max} with no change in K_m is the best answer. Pure noncompetitive inhibition lowers V_{max} without changing K_m .

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q5

past paper asked

Which of the following can restore the activity of an enzyme already irreversibly inhibited?

The original four options cannot reverse irreversible enzyme inactivation. Adding E prevents teaching an incorrect source key.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q12

Test yourself

1. What is feedback inhibition?

- A. An end product inhibits an earlier regulatory enzyme in its pathway
- B. A substrate permanently destroys its enzyme
- C. Every enzyme activates the next one
- D. A ribosome stops at a stop codon

2. A lower K_m generally indicates what about enzyme-substrate interaction?

- A. Higher apparent substrate affinity
- B. Lower V_{max} in every case
- C. Irreversible inhibition
- D. Loss of the active site

3. How does a pure competitive inhibitor affect Michaelis-Menten parameters?

- A. It decreases both K_m and V_{max}
- B. It decreases V_{max} without changing K_m
- C. It increases apparent K_m without changing V_{max}
- D. It increases V_{max}

Answers: 1 A · 2 A · 3 C

Bioenergetics and oxidative phosphorylation

Lippincott 6e, Chapter 6; teacher slides cited under each idea.

Free energy, reaction direction, and energetic coupling

A negative actual free-energy change (ΔG) means a reaction is thermodynamically favorable in the written direction; $\Delta G = 0$ at equilibrium. Actual ΔG depends on concentrations, so it can differ from standard ΔG° . Cells drive an endergonic step by coupling it to a sufficiently exergonic process such as ATP hydrolysis, making the summed ΔG negative.

1. ΔG predicts feasibility, not reaction rate
2. Actual concentrations can change the sign of ΔG
3. Free-energy changes of coupled reactions are additive
4. ATP is kinetically stable but has high phosphate-transfer potential

→ clinic: Energy failure during hypoxia

→ clinic: ATP-dependent ion-pump failure

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 6, bioenergetics and free energy (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Electron-transport chain organization and electron flow

Electrons from NADH enter at Complex I and those from succinate-linked FADH₂ enter at Complex II; both routes converge on coenzyme Q, Complex III, cytochrome c, and Complex IV. Oxygen is the terminal electron acceptor and is reduced to water. Complexes I, III, and IV pump protons, whereas Complex II does not.

1. ETC is in the inner mitochondrial membrane
2. CoQ and cytochrome c are mobile carriers
3. Complex II is also succinate dehydrogenase
4. O₂ accepts electrons at Complex IV

→ clinic: Cyanide and carbon-monoxide poisoning

→ clinic: Mitochondrial respiratory-chain disease

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 6, electron transport (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

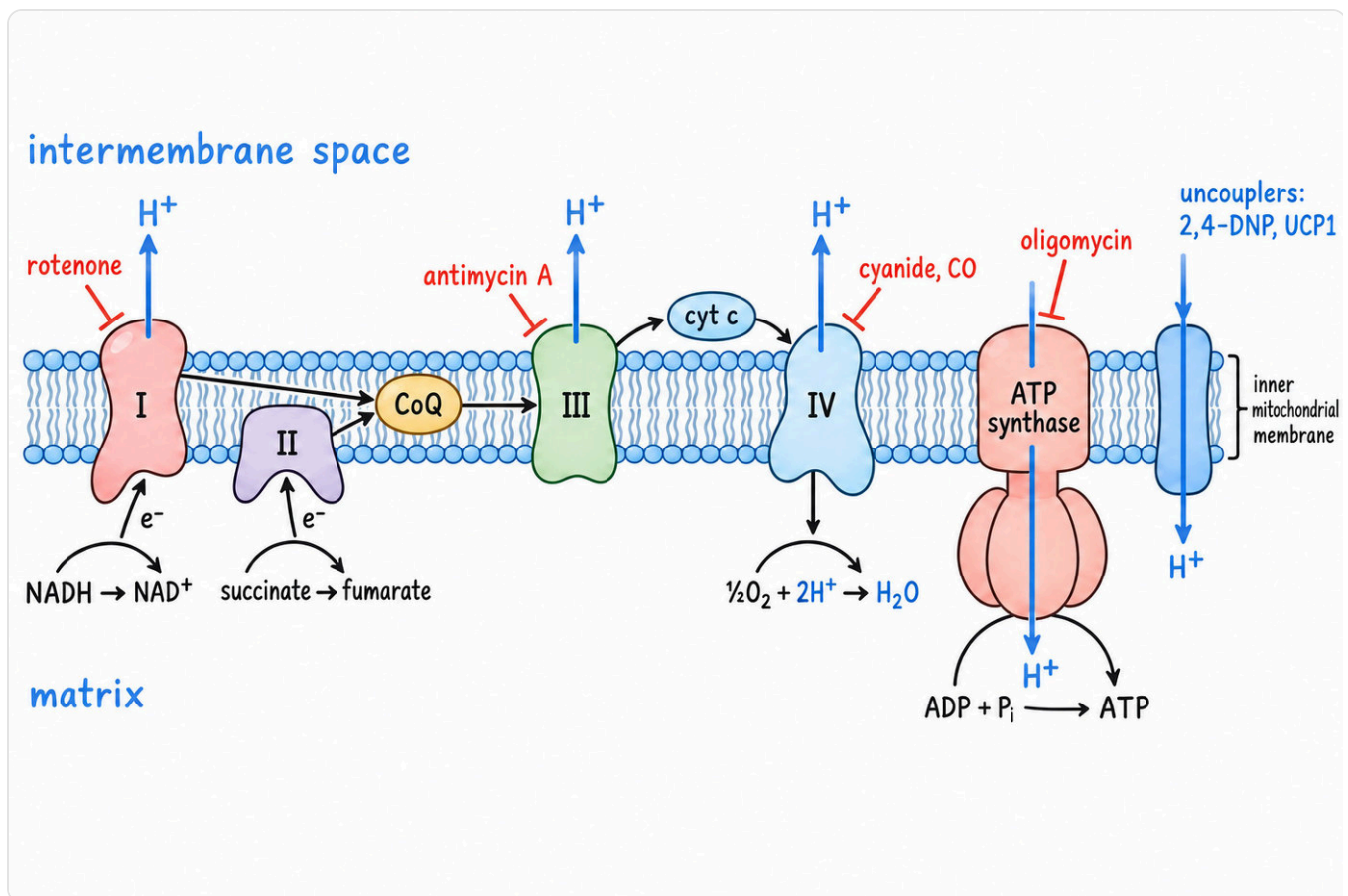


fig. 9 Electron transport across the inner mitochondrial membrane, proton pumping, ATP synthesis, inhibitors, and uncoupling.

chemiosmosis, ATP synthase, and oxidative phosphorylation

Electron transport pumps protons from the matrix into the intermembrane space, creating an electrochemical gradient. Proton return through the F_o portion of Complex V drives conformational changes in F_1 that synthesize ATP. Electron flow, proton pumping, and phosphorylation are therefore functionally coupled.

1. Proton-motive force is the coupling intermediate
2. F_o conducts protons
3. F_1 catalyzes ATP formation
4. ETC blockade prevents maintenance of the gradient

→ clinic: Oligomycin inhibition of ATP synthase

→ clinic: Tissue vulnerability in severe hypoxia

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 6, oxidative phosphorylation (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Respiratory-chain inhibitors versus uncouplers

A respiratory-chain inhibitor stops electron flow at a defined complex, decreases oxygen consumption, and prevents ATP generation. An uncoupler allows electron transport and oxygen use to continue or accelerate while dissipating the proton gradient as heat, so ATP production falls. Rotenone/amytal inhibit Complex I, antimycin A inhibits Complex III, and cyanide or carbon monoxide inhibit Complex IV.

1. Inhibitor: electron flow and oxygen consumption fall

2. Uncoupler: oxygen consumption rises while ATP falls
3. UCP1 supports physiologic thermogenesis
4. DNP is a dangerous synthetic uncoupler

→ clinic: Cyanide poisoning

→ clinic: Salicylate toxicity

→ clinic: Brown-fat thermogenesis

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 6, inhibitors and uncoupling (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

cytosolic NADH shuttles and reactive oxygen species

The inner mitochondrial membrane lacks a direct NADH transporter, so reducing equivalents enter through the malate-aspartate or glycerol-3-phosphate shuttle. The malate-aspartate shuttle preserves transfer to mitochondrial NADH and the higher ATP yield. Electron leakage can partially reduce oxygen to reactive oxygen species, opposed by superoxide dismutase, catalase, and glutathione peroxidase.

1. Shuttles move reducing equivalents, not NADH itself
2. Malate-aspartate is more energy efficient
3. ROS damage lipids, proteins, and DNA
4. Antioxidant enzymes limit mitochondrial oxidant injury

→ clinic: Oxidative stress

→ clinic: Mitochondrial disease

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 6, shuttles and ROS (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Dehydrogenases, flavins, and cytochrome a/a3

Dehydrogenases catalyze oxidation-reduction reactions by transferring hydrogen or electrons to acceptors such as NAD⁺ or FAD; molecular oxygen need not be a direct substrate. NADH supplies electrons to Complex I through FMN, whereas succinate dehydrogenase contains covalently attached FAD and constitutes Complex II. Complex IV contains cytochromes a and a₃ together with copper centers and transfers electrons ultimately to oxygen. Cytochrome c is a separate mobile carrier that carries electrons from Complex III to Complex IV.

1. Complex I uses FMN; succinate dehydrogenase/Complex II uses FAD.
2. Complex II transfers electrons to coenzyme Q without pumping protons.
3. Cytochromes a and a₃ belong to Complex IV, cytochrome c oxidase.
4. Cytochrome c transfers single electrons between Complexes III and IV.
5. Dehydrogenases transfer reducing equivalents without directly requiring O₂ as their acceptor.

Lippincott 6e, Chapter 6, pp. 73-76 (ETC carrier sequence and dehydrogenases), and Chapter 9, pp. 112-113 (succinate dehydrogenase-bound FAD). See also Chapter 28, p. 380 (flavin binding to enzymes).

ADP/ATP exchange and atractyloside inhibition

The inner mitochondrial membrane requires specific carriers for adenine nucleotides and inorganic phosphate. Adenine nucleotide translocase exchanges cytosolic ADP for ATP made in the matrix, supplying ATP synthase with substrate while exporting its product. Atractyloside inhibits this ADP/ATP carrier and therefore restricts oxidative phosphorylation by blocking nucleotide exchange. The phosphate carrier is a separate transport system, and neither transporter is the carnitine shuttle.

1. Adenine nucleotide translocase is also called the ADP/ATP antiporter.
2. One ADP enters the matrix as one ATP leaves.
3. Atractyloside blocks the adenine nucleotide carrier.
4. Phosphate transport and carnitine-acylcarnitine transport use different proteins.

Lippincott 6e, Chapter 6, pp. 78-79 (mitochondrial ATP/ADP and phosphate transport). · Supplement for atractyloside specificity: Identification of adenine nucleotide translocase 4 inhibitors by molecular docking (2013), Introduction: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3821932/>

cytochrome c release initiates intrinsic apoptosis

Cytochrome c normally functions in electron transport on the outer face of the inner mitochondrial membrane. Permeabilization of the outer mitochondrial membrane can release it from the intermembrane space into the cytosol. There, cytochrome c associates with Apaf-1 and other components to assemble the apoptosome, activating initiator caspase-9 and downstream executioner caspases. This links mitochondrial injury to the intrinsic pathway of programmed cell death.

1. The relevant redistribution is from the intermembrane space into the cytosol.
2. Cytosolic cytochrome c promotes apoptosome assembly with Apaf-1.
3. Caspase activation drives proteolysis in intrinsic apoptosis.

→ **clinic:** Mitochondrial outer-membrane permeabilization is an important control point in cell death.

Lippincott 6e, Chapter 6, pp. 79-80 (mitochondria and apoptosis). · Supplement for Apaf-1/caspase-9: Cytochrome c and dATP-dependent formation of Apaf-1/caspase-9 complex initiates an apoptotic protease cascade (Cell, 1997): <https://pubmed.ncbi.nlm.nih.gov/9390557/>

High-energy phosphate compounds versus glucose 6-phosphate

High-energy phosphate compounds have a large negative free-energy change of hydrolysis and can support favorable phosphoryl-group transfer. Phosphoenolpyruvate and phosphocreatine have high phosphoryl-transfer potential, and carbamoyl phosphate contains an energy-rich mixed anhydride linkage. Glucose 6-phosphate is an ordinary phosphate ester with a lower transfer potential and is not classified with these high-energy compounds. The term refers to the energetics of the complete reaction, not to energy being released simply by breaking an isolated bond.

1. Phosphoenolpyruvate is a high-energy enol phosphate.
2. Phosphocreatine provides a rapidly transferable phosphate reserve.
3. Carbamoyl phosphate contains a high-energy mixed anhydride.
4. Glucose 6-phosphate is the low-energy phosphate ester in this comparison.

Lippincott 6e, Chapter 6, pp. 72-73 (ATP and energy coupling); Chapter 8, pp. 97-103 (glucose phosphorylation and high-energy glycolytic intermediates); Chapter 19, pp. 253-254 (carbamoyl phosphate); Chapter 21, pp. 287-288 (creatine phosphate).

ATP accounting: glycolysis, shuttles, and complete glucose

oxidation

Glycolysis itself yields two net ATP and two cytosolic NADH per glucose, and the ATP obtained from those NADH depends on the shuttle and the P/O convention used. Lippincott 6e uses the historical approximation of three ATP per mitochondrial NADH and two per FADH₂, giving six additional ATP from the two glycolytic NADH when the malate-aspartate shuttle is assumed. Modern textbook calculations commonly use approximately 2.5 ATP per NADH and 1.5 per FADH₂. Under that convention, complete glucose oxidation yields about 32 ATP with the malate-aspartate shuttle or 30 ATP with the glycerol-3-phosphate shuttle. These are theoretical accounting estimates rather than fixed measurements in every cell.

1. Direct glycolytic yield: two net ATP per glucose.
2. Two glycolytic NADH: historical maximum six ATP; modern maximum five ATP.
3. Modern complete glucose oxidation: approximately 30–32 ATP.
4. Modern total includes four substrate-level ATP equivalents plus oxidative phosphorylation.
5. Always state the P/O ratio and cytosolic NADH shuttle when comparing totals.

Lippincott 6e, Chapter 6, pp. 77–79; Chapter 8, p. 104; Chapter 9, pp. 113–114 (historical ATP convention and pathway yields). · carbohydrate metabolism MD Kish.pdf, pp. 79–80 (TCA products and 32-ATP glucose calculation). · Hinkle et al., Mechanistic stoichiometry of mitochondrial oxidative phosphorylation, Biochemistry 1991;30:3576–3582, <https://pubmed.ncbi.nlm.nih.gov/2012815/> (experimental basis of approximately 2.5/1.5 P/O estimates).

Test yourself

1. Cyanide and carbon monoxide can rapidly stop oxidative phosphorylation by inhibiting which complex?

- | | |
|----------------------|---------------|
| A. ATP-citrate lyase | B. Complex IV |
| C. Complex I | D. Complex II |

2. Rotenone and amytal primarily inhibit which respiratory-chain complex?

- | | |
|---------------|----------------|
| A. Complex II | B. Complex III |
| C. Complex IV | D. Complex I |

3. What enzymes defend against ROS generated by the ETC?

- | | |
|---|--|
| A. Four electrons are required to reduce one O ₂ to two molecules of H ₂ O. | B. Ubiquinone, because it is ubiquitous (found in all biologic systems). |
| C. F is the Faraday constant, approximately 23.1 kcal/volt-mol. | D. Superoxide dismutase (SOD), catalase, and glutathione peroxidase. |

Answers: 1 B · 2 D · 3 D

carbohydrate structure and stereochemistry

Lippincott 6e, Chapter 7; teacher slides cited under each idea.

Disaccharides and bonds

Sugar	Building blocks / linkage	Reducing?
Maltose	Glucose- $\alpha(1\rightarrow4)$ -glucose	Yes
Cellobiose	Glucose- $\beta(1\rightarrow4)$ -glucose	Yes
Lactose	Galactose- $\beta(1\rightarrow4)$ -glucose	Yes
Sucrose	Glucose- $\alpha(1\leftrightarrow2)$ - β -fructose	No: both anomeric carbons linked
Trehalose	Glucose- $\alpha(1\leftrightarrow1)$ - α -glucose	No: both anomeric carbons linked
Amylopectin / glycogen	$\alpha(1\rightarrow4)$ chains, $\alpha(1\rightarrow6)$ branches	Glycogen is more highly branched
Cellulose / chitin	$\beta(1\rightarrow4)$ glucose / N-acetylglucosamine	Structural homopolymers

Monosaccharide classification and stereochemistry

Monosaccharides are classified by carbonyl type and carbon number. Epimers differ at one chiral carbon, while enantiomers are mirror images; D-glucose and D-galactose are C4 epimers, and glucose and mannose are C2 epimers. Cyclization creates an anomeric carbon, and alpha and beta forms differ at that carbon.

1. Aldose versus ketose
2. Epimer differs at one chiral center
3. Anomer differs at the anomeric carbon
4. D/L does not specify optical rotation

→ **clinic:** Galactose and fructose metabolic disorders

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 7, monosaccharide structure (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Glycosidic bonds and reducing sugars

A glycosidic bond links the anomeric carbon of a sugar to an oxygen or nitrogen on another molecule. A sugar is reducing when it retains a free anomeric carbon capable of opening to a reactive carbonyl. All monosaccharides are reducing; lactose and maltose are reducing, whereas sucrose is not because both anomeric carbons participate in the bond.

1. Free anomeric carbon permits reduction tests
2. Glycosidic linkages are named by alpha/beta and carbon positions
3. Sucrose is nonreducing
4. Benedict reagent detects reducing sugars

→ clinic: Reducing substances in urine

→ clinic: Carbohydrate qualification practical

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 7, glycosidic bonds (textbook-confirmed). · Carbohydrate Qualification Test Esmaili Jan 2026 — 21-page teacher practical deck (teacher-confirmed practical context). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Dietary disaccharides and carbohydrate digestion

Starch digestion begins with alpha-amylases, whereas final carbohydrate digestion occurs at the small-intestinal brush border. Lactase, sucrase-isomaltase, and maltase produce glucose, galactose, and fructose for absorption. Undigested carbohydrate retains water and is fermented by colonic bacteria, producing acidic diarrhea, gas, and bloating.

1. Lactose = glucose + galactose
2. Sucrose = glucose + fructose
3. Brush-border enzymes finish digestion
4. Lactase is commonly deficient

→ clinic: Lactose intolerance

→ clinic: Congenital sucrase-isomaltase deficiency

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 7, digestion of carbohydrates (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Intestinal monosaccharide absorption

Glucose and galactose enter enterocytes apically through sodium-dependent SGLT1, whereas fructose uses facilitated diffusion through GLUT5. All three leave basolaterally mainly through GLUT2. The sodium gradient that powers SGLT1 is maintained indirectly by the basolateral Na⁺/K⁺-ATPase.

1. SGLT1 transports glucose and galactose
2. GLUT5 transports fructose
3. GLUT2 supports basolateral exit
4. Na⁺/K⁺-ATPase maintains the driving gradient

→ clinic: Oral rehydration therapy

→ clinic: Glucose-galactose malabsorption

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 7, monosaccharide uptake (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

cyclic sugars, hemiacetals, and the ring oxygen

An aldose such as glucose cyclizes when its aldehyde reacts with a hydroxyl group on the same molecule, producing an intramolecular hemiacetal. A pyranose ring contains five carbons and one oxygen. Its ring oxygen is bonded to two carbons, so that portion of the ring is an ether-like C–O–C linkage, while the anomeric carbon carries both the ring oxygen and a hydroxyl group in the free sugar. Cyclization creates the anomeric carbon and allows alpha and beta anomers to form.

1. Glucopyranose is a cyclic hemiacetal.
2. A pyranose ring has six ring atoms: five C and one O.
3. The ring oxygen forms a C–O–C linkage.

4. The former carbonyl carbon becomes the anomeric carbon.

Lippincott 6e, Chapter 7, pp. 84–85, Figure 7.6 (cyclization and pyranose structure).

Starch, glycogen, and cellulose

Starch is the principal storage polysaccharide in plants and contains amylose and amylopectin. Amylose is an unbranched $\alpha(1\rightarrow4)$ -linked glucose polymer, whereas amylopectin also contains $\alpha(1\rightarrow6)$ branch points. Glycogen is the highly branched storage glucose polymer in animals. Cellulose is a structural plant polymer of $\beta(1\rightarrow4)$ -linked glucose, and human digestive enzymes cannot hydrolyze these beta linkages.

1. Plant glucose storage: starch.
2. Animal glucose storage: glycogen.
3. Amylose: $\alpha(1\rightarrow4)$ bonds without branches.
4. Amylopectin and glycogen: $\alpha(1\rightarrow4)$ chains with $\alpha(1\rightarrow6)$ branches.
5. Cellulose: $\beta(1\rightarrow4)$ -linked glucose; indigestible to human enzymes.

Lippincott 6e, Chapter 7, pp. 85–87; Chapter 11, pp. 126–128 (polysaccharides, starch digestion, and branching).

past paper asked

A patient gets abdominal pains and cramps, diarrhea and gas in intestines after consuming milk and dairy products. His problem is related to a defect in which sugar and enzyme below?

Lactose-lactase is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q37

past paper asked

All the following compounds possess a chiral carbon, except

Dihydroxyacetone is the best answer. Dihydroxyacetone has no stereogenic carbon.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q43

past paper asked

Which of the following are anomeric isomers?

α -D-Glucose and β -D-glucose is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q46

past paper asked

Which of the following is a homo-polysaccharide?

Cellulose is the best answer. Cellulose is a homopolymer of glucose.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q56

past paper asked

The enzyme which hydrolyses starch to maltose is

Amylase is the best answer. Amylase cleaves $\alpha(1\rightarrow4)$ bonds in starch, producing maltose and oligosaccharides.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q62

past paper asked

Toothpaste and sugar-free gum are probably sweetened with a sugar alcohol made by reducing the aldehyde carbon of xylose to an alcohol. The resulting compound would be named

Xylitol is the best answer. Reduction of xylose yields the sugar alcohol xylitol.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q64

past paper asked

In which of the following disaccharides the glycosidic linkage is in between anomeric carbons of both monosaccharides?

Sucrose is the best answer. Sucrose links both anomeric carbons, making it nonreducing.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q66

past paper asked

The phenomenon of mutarotation is due to which of the following?

Conversion of α -D-Glucose to β -D-Glucose is the best answer. Mutarotation is interconversion of α and β anomers through the open-chain form.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q28

past paper asked

Which of the following is a deoxy sugar?

L-Fucose is the best answer. L-Fucose is 6-deoxy-L-galactose.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q31

past paper asked

Galactonic acid is caused by oxidation of which factor in galactose?

Aldehyde group is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior. Oxidation of the aldehyde group gives the aldonic acid. An aldehyde contains a carbonyl group, so both the broad and specific choices describe that same site.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q32

past paper asked

The enzyme which hydrolyses Galactose -Glucose linkage is

Lactase is the best answer. Lactase hydrolyzes the $\beta(1 \rightarrow 4)$ galactose-glucose bond of lactose.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q48

past paper asked

Which option is correct for Cellobiose?

It has $\beta 1 \rightarrow 4$ glycosidic bond is the best answer. Cellobiose consists of glucose residues joined by a $\beta(1 \rightarrow 4)$ glycosidic bond.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q50

past paper asked

What units make up Lactose with what bonds?

β -galactose+glucose, $\beta 1 \rightarrow 4$ is the best answer. Lactose is β -galactose linked $\beta(1 \rightarrow 4)$ to glucose.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q52

past paper asked

How many spatial isomers does the linear form of aldohexoses have?

16 is the best answer. An aldohexose has four stereocenters, giving $2^4 = 16$ stereoisomers.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q54

past paper asked

Which of the following produces N-acetyl glucosamine upon hydrolysis?

Chitin is the best answer. Chitin is a polymer of $\beta(1 \rightarrow 4)$ -linked N-acetylglucosamine.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q62

past paper asked

Amylopectin has:

$\alpha(1 \rightarrow 4)$ and $\alpha(1 \rightarrow 6)$ linkage is the best answer. Amylopectin contains $\alpha(1 \rightarrow 4)$ glucose chains with $\alpha(1 \rightarrow 6)$ branch points.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q61

past paper asked

Which of the following is a pair of non-reducing carbohydrate?

Sucrose and trehalose is the best answer. Sucrose and trehalose have both anomeric carbons tied up in their glycosidic bonds, so neither is reducing.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q46

past paper asked

Which of the following disaccharides contains a beta-glycosidic bond?

Lactose and cellobiose is the best answer. Monosaccharide composition and glycosidic linkage define carbohydrate behavior. The intended comparison is lactose/cellobiose (both beta-1,4). Sucrose also has a beta-D-fructofuranosyl anomeric linkage, so the literal wording also allows lactose/sucrose.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q49

past paper asked

In polysaccharides, monosaccharides are joined by:

Glycosidic bonds is the best answer. Monosaccharides are joined to one another by glycosidic bonds.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q23

past paper asked

The enzyme which hydrolyses Glucose-Fructose linkage is....

Sucrase is the best answer. Sucrase hydrolyzes the glucose-fructose glycosidic bond of sucrose.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q56

past paper asked

Structural polysaccharides include:

Cellulose and chitin is the best answer. Cellulose and chitin are structural polysaccharides; glycogen and starch are storage polysaccharides.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q63

Test yourself

1. What transporter is responsible for fructose uptake into intestinal mucosal cells and what kind of transport is it?
 - A. GLUT-5, which mediates energy- and sodium-independent facilitated diffusion of fructose.
 - B. Lactose: galactose + glucose; Sucrose: glucose + fructose; Maltose: glucose + glucose.
 - C. They are transmembrane proteins of the brush border (luminal surface) of intestinal mucosal cells.
 - D. Fructose is a ketose and often exists as a five-membered furanose ring in solution.
2. In what two anatomic sites does most dietary carbohydrate digestion occur?
 - A. About 36% of glucose in solution is in the α form at equilibrium.
 - B. Glucose and mannose are C-2 epimers because they differ only at carbon 2.
 - C. In the mouth and in the intestinal lumen (mainly small intestine).
 - D. They are C-4 epimers: they differ only in configuration around carbon 4.
3. Give three important dietary disaccharides and their monosaccharide components.
 - A. Maltase cleaves $\alpha(1\rightarrow4)$ glycosidic bonds in maltose and maltotriose, producing glucose.
 - B. They are mostly polysaccharides and disaccharides; very little monosaccharide is present in such diets.
 - C. Lactose: galactose + glucose; Sucrose: glucose + fructose; Maltose: glucose + glucose.
 - D. GLUT-5, which mediates energy- and sodium-independent facilitated diffusion of fructose.

Answers: 1 A · 2 C · 3 C

Glycolysis

Lippincott 6e, Chapter 8; teacher slides cited under each idea.

practice-bank topic · outside the confirmed Term 1 syllabus, taught because the practice questions test it

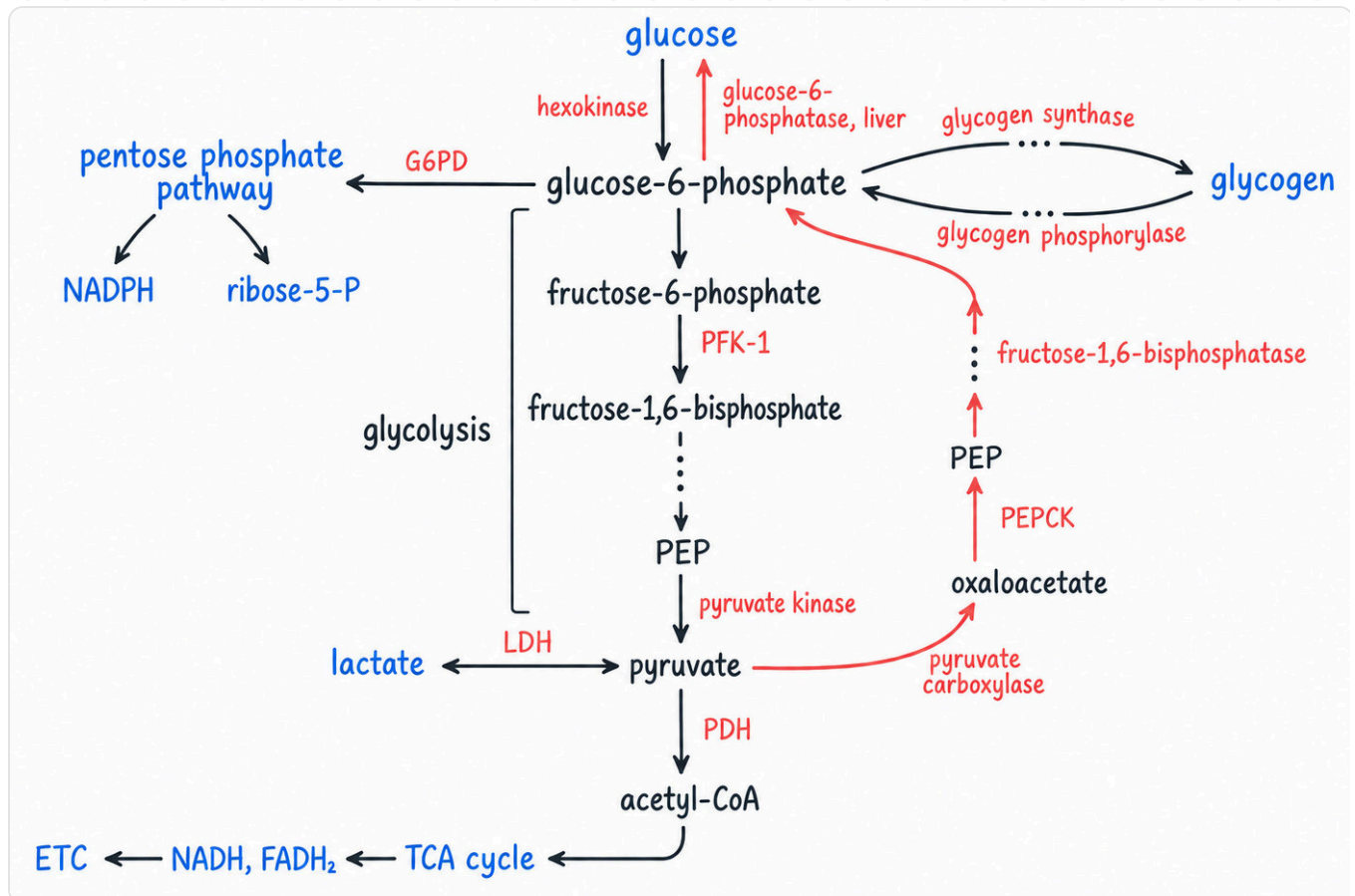


Fig. 10 Glucose-6-phosphate links glycogen turnover, the pentose phosphate pathway, glycolysis, and gluconeogenesis.

Glycolysis: purpose, phases, location, and net yield

Glycolysis is a cytosolic pathway converting one glucose to two pyruvate through an investment phase and a payoff phase. Net substrate-level yield is two ATP and two NADH per glucose. Under anaerobic conditions, pyruvate is reduced to lactate to regenerate NAD⁺, so the net energy yield remains two ATP.

1. Occurs in cytosol
2. One glucose yields two pyruvate
3. Net 2 ATP and 2 NADH aerobically
4. Lactate formation regenerates NAD⁺

→ clinic: Exercise-associated lactate production

→ clinic: RBC dependence on glycolysis

Hexokinase versus glucokinase

Hexokinases I–III have low K_m , are active at low glucose, and are inhibited by glucose 6-phosphate in most tissues. Glucokinase in liver and pancreatic beta cells has a higher K_m and V_{max} and is not directly inhibited by glucose 6-phosphate. Its properties allow liver to take up excess postprandial glucose and beta cells to sense rising glucose.

1. Hexokinase: low K_m , low V_{max}
2. Glucokinase: high K_m , high V_{max}
3. Glucokinase is in liver and beta cells
4. Both trap glucose as glucose 6-phosphate

→ clinic: MODY2 from glucokinase variants

→ clinic: Glucose sensing by pancreatic beta cells

Irreversible glycolytic steps and reciprocal regulation

Hexokinase/glucokinase, PFK-1, and pyruvate kinase catalyze the major irreversible glycolytic steps. PFK-1 is activated by AMP and fructose 2,6-bisphosphate and inhibited by ATP and citrate. In liver, glucagon lowers fructose 2,6-bisphosphate and phosphorylates pyruvate kinase, suppressing glycolysis during fasting.

1. PFK-1 is the committed and rate-limiting step
2. Fructose 2,6-bisphosphate activates PFK-1
3. ATP and citrate inhibit PFK-1
4. Glucagon inhibits hepatic glycolysis

→ clinic: Fed-fast transition

→ clinic: Cancer-associated high glycolytic flux

Anaerobic glycolysis, erythrocytes, and 2,3-BPG

Mature erythrocytes lack mitochondria, so glycolysis is their only ATP source and lactate is their end product. The Rapoport-Luebering shunt diverts 1,3-BPG to 2,3-BPG, lowering hemoglobin oxygen affinity at the cost of bypassing an ATP-generating step. Pyruvate-kinase deficiency therefore causes chronic hemolytic anemia from inadequate RBC ATP.

1. RBCs cannot perform TCA or oxidative phosphorylation
2. LDH regenerates NAD^+
3. 2,3-BPG promotes oxygen unloading
4. Pyruvate-kinase deficiency causes hemolysis

→ clinic: Hemolytic anemia

→ clinic: Adaptation to hypoxia and altitude

→ clinic: Lactic acidosis

Carbohydrate Metabolism, MD Kish — pyruvate fate and tissue-use slides (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 8 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8-13 remain excluded unless separately confirmed.

Major metabolic fates of pyruvate

Pyruvate can become lactate through LDH, acetyl-CoA through PDH, oxaloacetate through pyruvate carboxylase, or alanine by transamination. Oxygen availability, mitochondrial capacity, hormonal state, and tissue determine which fate dominates. The branch point connects glycolysis to TCA oxidation, gluconeogenesis, amino-acid transport, and anaerobic NAD⁺ regeneration.

1. LDH: pyruvate to lactate
2. PDH: pyruvate to acetyl-CoA
3. Pyruvate carboxylase: pyruvate to oxaloacetate
4. ALT interconverts pyruvate and alanine

→ clinic: Hypoxic lactic acidosis

→ clinic: Thiamine-deficiency lactic acidosis

→ clinic: Glucose-alanine cycle

Carbohydrate Metabolism, MD Kish — pp. 22-26 and 61-82 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8-13 remain excluded unless separately confirmed.

Hexokinase substrate specificity and glucose trapping

Hexokinases I-III transfer a phosphate from ATP to glucose and can also phosphorylate several other hexoses. Their low K_m permits efficient glucose phosphorylation even at low glucose concentrations, while their low V_{max} limits the rate of phosphate trapping. Accumulating glucose 6-phosphate inhibits these isozymes when its further use slows. Glucokinase has a higher K_m and V_{max} and is not directly inhibited by glucose 6-phosphate, enabling hepatic handling of a carbohydrate-rich meal.

1. Hexokinases I-III have broad hexose substrate specificity.
2. Hexokinases I-III: low K_m , low V_{max} .
3. Glucose 6-phosphate directly inhibits hexokinases I-III.
4. Phosphorylation traps glucose as intracellular glucose 6-phosphate.
5. Glucokinase supports liver glucose uptake at high glucose concentrations.

Lippincott 6e, Chapter 8, pp. 98-99 (hexokinase and glucokinase).

PFK-1 chemistry, PFK-2, and hepatic glycolytic regulation

PFK-1 uses ATP to irreversibly phosphorylate fructose 6-phosphate to fructose 1,6-bisphosphate, the committed and rate-limiting reaction of glycolysis. Fructose 2,6-bisphosphate is its most potent allosteric activator and is synthesized from fructose 6-phosphate by the distinct enzyme PFK-2. The hepatic bifunctional PFK-2/FBPase-2 protein makes or degrades this regulatory metabolite. During fasting, glucagon raises cAMP and activates PKA, whose phosphorylation favors the hepatic FBPase-2 activity and lowers fructose 2,6-bisphosphate. PKA also phosphorylates and inhibits hepatic pyruvate kinase, helping direct phosphoenolpyruvate toward gluconeogenesis.

1. PFK-1: fructose 6-phosphate + ATP $\rightarrow$ fructose 1,6-bisphosphate + ADP.
2. PFK-2 makes fructose 2,6-bisphosphate, an allosteric regulator.
3. Fructose 2,6-bisphosphate activates PFK-1 and inhibits FBPase-1.
4. Glucagon-cAMP-PKA suppresses hepatic glycolysis.
5. Hepatic pyruvate kinase is inhibited by phosphorylation.

Lippincott 6e, Chapter 8, pp. 99-102; Chapter 10, pp. 120-123 (PFK regulation and hepatic covalent control).

Glycolytic enzymes and substrate-level ATP formation

Human glycolysis uses hexokinase or glucokinase, phosphoglucose isomerase, PFK-1, aldolase, and triose phosphate isomerase in its investment phase. The payoff phase uses glyceraldehyde-3-phosphate dehydrogenase, phosphoglycerate kinase, phosphoglycerate mutase, enolase, and pyruvate kinase; glucose oxidase is not a glycolytic enzyme. Aldolase cleaves intracellular fructose 1,6-bisphosphate into two triose phosphates and is not a pancreatic digestive enzyme. Oxidation of glyceraldehyde 3-phosphate forms the high-energy acyl phosphate 1,3-bisphosphoglycerate, which donates phosphate to ADP through phosphoglycerate kinase. Pyruvate kinase makes ATP from phosphoenolpyruvate, giving four ATP produced per glucose before subtracting the two ATP invested, for a net yield of two.

1. 1,3-Bisphosphoglycerate contains a high-energy acyl-phosphate bond at C1.
2. Phosphoenolpyruvate contains a high-energy enol-phosphate bond.
3. Phosphoglycerate kinase and pyruvate kinase generate ATP.
4. Glycolysis forms four gross ATP and two net ATP per glucose.
5. Glucose oxidase is absent from human glycolysis.
6. Aldolase functions inside cells; pancreatic amylase, lipase, and trypsin act in digestion.

Lippincott 6e, Chapter 8, pp. 98-104 (glycolytic reactions and ATP accounting); Chapter 7, pp. 86-87; Chapter 15, pp. 174-175; Chapter 19, pp. 248-249 (pancreatic digestive enzymes). · carbohydrate metabolism MD Kish.pdf, pp. 10-20 (investment and payoff reactions).

Glycolysis, oxygen availability, and circulatory failure

Glycolysis oxidizes glucose to pyruvate or lactate, supplies ATP and biosynthetic intermediates, and accepts carbon from other dietary sugars through shared intermediates. When mitochondrial oxidative capacity and oxygen supply are adequate, glycolytic pyruvate can proceed to oxidation and cytosolic NADH can be reoxidized through mitochondrial shuttles. When oxygen delivery fails or mitochondria are absent, lactate dehydrogenase reduces pyruvate to lactate while oxidizing NADH to NAD⁺, sustaining glycolysis. Circulatory collapse therefore impairs oxidative ATP production and increases reliance on the two net ATP supplied by anaerobic glycolysis. Lactate accumulation accompanies the metabolic acidosis of severe hypoperfusion, although the lactate dehydrogenase reaction itself consumes a proton rather than producing one.

1. With mitochondria and adequate oxygen, the direct glycolytic end product is pyruvate.
2. LDH reaction: pyruvate + NADH + H⁺ $\rightleftharpoons$ lactate + NAD⁺.

3. Anaerobic glycolysis regenerates NAD^+ and yields two net ATP per glucose.
4. RBCs lack mitochondria and obligatorily form lactate.
5. Hypoperfusion limits oxygen-dependent oxidative phosphorylation.
6. Shock can cause hyperlactatemia with metabolic acidosis.

→ clinic: Lactic acidosis in circulatory shock

→ clinic: Exercise-associated lactate production

Lippincott 6e, Chapter 8, pp. 96, 103–104, Figure 8.21 (aerobic versus anaerobic glycolysis, LDH, and circulatory collapse).

GLUT2 and insulin-responsive GLUT4

GLUT proteins transport sugars by facilitated diffusion down their concentration gradients without directly consuming ATP. GLUT2 permits bidirectional glucose movement in liver, enabling uptake after a meal and release during fasting, and its membrane availability is not dependent on insulin-stimulated recruitment. Lippincott 6e also lists GLUT2 in kidney and pancreatic beta cells. Insulin instead increases surface GLUT4 in skeletal muscle and adipose tissue, increasing glucose uptake. The beta-cell association is the textbook exam convention; human beta cells predominantly express GLUT1 and have much less GLUT2 than rodent beta cells.

1. GLUT2 supports glucose entry or exit according to the concentration gradient.
2. Liver GLUT2 transport is insulin independent.
3. Lippincott 6e lists GLUT2 in liver, kidney, and pancreatic beta cells.
4. Insulin recruits GLUT4 in skeletal muscle and adipose tissue.
5. Human beta-cell glucose transport predominantly uses GLUT1; glucokinase remains important for glucose sensing.

Lippincott 6e, Chapter 8, pp. 96–97 (GLUT tissue distribution and facilitated diffusion); Chapter 23, p. 311 (GLUT4 recruitment). · Human-versus-rodent qualification: De Vos et al., Human and rat beta cells differ in glucose transporter but not in glucokinase gene expression, Journal of Clinical Investigation 1995;96:2489–2495: <https://pubmed.ncbi.nlm.nih.gov/7593639/>

Pyruvate dehydrogenase and the TCA cycle

Lippincott 6e, Chapter 9; teacher slides cited under each idea.

practice-bank topic · outside the confirmed Term 1 syllabus, taught because the practice questions test it

Pyruvate dehydrogenase reaction, cofactors, and regulation

Mitochondrial PDH oxidatively decarboxylates pyruvate to acetyl-CoA, CO₂, and NADH. Its five cofactors are TPP, lipoate, CoA, FAD, and NAD⁺. PDH kinase phosphorylation inactivates the complex, whereas phosphatase dephosphorylation activates it; high acetyl-CoA/NADH favor inhibition and muscle Ca²⁺ favors activation.

1. Irreversible link to acetyl-CoA
2. Five cofactors: TPP, lipoate, CoA, FAD, NAD⁺
3. Phosphorylation turns PDH off
4. Ca²⁺ activates PDH phosphatase in muscle

→ clinic: Thiamine-deficiency lactic acidosis

→ clinic: Arsenic inhibition through lipoate

→ clinic: PDH deficiency

Carbohydrate Metabolism, MD Kish — pp. 61-82, PDH and TCA (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 9 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8-13 remain excluded unless separately confirmed.

TCA-cycle sequence, carbon loss, and energy yield

Acetyl-CoA condenses with oxaloacetate to form citrate, and the cycle regenerates oxaloacetate while releasing two CO₂. Each acetyl-CoA yields three NADH, one FADH₂, and one GTP. Succinate dehydrogenase is both a TCA enzyme and respiratory Complex II, and succinyl-CoA synthetase performs substrate-level phosphorylation.

1. Citrate synthase begins the cycle
2. Two oxidative decarboxylations release CO₂
3. Yield: 3 NADH, 1 FADH₂, 1 GTP
4. Succinate dehydrogenase is Complex II

→ clinic: Energy failure in mitochondrial disease

→ clinic: Oncometabolites from TCA-enzyme defects

Carbohydrate Metabolism, MD Kish — pp. 61-82 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8-13 remain excluded unless separately confirmed.

TCA regulation and amphibolic function

Citrate synthase is regulated by substrate availability and inhibited by citrate. Isocitrate dehydrogenase is activated by ADP and Ca^{2+} and inhibited by ATP and NADH; α -ketoglutarate dehydrogenase is activated by Ca^{2+} and inhibited by NADH and succinyl-CoA, so Ca^{2+} stimulates oxidative metabolism in contracting muscle. TCA intermediates also supply biosynthetic pathways, making the cycle amphibolic. Anaplerotic reactions, especially biotin-dependent pyruvate carboxylase, replenish withdrawn oxaloacetate.

1. ADP activates isocitrate dehydrogenase
2. Ca^{2+} stimulates muscle oxidative flux
3. Intermediates support biosynthesis
4. Pyruvate carboxylase is a major anaplerotic enzyme

→ clinic: Exercise metabolism

→ clinic: Biotin-dependent carboxylase function

Carbohydrate Metabolism, MD Kish — pp. 61-82 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8-13 remain excluded unless separately confirmed.

Clinically important PDH/TCA cofactor defects and toxins

Thiamine deficiency lowers TPP-dependent PDH and α -ketoglutarate-dehydrogenase activity, diverting pyruvate toward lactate. Arsenite binds reduced lipoate and inhibits both multienzyme complexes. Fluoroacetate is converted to fluorocitrate, which blocks aconitase and arrests TCA flux.

1. TPP deficiency promotes lactic acidosis
2. Arsenic targets lipoate
3. Fluorocitrate inhibits aconitase
4. PDH and α -ketoglutarate dehydrogenase share five cofactors

→ clinic: Wernicke encephalopathy and lactic acidosis

→ clinic: Arsenic poisoning

→ clinic: Fluoroacetate toxicity

Carbohydrate Metabolism, MD Kish — PDH/TCA clinical correlations, pp. 61-82 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8-13 remain excluded unless separately confirmed.

Mitochondrial oxidation and citrate export for lipid synthesis

Pyruvate dehydrogenase converts pyruvate to acetyl-CoA in the mitochondrial matrix. Most TCA-cycle enzymes are also in the matrix, whereas succinate dehydrogenase is embedded in the inner mitochondrial membrane as respiratory Complex II. In the fed state, acetyl units can leave mitochondria as citrate because acetyl-CoA itself cannot cross the inner membrane. Cytosolic ATP-citrate lyase cleaves exported citrate to acetyl-CoA and oxaloacetate, supplying acetyl-CoA for fatty-acid and cholesterol synthesis.

1. PDH is located in the mitochondrial matrix.
2. Most TCA reactions occur in the matrix.
3. Succinate dehydrogenase is the inner-membrane exception.
4. Citrate exports mitochondrial acetyl units to the cytosol.

5. Cytosolic acetyl-CoA supplies fatty-acid and cholesterol synthesis.

Lippincott 6e, Chapter 9, pp. 109–113; Chapter 16, pp. 183–186; Chapter 18, pp. 220–221 (compartments, citrate export, and lipid synthesis). · carbohydrate metabolism MD Kish.pdf, pp. 63, 76 (TCA localization and membrane-bound succinate dehydrogenase).

Test yourself

1. Which TCA enzyme is a major regulated step activated by ADP and calcium?
A. TPP, lipoate, CoA, FAD, and NAD⁺
B. Citrate synthase
C. Alpha-ketoglutarate dehydrogenase
D. Isocitrate dehydrogenase
2. How many NADH, FADH₂, and GTP are produced per acetyl-CoA oxidized in the TCA cycle?
A. Citrate synthase
B. Three NADH, one FADH₂, and one GTP
C. Rapid consumption of oxaloacetate by citrate synthase
D. Lipoic acid
3. Why is mitochondrial citrate exported to the cytosol in the well-fed state?
A. To provide acetyl-CoA for fatty-acid and cholesterol synthesis
B. Citrate synthase
C. Lipoic acid
D. Aconitase

Answers: 1 D · 2 B · 3 A

Gluconeogenesis

Lippincott 6e, Chapter 10; teacher slides cited under each idea.

practice-bank topic · outside the confirmed Term 1 syllabus, taught because the practice questions test it

Purpose, organs, and substrates of gluconeogenesis

Gluconeogenesis maintains blood glucose for glucose-dependent tissues during fasting. Liver is the major early site, while renal cortex contributes increasingly during prolonged fasting. Major precursors are lactate, glycerol, and glucogenic amino acids; even-chain fatty acids cannot provide net glucose.

1. Liver dominates early fasting
2. Kidney contribution rises with prolonged fasting
3. Lactate, glycerol, and alanine are major substrates
4. Even-chain fatty acids are not net glucogenic

→ clinic: Fasting hypoglycemia

→ clinic: Renal contribution during prolonged fasting

Carbohydrate Metabolism, MD Kish — pp. 38–60 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Gluconeogenic bypass reactions and energy cost

Pyruvate carboxylase and PEP carboxykinase bypass pyruvate kinase; fructose-1,6-bisphosphatase bypasses PFK-1; and glucose-6-phosphatase bypasses hexokinase/glucokinase. Producing one glucose from two pyruvate consumes four ATP, two GTP, and two NADH. Glucose-6-phosphatase in liver and kidney permits release of free glucose, whereas muscle lacks it.

1. Pyruvate carboxylase uses ATP and biotin
2. PEPCK uses GTP
3. FBPase-1 bypasses PFK-1
4. Glucose-6-phosphatase enables glucose export

→ clinic: Von Gierke disease

→ clinic: Muscle cannot export glycogen-derived glucose

Carbohydrate Metabolism, MD Kish — pp. 38–60 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Reciprocal regulation of glycolysis and gluconeogenesis

High acetyl-CoA activates pyruvate carboxylase and inhibits PDH, directing pyruvate toward glucose production during fatty-acid oxidation. AMP and fructose 2,6-bisphosphate inhibit fructose-1,6-bisphosphatase. Glucagon activates PKA, shifts the hepatic PFK-2/FBPase-2 enzyme toward FBPase-2, lowers fructose 2,6-bisphosphate, and therefore favors gluconeogenesis.

1. Acetyl-CoA activates pyruvate carboxylase
2. AMP inhibits FBPase-1
3. F2,6BP inhibits FBPase-1 and activates PFK-1
4. Glucagon lowers hepatic F2,6BP

→ clinic: Hormonal fasting response

→ clinic: Metabolic effects of glucagon

Carbohydrate Metabolism, MD Kish — pp. 55–62 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Cori cycle, glucose-alanine cycle, and ethanol hypoglycemia

In the Cori cycle, liver converts lactate from RBCs or exercising muscle back to glucose. In the glucose-alanine cycle, alanine carries both carbon and amino nitrogen from muscle to liver. Ethanol oxidation raises hepatic NADH, shifting pyruvate to lactate and oxaloacetate to malate, thereby impairing gluconeogenesis and risking fasting hypoglycemia.

1. Cori cycle recycles lactate
2. Alanine carries carbon plus nitrogen
3. Glycerol enters through DHAP
4. High NADH inhibits gluconeogenesis

→ clinic: Alcohol-associated fasting hypoglycemia

→ clinic: Lactic acidosis

→ clinic: Exercise metabolism

Carbohydrate Metabolism, MD Kish — pp. 38–60 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Test yourself

1. How does glucagon lower hepatic fructose 2,6-bisphosphate?

- | | |
|--|----------------------------|
| A. Dihydroxyacetone phosphate | B. Liver and kidney cortex |
| C. PKA phosphorylates the bifunctional PFK-2/FBPase-2 enzyme | D. Glucose-6-phosphatase |

2. How does high mitochondrial acetyl-CoA regulate pyruvate metabolism during fasting?

- | | |
|--------------------------------|---|
| A. Fructose-1,6-bisphosphatase | B. Converts lactate to glucose |
| C. Dihydroxyacetone phosphate | D. It activates pyruvate carboxylase and inhibits PDH |

3. What nitrogen-carrying gluconeogenic substrate travels from muscle to liver in the glucose-alanine cycle?

- | | |
|--|--|
| A. Alanine | B. Maintain blood glucose for glucose-dependent tissues |
| C. PKA phosphorylates the bifunctional PFK-2/FBPase-2 enzyme | D. Oxaloacetate cannot cross the inner mitochondrial membrane directly |

Answers: 1 C · 2 D · 3 A

Glycogen metabolism

Lippincott 6e, Chapter 11; teacher slides cited under each idea.

practice-bank topic · outside the confirmed Term 1 syllabus, taught because the practice questions test it

Glycogen roles and synthesis

Liver glycogen buffers blood glucose, whereas muscle glycogen supplies local ATP during contraction. Glycogenin creates the primer, UDP-glucose provides activated glucose, glycogen synthase forms alpha-1,4 bonds, and branching enzyme creates alpha-1,6 branches. Branching increases solubility and the number of nonreducing ends available for rapid synthesis or breakdown.

1. Liver supports blood glucose
2. Muscle glycogen is for muscle itself
3. UDP-glucose is the activated donor
4. Glycogen synthase makes alpha-1,4 bonds
5. Branching enzyme makes alpha-1,6 bonds

→ clinic: Exercise fuel

→ clinic: Fasting glucose maintenance

Carbohydrate Metabolism, MD Kish — glycogen sections, including pp. 36–37 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 11 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Glycogenolysis and debranching

Glycogen phosphorylase uses pyridoxal phosphate to cleave alpha-1,4 bonds and release glucose 1-phosphate. Debranching enzyme has transferase and alpha-1,6-glucosidase activities, and the branch-point glucose is released as free glucose. Phosphoglucomutase converts glucose 1-phosphate to glucose 6-phosphate for glycolysis or hepatic glucose release.

1. Phosphorylase releases glucose 1-phosphate
2. PLP is the phosphorylase cofactor
3. Debranching enzyme has two activities
4. Alpha-1,6 cleavage releases free glucose

→ clinic: Cori disease

→ clinic: McArdle disease

Carbohydrate Metabolism, MD Kish — pp. 36–37 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Hormonal and reciprocal regulation of glycogen

Glucagon in liver and epinephrine in liver or muscle signal through cAMP/PKA to activate phosphorylase kinase and promote glycogenolysis. Insulin activates protein phosphatase 1, which dephosphorylates and activates glycogen synthase while inactivating phosphorylase. Skeletal muscle lacks a glucagon receptor, so glucagon does not directly mobilize muscle glycogen.

1. Phosphorylation favors glycogen breakdown
2. Dephosphorylation favors synthesis
3. Insulin acts through protein phosphatase 1
4. Glucagon targets liver, not skeletal muscle

→ clinic: Fed-fast transition

→ clinic: Epinephrine response during exercise

Carbohydrate Metabolism, MD Kish — glycogen regulation section (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 11 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8-13 remain excluded unless separately confirmed.

Pattern recognition in glycogen-storage diseases

Von Gierke disease causes severe fasting hypoglycemia and lactic acidosis from glucose-6-phosphatase deficiency. Pompe disease is lysosomal acid alpha-glucosidase deficiency with cardiomyopathy, while McArdle disease is muscle phosphorylase deficiency with exercise intolerance and absent lactate rise. Cori disease causes limit-dextrin accumulation, and Andersen disease produces poorly branched, insoluble glycogen.

1. Von Gierke: glucose-6-phosphatase
2. Pompe: lysosomal acid alpha-glucosidase
3. McArdle: muscle phosphorylase
4. Cori: debranching enzyme
5. Andersen: branching enzyme

→ clinic: Infantile cardiomyopathy

→ clinic: Exercise intolerance

→ clinic: Fasting hypoglycemia

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 11, glycogen-storage diseases (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8-13 remain excluded unless separately confirmed.

cytosolic glycogen synthesis and granule organization

Glycogen synthesis occurs in the cytosol, particularly in liver and skeletal muscle, which contain the largest stores. Glycogen granules contain the branched glucose polymer together with enzymes that synthesize and degrade it. Glycogenin initiates a primer, glycogen synthase extends alpha(1→4) chains with glucose from UDP-glucose, and branching enzyme creates alpha(1→6) linkages. This cytosolic pathway is distinct from the minor lysosomal degradation pathway that is impaired in Pompe disease.

1. Main glycogen stores: skeletal muscle and liver.
2. Glycogen synthesis is cytosolic.
3. Glycogen granules contain metabolic enzymes.

4. UDP-glucose is the activated donor for chain extension.

Lippincott 6e, Chapter 11, pp. 126–130 (glycogen granules, cytosolic synthesis, and lysosomal degradation).

 Test yourself

1. Which two activities are present in glycogen debranching enzyme?
 - A. Calcium-calmodulin activates phosphorylase kinase
 - B. Acid alpha-glucosidase
 - C. Transferase and alpha-1,6-glucosidase
 - D. Protein phosphatase 1
2. Which vitamin-derived cofactor is required by glycogen phosphorylase?
 - A. Free glucose
 - B. Pyridoxal phosphate
 - C. UDP-glucose
 - D. Glycogen debranching enzyme
3. Through which second messenger does beta-adrenergic stimulation promote glycogen breakdown?
 - A. Pyridoxal phosphate
 - B. Cyclic AMP and protein kinase A
 - C. It favors dephosphorylation and inactivation
 - D. Andersen disease

Answers: 1 C · 2 B · 3 B

Fructose, galactose and lactose

Lippincott 6e, Chapter 12; teacher slides cited under each idea.

practice-bank topic · outside the confirmed Term 1 syllabus, taught because the practice questions test it

Fructose metabolism and hereditary fructose intolerance

Hepatic fructokinase converts fructose to fructose 1-phosphate, which aldolase B cleaves into triose products downstream of PFK-1. Fructokinase deficiency causes benign essential fructosuria. Aldolase-B deficiency causes hereditary fructose intolerance: fructose 1-phosphate traps phosphate and inhibits glycogenolysis and gluconeogenesis, producing vomiting and hypoglycemia after fructose or sucrose intake.

1. Fructokinase makes fructose 1-phosphate
2. Fructokinase deficiency is usually benign
3. Aldolase-B deficiency causes HFI
4. Avoid fructose, sucrose, and sorbitol in HFI

→ clinic: Hereditary fructose intolerance

→ clinic: Essential fructosuria

→ clinic: Fructose-driven hepatic lipogenesis

Carbohydrate Metabolism, MD Kish — pp. 27–35 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Galactose metabolism and galactosemias

Galactokinase forms galactose 1-phosphate, and GALT transfers UMP from UDP-glucose to galactose 1-phosphate, generating glucose 1-phosphate and UDP-galactose. GALT deficiency causes classic galactosemia with liver injury, cataracts, and increased E. coli sepsis risk. Isolated galactokinase deficiency mainly causes cataracts from galactitol accumulation.

1. Galactokinase is the first enzyme
2. GALT deficiency is classic galactosemia
3. Galactitol causes cataracts
4. Remove galactose and lactose from the diet

→ clinic: Neonatal E. coli sepsis

→ clinic: Cataracts

→ clinic: Liver dysfunction

Carbohydrate Metabolism, MD Kish — pp. 27–35 and 83–90 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Disaccharidases and monosaccharide transport

Brush-border lactase, sucrase-isomaltase, and maltase hydrolyze dietary disaccharides. Glucose and galactose enter through SGLT1, while fructose enters through GLUT5. Lactase deficiency leaves lactose osmotically active and fermentable, causing bloating, gas, and acidic diarrhea.

1. Disaccharidases are on enterocyte brush border
2. SGLT1 handles glucose and galactose
3. GLUT5 handles fructose
4. Lactase deficiency causes osmotic diarrhea

→ clinic: Lactose intolerance

→ clinic: Congenital glucose-galactose malabsorption

Carbohydrate Metabolism, MD Kish — pp. 27–35 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Polyol pathway and sorbitol injury

Aldose reductase uses NADPH to reduce glucose to sorbitol. Liver and seminal vesicles can oxidize sorbitol to fructose through sorbitol dehydrogenase, but lens, retina, Schwann cells, and kidney have limited activity. During hyperglycemia, trapped sorbitol produces osmotic stress and NADPH consumption may weaken antioxidant defense.

1. Aldose reductase consumes NADPH
2. Sorbitol dehydrogenase makes fructose
3. Sorbitol accumulates in selected insulin-independent tissues
4. Osmotic and oxidative injury contribute to complications

→ clinic: Diabetic cataracts

→ clinic: Neuropathy

→ clinic: Retinopathy

Carbohydrate Metabolism, MD Kish — carbohydrate-entry and clinical sections (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Test yourself

1. Deficiency of fructokinase produces which disorder?

- | | |
|--------------------------|---------------------------|
| A. Glucose and galactose | B. Essential fructosuria |
| C. Glucose and fructose | D. Sorbitol dehydrogenase |

2. Which enzyme is deficient in classic galactosemia?

- | | |
|--|---|
| A. Glucose and fructose | B. These tissues have limited sorbitol dehydrogenase activity |
| C. Galactose-1-phosphate uridyltransferase | D. Fructose 6-phosphate |

3. What is the characteristic major manifestation of isolated galactokinase deficiency?

- | | |
|---------------------------|------------------------------|
| A. Sorbitol dehydrogenase | B. UDP-galactose 4-epimerase |
| C. Fructokinase | D. Infantile cataracts |

QNSWRS: 1 B · 2 C · 3 D

Pentose phosphate pathway and NADPH

Lippincott 6e, Chapter 13; teacher slides cited under each idea.

practice-bank topic · outside the confirmed Term 1 syllabus, taught because the practice questions test it

Oxidative pentose-phosphate pathway and G6PD regulation

The cytosolic oxidative PPP irreversibly converts glucose 6-phosphate to ribulose 5-phosphate, CO₂, and NADPH. Glucose-6-phosphate dehydrogenase catalyzes the regulated first step. A high NADP⁺/NADPH ratio activates pathway flux so NADPH production rises when reducing power is needed.

1. Oxidative phase is irreversible
2. G6PD is rate limiting
3. NADPH and ribulose 5-phosphate are products
4. NADP⁺ stimulates G6PD

→ clinic: G6PD deficiency

→ clinic: Reductive biosynthesis

Carbohydrate Metabolism, MD Kish — pp. 91–103 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Nonoxidative PPP and carbon rearrangement

The nonoxidative PPP reversibly interconverts sugars containing three to seven carbons. Transketolase transfers two-carbon units and requires TPP, whereas transaldolase transfers three-carbon units. The pathway can supply ribose 5-phosphate for nucleotides or return carbons to glycolysis as fructose 6-phosphate and glyceraldehyde 3-phosphate.

1. Nonoxidative reactions are reversible
2. Transketolase requires TPP
3. Transaldolase transfers three carbons
4. Ribose 5-phosphate supports nucleotide synthesis

→ clinic: Thiamine status assessed by transketolase

→ clinic: Rapidly dividing cells need ribose

Carbohydrate Metabolism, MD Kish — pp. 91–103 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

NADPH, glutathione, and G6PD-deficient hemolysis

RBCs rely on the PPP as their only NADPH source. NADPH allows glutathione reductase to regenerate reduced glutathione, which removes peroxides through glutathione peroxidase. In G6PD deficiency, infection, fava beans, or oxidant drugs can cause Heinz-body formation and splenic removal of inclusions, producing bite cells and episodic hemolysis.

1. RBC PPP supplies NADPH
2. Reduced glutathione neutralizes peroxides
3. Oxidant stress precipitates hemolysis
4. Heinz bodies lead to bite cells

→ clinic: Acute hemolytic anemia

→ clinic: Neonatal jaundice

→ clinic: Drug-triggered oxidant injury

Carbohydrate Metabolism, MD Kish — pp. 91–103 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

Other major uses of NADPH

NADPH provides reducing power for fatty-acid and steroid synthesis, cytochrome-P450 monooxygenase reactions, glutathione reduction, and the phagocyte respiratory burst. NADPH oxidase generates superoxide for microbial killing. Defective NADPH oxidase causes chronic granulomatous disease and an abnormal dihydrorhodamine test.

1. NADPH drives reductive biosynthesis
2. P450 reactions require NADPH
3. NADPH oxidase initiates respiratory burst
4. DHR testing evaluates oxidase activity

→ clinic: Chronic granulomatous disease

→ clinic: Drug metabolism

→ clinic: Steroid synthesis

Carbohydrate Metabolism, MD Kish — pp. 91–103 (local teacher-authored deck available; August 25 exam inclusion explicitly unconfirmed/excluded by exam-scope.md). · exam-scope.md (updated 2026-08-20): Chapters 8–13 remain excluded unless separately confirmed.

PPP products: NADPH, ribulose, and ribose

The pentose phosphate pathway supplies NADPH for reductive reactions and ribose 5-phosphate for nucleotide synthesis. Its oxidative phase produces two NADPH, carbon dioxide, and ribulose 5-phosphate from glucose 6-phosphate. Nonoxidative reactions can isomerize ribulose 5-phosphate to ribose 5-phosphate or rearrange pentose carbons into glycolytic intermediates. The pathway directly produces no ATP, and the nonoxidative reactions can also supply ribose from glycolytic intermediates when nucleotide demand exceeds NADPH demand.

1. Whole-pathway purposes: NADPH and ribose 5-phosphate production.
2. Direct oxidative pentose product: ribulose 5-phosphate.
3. Two NADPH and one CO₂ form per oxidative passage of glucose 6-phosphate.
4. Ribose 5-phosphate is used in nucleotides.
5. Nonoxidative reactions are reversible.

Lippincott 6e, Chapter 13, pp. 145–147 (oxidative and nonoxidative reactions). · carbohydrate metabolism MD Kish.pdf, pp. 94–96 (NADPH and ribose products).

Test yourself

1. What are the two major products of the oxidative pentose phosphate pathway?

- A. Glucose-6-phosphate dehydrogenase
- B. Thiamine pyrophosphate
- C. NADPH and ribulose 5-phosphate
- D. NADPH-oxidase generation of superoxide

2. Which glycolytic intermediates can be produced by the nonoxidative pentose phosphate pathway?

- A. Fructose 6-phosphate and glyceraldehyde 3-phosphate
- B. Infection, fava beans, and oxidant drugs
- C. Splenic removal of Heinz-body inclusions
- D. They reversibly interconvert sugars with three to seven carbons

3. Which cofactor is required by transketolase?

- A. The cytosol
- B. Glucose-6-phosphate dehydrogenase
- C. NADPH-oxidase generation of superoxide
- D. Thiamine pyrophosphate

Answers: 1 C · 2 A · 3 D

Glycoconjugates

Lippincott 6e, Chapter 14; teacher slides cited under each idea.

Glycosaminoglycan structure and major examples

Glycosaminoglycans are long repeating disaccharide chains whose sulfate and carboxyl groups attract cations and water. Their hydrated gels resist compression and support extracellular matrix. Hyaluronan is unusual because it is unsulfated, is not covalently attached to a core protein, and is synthesized at the plasma membrane.

1. Most GAGs are highly negatively charged
2. Hydration provides resilience
3. Hyaluronan is unsulfated
4. Keratan sulfate lacks uronic acid

→ clinic: Cartilage mechanics

→ clinic: Heparin anticoagulation

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 14, glycosaminoglycans (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Proteoglycans versus glycoproteins

A proteoglycan contains a core protein bearing long GAG chains, so carbohydrate is usually the dominant mass. Glycoproteins typically contain shorter, branched oligosaccharides and relatively more protein. Proteoglycans chiefly organize hydrated extracellular matrix, while glycoproteins have diverse roles in receptors, adhesion, immunity, and secretion.

1. Proteoglycan = core protein plus GAG chains
2. Glycoprotein glycans are shorter and branched
3. Proteoglycans are carbohydrate-rich
4. Both are formed through specific glycosylation pathways

→ clinic: Basement membranes

→ clinic: Cell-surface receptors and immune proteins

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 14, proteoglycans and glycoproteins (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

N-linked glycosylation and mannose-6-phosphate targeting

N-linked oligosaccharides attach to asparagine and begin assembly on dolichol in the ER before Golgi processing. Lysosomal hydrolases acquire mannose-6-phosphate in the Golgi, which directs them to lysosomes. Failure to add the tag causes I-cell disease, with hydrolases secreted extracellularly and undegraded substrates accumulating in cells.

1. N-linked glycans attach to asparagine
2. Dolichol supports ER assembly
3. Mannose-6-phosphate targets lysosomal enzymes

4. I-cell disease is a targeting defect

→ clinic: I-cell disease

→ clinic: Congenital disorders of glycosylation

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 14, glycoprotein synthesis and lysosomal targeting (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Mucopolysaccharidoses: Hurler versus Hunter

Lysosomal defects in GAG degradation cause mucopolysaccharidoses with coarse facial features, organomegaly, skeletal abnormalities, and developmental effects. Hurler syndrome is autosomal recessive alpha-L-iduronidase deficiency and typically has corneal clouding. Hunter syndrome is X-linked iduronate-sulfatase deficiency and generally lacks corneal clouding.

1. Both accumulate dermatan and heparan sulfate
2. Hurler is autosomal recessive
3. Hunter is X-linked
4. Corneal clouding favors Hurler

→ clinic: Hurler syndrome

→ clinic: Hunter syndrome

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 14, mucopolysaccharidoses (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

past paper asked

Which amino acid is involved in N-glycosidic bond formation?

Asparagine is the best answer. N-linked oligosaccharides attach to the amide nitrogen of asparagine.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q37

past paper asked

All of these polysaccharides are homopolysaccharide except?

Heparin is the best answer. Heparin is a heteropolysaccharide with repeating disaccharide units.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q40

past paper asked

Oligosaccharides linked to proteins are called:

Glycoproteins is the best answer. Proteins bearing covalently attached oligosaccharides are glycoproteins.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q52

past paper asked

Which of the following has more than one type of sugar in its structure?

Hyaluronic acid is the best answer. Hyaluronic acid is a repeating disaccharide of glucuronic acid and N-acetylglucosamine.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q15

Test yourself

1. Why do most glycosaminoglycans bind large amounts of water?

- | | |
|---|--|
| A. They contain only neutral triglycerides | B. They form peptide bonds with oxygen |
| C. They contain many negatively charged sulfate and carboxyl groups | D. They are entirely hydrophobic |

2. N-linked oligosaccharides are attached to which amino-acid side chain?

- | | |
|---------------|------------|
| A. Threonine | B. Glycine |
| C. Asparagine | D. Serine |

3. Which glycosaminoglycan lacks a uronic acid residue?

- | | |
|--------------------|------------------------|
| A. Keratan sulfate | B. Dermatan sulfate |
| C. Heparan sulfate | D. Chondroitin sulfate |

Answers: 1 C · 2 C · 3 A

Dietary lipids and absorption

Lippincott 6e, Chapter 15; teacher slides cited under each idea.

Lipid classes and fatty-acid structure

Fatty acids contain a carboxyl group and a hydrocarbon chain; saturated fatty acids have no carbon-carbon double bonds, whereas unsaturated fatty acids contain one or more. A cis double bond introduces a kink that reduces packing and generally lowers melting point, while greater chain length and saturation raise melting point. Triacylglycerols and cholesteryl esters are simple storage or transport lipids; phospholipids and glycolipids contain additional polar groups and are major membrane lipids.

1. Saturated fatty acids contain no carbon-carbon double bonds
2. Cis double bonds disrupt packing and lower melting point
3. Longer and more saturated chains generally melt at higher temperatures
4. Triacylglycerol is a concentrated hydrophobic energy store
5. Complex lipids include additional phosphate, carbohydrate, or nitrogen-containing groups

→ **clinic:** Membrane fluidity depends partly on fatty-acid chain length and unsaturation

→ **clinic:** Essential omega-6 and omega-3 fatty acids must be obtained from diet

Lipid structure Esmaeili.pdf, 96-page teacher deck, early fatty-acid and lipid-classification sections. · biochemistry-telegram-scope.md: lipid structure, classification, fatty acids, complex lipids, and lipoproteins are confirmed first-semester teacher material.

Enzymatic digestion of dietary lipids

Lingual and gastric lipases begin TAG digestion and are especially useful for short- and medium-chain TAG and in neonates. Pancreatic lipase with colipase performs most adult TAG digestion, hydrolyzing the 1 and 3 positions to yield 2-monoacylglycerol and free fatty acids. Pancreatic esterases and phospholipases digest cholesterol esters and phospholipids.

1. Acid lipases start digestion
2. Pancreatic lipase is the main adult TAG enzyme
3. Colipase anchors lipase at the lipid-water interface
4. TAG digestion yields 2-MAG plus free fatty acids

→ **clinic:** Pancreatic insufficiency

→ **clinic:** Orlistat inhibition of lipases

Lipid and Lipoprotein Metabolism 1, Fataneh Esmaeili — pp. 4-21 (teacher-confirmed). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 15 (cross-check). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Bile salts and emulsification

Conjugated bile salts are amphipathic cholesterol derivatives released into the duodenum. Together with intestinal mixing and phospholipids, they emulsify large lipid droplets into smaller particles with greater surface area. Bile salts also remain with digestion products to form mixed micelles for delivery across the unstirred water layer.

1. Bile salts are made in liver and stored in gallbladder
2. Conjugation increases amphipathic behavior
3. Emulsification increases enzyme-accessible area
4. Bile flow is required for fat-soluble vitamin absorption

→ clinic: Cholestasis

→ clinic: Fat-soluble vitamin deficiency

→ clinic: Gallstone disease

Lipid and Lipoprotein Metabolism 1, Fataneh Esmaili — pp. 17–24 (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

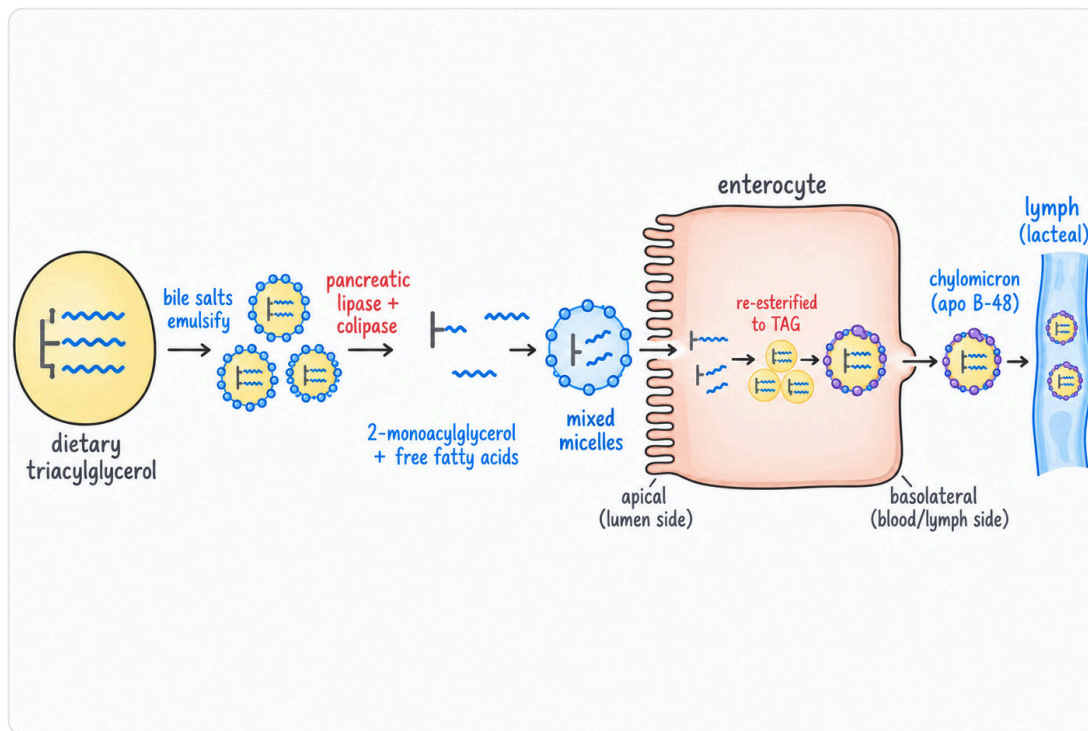


Fig. 11 Dietary triacylglycerol is digested, absorbed, re-esterified, and exported in chylomicrons through intestinal lymph.

Mixed micelles, enterocyte uptake, and re-esterification

Mixed micelles solubilize long-chain fatty acids, 2-monoacylglycerol, cholesterol, lysophospholipids, and fat-soluble vitamins and carry them to the enterocyte surface. The lipids leave the micelle and enter the cell; the micelle itself is not absorbed. Enterocytes activate long-chain fatty acids and re-esterify complex lipids in smooth ER, whereas short- and medium-chain fatty acids can enter portal blood bound to albumin.

1. Micelles cross the unstirred water layer
2. Micelles are not endocytosed intact
3. Long-chain lipids are re-esterified
4. Short/medium chains can use portal blood directly

→ clinic: Fat malabsorption

→ clinic: Use of medium-chain TAG in selected malabsorption disorders

chylomicron assembly, clearance, and steatorrhea

Enterocytes package re-esterified TAG with apoB-48 through microsomal TAG-transfer protein to form chylomicrons, which enter lymph before blood. In plasma they acquire apoC-II and apoE from HDL; apoC-II activates capillary LPL, and apoE directs remnant uptake by liver. Defective digestion, bile delivery, pancreatic secretion, or mucosal absorption can cause steatorrhea and fat-soluble vitamin deficiency.

1. ApoB-48 and MTP are required for assembly
2. Chylomicrons travel through lymph
3. ApoC-II activates LPL
4. ApoE supports remnant uptake

→ clinic: Familial chylomicronemia

→ clinic: Cystic fibrosis

→ clinic: Steatorrhea

Lipid and Lipoprotein Metabolism 1, Fataneh Esmaili — pp. 27–42 (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Drug targets in intestinal lipid handling

Orlistat inhibits gastric and pancreatic lipases, reducing hydrolysis of dietary triacylglycerol and therefore decreasing fat absorption. The resulting unabsorbed fat can cause oily stools and reduced absorption of fat-soluble vitamins. Ezetimibe acts at the intestinal brush border, inhibiting the NPC1L1 sterol transporter and reducing cholesterol absorption. These targets explain why lipase inhibition affects triacylglycerol digestion whereas NPC1L1 inhibition acts on sterol uptake.

1. Orlistat inhibits gastric and pancreatic lipases.
2. Less TAG digestion means less formation of absorbable fatty acids and 2-monoacylglycerol.
3. Ezetimibe inhibits NPC1L1-mediated intestinal cholesterol uptake.
4. NPC1L1 inhibition is distinct from HMG-CoA reductase inhibition by statins.

→ clinic: Orlistat can produce steatorrhea.

Lippincott 6e, Chapter 15, pp. 175–178 (lipase inhibition and reduced intestinal cholesterol absorption). · Lipid metabolism 1.pdf, p. 25 (ezetimibe and intestinal cholesterol absorption); p. 14 names acid lipases, but its orlistat wording is superseded here by the textbook gastric/pancreatic targets. · FDA, ZETIA prescribing information, section 12.1 (NPC1L1 target): https://www.accessdata.fda.gov/drugsatfda_docs/label/2011/021445s030lbl.pdf

Test yourself

1. Where are chylomicrons assembled and what protein is essential for assembly?

- A. Assembled in **ER**; **microsomal triglyceride transfer protein (MTP)** is essential.
- B. Excess fat (and fat-soluble vitamins/essential FAs) in feces due to malabsorption.
- C. Increased lipid in feces due to malabsorption: includes fats + fat-soluble vitamins + essential fatty acids.
- D. It's secreted in **high concentration** (2-3% of pancreatic protein) and is **highly catalytically efficient**.

2. Why do cis double bonds generally lower the melting point of a fatty acid?

- A. They introduce bends that hinder tight packing
- B. They remove all hydrophobicity
- C. They convert the fatty acid into a sterol
- D. They create peptide bonds

3. Which pairing correctly distinguishes simple from complex lipids in the teacher classification?

- A. Phospholipids are simple; steroids are complex
- B. Gangliosides are simple; triacylglycerols are complex
- C. Triacylglycerols are simple; phospholipids are complex
- D. All lipids are placed in one class

Answers: 1 A · 2 A · 3 C

Fatty acids, triacylglycerols and ketones

Lippincott 6e, Chapter 16; teacher slides cited under each idea.

Fatty-acid activation and the carnitine shuttle

Acyl-CoA synthetase activates a fatty acid to fatty acyl-CoA at a cost equivalent to two ATP. Long-chain acyl groups enter the mitochondrial matrix through CPT-I, the carnitine-acylcarnitine translocase, and CPT-II. Malonyl-CoA inhibits CPT-I, preventing simultaneous synthesis and oxidation of fatty acids.

1. Activation forms fatty acyl-CoA
2. CPT-I is the regulated entry step
3. Carnitine carries long-chain acyl groups
4. Malonyl-CoA inhibits CPT-I

→ clinic: Primary carnitine deficiency

→ clinic: CPT deficiency

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

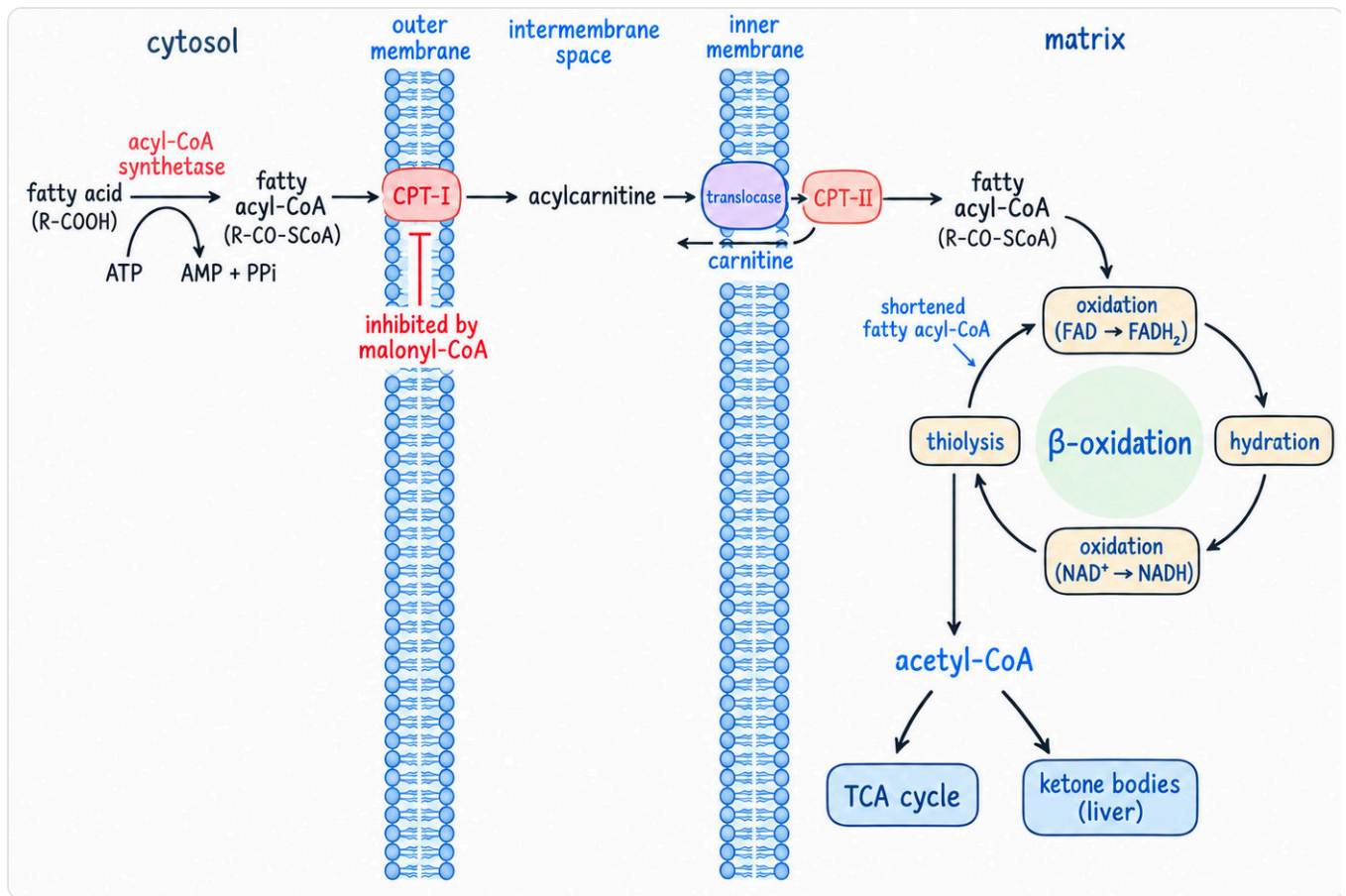


fig. 12 The carnitine shuttle transports long-chain fatty acyl groups into the mitochondrial matrix for β -oxidation.

Mitochondrial beta-oxidation and odd-chain fatty acids

Each beta-oxidation cycle performs oxidation, hydration, oxidation, and thiolysis, producing $FADH_2$, $NADH$, acetyl-CoA, and a two-carbon-shorter acyl-CoA. Even-chain fatty acids end as acetyl-CoA and cannot produce net glucose. Odd-chain fatty acids also yield propionyl-CoA, which becomes succinyl-CoA through biotin- and B12-dependent steps.

1. Occurs mainly in mitochondrial matrix
2. Four recurring reactions
3. Produces $NADH$, $FADH_2$, and acetyl-CoA
4. Propionyl-CoA is glucogenic through succinyl-CoA

→ clinic: Methylmalonic acidemia

→ clinic: Propionic acidemia

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16, fatty-acid oxidation (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Ketogenesis and ketone-body utilization

During fasting, hepatic mitochondrial beta-oxidation supplies acetyl-CoA while oxaloacetate is diverted toward gluconeogenesis, favoring ketogenesis. Mitochondrial HMG-CoA synthase is rate limiting for formation of acetoacetate and beta-hydroxybutyrate. Extrahepatic mitochondria use thiophorase to oxidize ketones, but liver lacks thiophorase and RBCs lack mitochondria.

1. Liver makes but cannot use ketone bodies

2. HMG-CoA synthase is rate limiting
3. Brain use rises during prolonged fasting
4. Excess ketones cause high-anion-gap acidosis

→ clinic: Diabetic ketoacidosis

→ clinic: Starvation adaptation

→ clinic: Hypoketotic hypoglycemia

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16, ketone bodies (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Fatty-acid synthesis and reciprocal control

Cytosolic acetyl-CoA carboxylase uses biotin and ATP to form malonyl-CoA, the committed and rate-limiting step. Fatty-acid synthase uses acetyl and malonyl units plus NADPH to make palmitate. Insulin and citrate activate synthesis, whereas glucagon, epinephrine, AMPK, and long-chain acyl-CoA oppose it; malonyl-CoA simultaneously blocks CPT-I.

1. ACC makes malonyl-CoA
2. Biotin is required
3. FAS mainly produces palmitate
4. NADPH supplies reducing power
5. Malonyl-CoA coordinates synthesis with oxidation

→ clinic: Hepatic steatosis

→ clinic: Metabolic syndrome

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16, fatty-acid synthesis (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Triacylglycerol synthesis and adipose lipolysis

TAG synthesis esterifies three fatty acyl-CoA molecules to glycerol 3-phosphate; adipocytes depend strongly on glucose-derived DHAP for their glycerol backbone. During fasting, adipose lipases release free fatty acids and glycerol. Fatty acids travel on albumin to oxidative tissues, whereas glycerol goes mainly to liver because adipose tissue has little glycerol kinase.

1. Glycerol 3-phosphate supplies the TAG backbone
2. Insulin promotes adipose TAG storage
3. Fasting activates lipolysis
4. Released fatty acids bind albumin
5. Glycerol is handled mainly by liver

→ clinic: Fasting hyperlipolysis

→ clinic: Insulin resistance

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16, TAG metabolism (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Fatty-acid oxidation disorders and hypoketotic hypoglycemia

Fatty-acid oxidation supports fasting ATP, hepatic gluconeogenesis, and ketone production. MCAD deficiency impairs medium-chain oxidation and commonly presents after fasting or illness with hypoketotic hypoglycemia, vomiting, lethargy, and dicarboxylic aciduria. Carnitine or shuttle defects similarly limit long-chain oxidation and can affect muscle and heart.

1. Low glucose plus low ketones suggests FA-oxidation failure
2. MCAD is autosomal recessive
3. Avoid fasting
4. Dicarboxylic acids reflect omega oxidation

→ clinic: MCAD deficiency

→ clinic: Carnitine deficiency

→ clinic: Sudden decompensation during illness

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 16, oxidation disorders (textbook concept; this chapter is confirmed for August 25 by exam-scope.md).
· exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

carnitine synthesis from lysine and methionine

Carnitine supports mitochondrial entry of long-chain fatty acyl groups and is obtained from food as well as endogenous synthesis. Lysine supplies the carbon skeleton, while methionine-derived methyl groups generate the trimethyllysine precursor. Liver and kidney complete carnitine synthesis, whereas skeletal and cardiac muscle rely on uptake from the circulation. A carnitine deficiency therefore impairs use of long-chain fatty acids as fuel even when fatty acids are available.

1. Lysine provides carnitine's carbon skeleton.
2. Methionine, through SAM, supplies methyl groups.
3. Liver and kidney synthesize carnitine; muscle depends on uptake.
4. The carnitine shuttle transfers long-chain acyl groups into mitochondria.

→ clinic: A defective carnitine transporter can impair cardiac and skeletal-muscle fatty-acid oxidation.

Lippincott 6e, Chapter 16, pp. 190–192; Chapter 20, p. 264. · Supplemental primary isotope evidence for the distinction between carbon skeleton and methyl donors: Cox and Hoppel, Biosynthesis of carnitine and 4-N-trimethylaminobutyrate from lysine, Biochemical Journal 136 (1973), pp. 1075–1082, <https://pubmed.ncbi.nlm.nih.gov/4786529/>.

citrate export, malic enzyme, and the acyl carrier arm

Fatty-acid synthesis occurs in the cytosol, but much of its acetyl-CoA is generated in mitochondria. Acetyl units are exported as citrate, and ATP-citrate lyase releases cytosolic acetyl-CoA and oxaloacetate. Cytosolic malic enzyme converts malate to pyruvate and CO₂ while generating NADPH; the pentose phosphate pathway is another major NADPH source. Fatty-acid synthase carries growing acyl chains on the thiol of its 4-prime-phosphopantetheine arm, derived from pantothenic acid. In the fed state, active ACC supplies malonyl-CoA for synthesis and simultaneously suppresses CPT-1-mediated fatty-acid entry into mitochondria.

1. Citrate crosses the inner membrane; acetyl-CoA does not diffuse across it.
2. ATP-citrate lyase produces cytosolic acetyl-CoA.
3. Malic enzyme produces pyruvate, CO₂, and NADPH from malate.
4. The ACP phosphopantetheine arm derives from vitamin B5.

5. Citrate activates ACC; malonyl-CoA supplies FAS and inhibits CPT-I.

Lippincott 6e, Chapter 16, pp. 183–187 (citrate export, ACC, fatty-acid synthase, and malic enzyme), and Chapter 28, p. 381 (pantothenate).

Smooth-ER desaturation and essential fatty acids

Human long-chain fatty-acid desaturases are located in the smooth endoplasmic reticulum and introduce cis double bonds. The reactions use oxygen and reducing equivalents transferred through cytochrome b5. Humans can introduce delta-9 and certain double bonds nearer the carboxyl end, but cannot introduce new double bonds at carbons 10 or beyond toward the methyl end. Consequently, linoleic acid and alpha-linolenic acid must be supplied in the diet.

1. The principal site of long-chain fatty-acid desaturation is smooth ER.
2. Delta numbering begins at the carboxyl carbon.
3. Humans lack delta-12 and delta-15 desaturases.
4. Linoleic acid is omega-6; alpha-linolenic acid is omega-3.

Lippincott 6e, Chapter 16, pp. 182–183 and 187–188 (essential fatty acids and desaturation).

Beta-oxidation reaction accounting and ATP conventions

Each standard saturated-fatty-acid beta-oxidation cycle produces one FADH₂ and one NADH through two dehydrogenations, with a water-consuming hydration and a thiolytic cleavage between or after them. Seven cycles convert palmitoyl-CoA into eight acetyl-CoA, seven NADH, and seven FADH₂. Using approximately 2.5 ATP per NADH, 1.5 per FADH₂, and 10 per acetyl-CoA gives 108 ATP before activation and 106 ATP net from free palmitate. Lippincott 6e uses the older 3/2 convention and therefore reports 129 ATP net, so the convention must be stated. Palmitoleate has a pre-existing cis double bond whose processing bypasses one FAD-dependent dehydrogenation, lowering yield relative to palmitate.

1. A standard cycle gives one NADH, one FADH₂, and an acetyl-CoA; final cleavage gives two acetyl-CoA.
2. Enoyl-CoA hydratase consumes water.
3. Fatty-acid activation to acyl-CoA costs two ATP equivalents.
4. Modern palmitate calculation: $8 \times 10 + 7 \times 2.5 + 7 \times 1.5 - 2 = 106$ ATP.
5. The sixth-edition legacy calculation gives 129 ATP net.
6. A pre-existing double bond can reduce FADH₂ production during oxidation.

Lippincott 6e, Chapter 16, pp. 192–195 (reaction sequence, legacy ATP yield, and unsaturated fatty acids). · Supplement for modern P/O accounting: Hinkle et al., Mechanistic stoichiometry of mitochondrial oxidative phosphorylation, Biochemistry 1991;30:3576–3582, <https://pubmed.ncbi.nlm.nih.gov/2012815/> (experimental basis of approximately 2.5/1.5 P/O estimates; the 106-ATP total is calculated from these ratios).

Odd-chain carbon fate and limits of energy comparisons

Odd-chain fatty acids undergo beta-oxidation until the final three-carbon fragment is propionyl-CoA. Biotin-dependent propionyl-CoA carboxylase consumes ATP to form methylmalonyl-CoA, which is rearranged by a vitamin B12-dependent mutase to succinyl-CoA after epimerization. This terminal fragment is glucogenic and must be accounted for separately from an acetyl-CoA. In the common calculation that stops the terminal fragment at oxaloacetate, succinyl-CoA oxidation yields five ATP equivalents and the preceding carboxylation costs one, leaving four rather than the ten from oxidation of an acetyl-CoA. That four-ATP value is not complete oxidation of all propionyl carbons, so an odd-versus-even comparison must specify chain length and the endpoint. Even-chain fatty acids yield acetyl-CoA, which cannot reverse the PDH reaction to make pyruvate and cannot yield net glucose through the TCA cycle because two carbons are lost as CO₂ for each acetyl unit entering.

1. Pentadecanoate (C15:0) yields six acetyl-CoA plus one propionyl-CoA.

2. Propionyl-CoA carboxylase uses biotin and ATP.
3. Methylmalonyl-CoA mutase uses adenosylcobalamin (B12).
4. Succinyl-CoA can contribute carbon to gluconeogenesis.
5. Energy accounting depends on whether residual carbon stops at oxaloacetate or is fully oxidized.
6. Even-chain acetyl-CoA cannot give net glucose: PDH is irreversible and the TCA cycle has no net carbon gain.

Lippincott 6e, Chapter 16, pp. 193-195 (odd-chain oxidation); Chapter 9, pp. 112-114 (succinyl-CoA to oxaloacetate reactions). · The four-ATP partial-pathway figure is calculated from the stated reactions using modern P/O ratios, not quoted as the textbook yield.

ATGL initiates adipose lipolysis

Adipose triglyceride lipase initiates mobilization of stored triacylglycerol by releasing its first fatty acid and forming diacylglycerol. Hormone-sensitive lipase preferentially acts on the resulting diacylglycerol, and monoacylglycerol lipase completes hydrolysis. Catecholamine signaling raises cAMP and activates protein kinase A, which promotes access to lipid droplets and activates hormone-sensitive lipase. Low insulin during fasting removes suppression of lipolysis, whereas high insulin favors fat storage. The released nonesterified fatty acids travel in blood bound to albumin.

1. ATGL: triacylglycerol to diacylglycerol.
2. HSL preferentially hydrolyzes diacylglycerol.
3. Monoacylglycerol lipase completes glycerol release.
4. Catecholamines promote cAMP/PKA signaling; insulin opposes lipolysis.
5. Plasma nonesterified fatty acids are albumin-bound.

Lippincott 6e, Chapter 16, pp. 189-191 (mobilization of stored fats and lipase regulation). · Lipid metabolism 1.pdf, pp. 8 and 32-34 (fed/fasted state and fatty-acid/glycerol transport).

Ketone-body identities and the fasting hormonal signal

The ketone bodies are acetoacetate, beta-hydroxybutyrate, and acetone, although beta-hydroxybutyrate is chemically a hydroxy acid. Low insulin with high glucagon favors hepatic fatty-acid oxidation and ketone formation during fasting. Fatty-acid-derived acetyl-CoA accumulates as oxaloacetate is directed toward gluconeogenesis, and mitochondrial HMG-CoA synthase controls the rate of ketogenesis. Acetoacetate and beta-hydroxybutyrate supply extrahepatic oxidative tissues, whereas acetone is a volatile side product released in breath. The liver produces but cannot consume ketone bodies because it lacks succinyl-CoA:acetoacetate CoA transferase, also called thiophorase or SCOT.

1. Ketone bodies: acetoacetate, beta-hydroxybutyrate, and acetone.
2. Fasting favors ketogenesis through a low insulin-to-glucagon ratio.
3. Acetoacetate is reduced to beta-hydroxybutyrate using NADH.
4. Spontaneous acetoacetate decarboxylation produces acetone.
5. Mitochondrial HMG-CoA synthase is the rate-limiting enzyme.
6. Hepatocytes lack thiophorase (SCOT); RBCs lack the mitochondria needed for ketone oxidation.

→ **clinic:** Excessive ketone acid production contributes to ketoacidosis.

Lippincott 6e, Chapter 16, pp. 195-198 (ketone synthesis and use), and Chapter 24, pp. 329-331 (hepatic fasting metabolism).

past paper asked

Which of the following is an essential fatty acid?

Linoleic acid is the best answer. Linoleic and α -linolenic acids are essential because humans cannot introduce the required distal double bonds.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q41

past paper asked

Which of the following fatty acids has the lowest melting point?

Eicosapentaenoic acid is the best answer. More cis double bonds introduce more kinks and lower fatty-acid melting temperature.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q58

past paper asked

Which of the following fatty acid has 16 carbon atoms?

Palmitic acid is the best answer. Palmitic acid is the 16-carbon saturated fatty acid 16:0.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q2

past paper asked

Which of the following statements concerning fatty acids is correct?

One of them is a precursor of prostaglandins is the best answer. Eicosanoids are local signals derived mainly from arachidonic acid.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q18

Test yourself

1. Glycerol-3-phosphate (G3P): role in TAG synthesis.

- A. Hydrolytic release of **fatty acids + glycerol** from stored **TAG**.
- C. Initial **acceptor** of fatty acids during TAG synthesis.

B. Inactivation of ACC $\rightarrow$ decreased malonyl-CoA and fatty acid synthesis.

D. **ATGL**: TAG $\rightarrow$ DAG; **HSL**: DAG $\rightarrow$ MAG; **MAG lipase**: MAG $\rightarrow$ glycerol + FA.

Answers: 1 C

complex lipids and eicosanoids

Lippincott 6e, Chapter 17; teacher slides cited under each idea.

Phospholipid structure, membranes, and PIP₂ signaling

Phospholipids are amphipathic molecules that form membrane bilayers and also serve as reservoirs for signaling molecules. Receptor-activated phospholipase C cleaves PIP₂ to IP₃ and DAG. IP₃ releases ER Ca²⁺, while DAG activates protein kinase C.

1. Polar head and hydrophobic tails
2. Phospholipids form bilayers
3. PLC cleaves PIP₂
4. IP₃ raises cytosolic Ca²⁺
5. DAG activates PKC

→ clinic: Signal-transduction disorders

→ clinic: GPI-anchor deficiency in PNH

Lipid Structure Esmaeili — 96-page teacher deck, phospholipid section (teacher-confirmed). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 17 (mechanism cross-check). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

DPPC, pulmonary surfactant, and neonatal respiratory distress

Type II pneumocytes secrete surfactant rich in dipalmitoyl phosphatidylcholine, which lowers alveolar surface tension and prevents collapse at end-expiration. Production rises late in gestation. Prematurity can therefore cause neonatal respiratory distress syndrome, treated or prevented with antenatal glucocorticoids and exogenous surfactant as appropriate.

1. DPPC is the major surface-active phospholipid
2. Made by type II pneumocytes
3. Reduces surface tension
4. Deficiency causes neonatal RDS

→ clinic: Prematurity

→ clinic: Neonatal respiratory distress syndrome

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 17, surfactant (confirmed August 25 scope in tracked bank). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Sphingolipid architecture and sphingolipidoses

Ceramide is the core of sphingomyelin and glycosphingolipids. Lysosomal hydrolase defects cause substrate-specific accumulation: hexosaminidase A deficiency causes Tay-Sachs, sphingomyelinase deficiency causes Niemann-Pick A/B, and glucocerebrosidase deficiency causes Gaucher disease. Clinical pattern recognition should be linked to the accumulated lipid and missing enzyme rather than memorized as isolated names.

1. Ceramide is the common sphingolipid core
2. Tay-Sachs: GM2 and hexosaminidase A
3. Niemann-Pick A/B: sphingomyelinase
4. Gaucher: glucocerebrosidase

→ clinic: Cherry-red macula

→ clinic: Hepatosplenomegaly

→ clinic: Neurodegeneration

Lipid Structure Esmaeili — sphingolipid section (teacher-confirmed class). · Tracked comprehensive bank — Chapter 17, PDF pp. 374-405 (marked confirmed August 25 scope). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Arachidonic acid, cyclooxygenase products, and NSAIDs

Phospholipase A2 releases arachidonic acid from membrane phospholipids. Cyclooxygenases convert it toward prostaglandins, prostacyclin, and thromboxanes; NSAIDs inhibit COX and therefore reduce these mediators. Glucocorticoids act further upstream by inducing lipocortin and reducing phospholipase-A2-mediated release.

1. Arachidonate is stored in membrane phospholipids
2. COX produces prostanoid precursors
3. NSAIDs inhibit COX
4. Glucocorticoids reduce arachidonate release

→ clinic: Pain and inflammation

→ clinic: Aspirin antiplatelet effect

→ clinic: Gastric and renal NSAID toxicity

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 17, eicosanoids (confirmed August 25 scope in tracked bank). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Lipoxygenase, leukotrienes, and aspirin-exacerbated asthma

The 5-lipoxygenase pathway converts arachidonic acid into leukotrienes. Cysteinyl leukotrienes promote bronchoconstriction, mucus production, and vascular permeability. COX inhibition does not block this branch and can divert substrate toward leukotrienes, contributing to aspirin-exacerbated respiratory disease.

1. 5-LOX produces leukotrienes
2. Leukotrienes are not blocked by NSAIDs
3. Cysteinyl leukotrienes constrict bronchi
4. Montelukast blocks their receptor

→ clinic: Asthma

→ clinic: Aspirin-exacerbated respiratory disease

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 17, leukotrienes (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

past paper asked

An example of glycerophospholipid involved in cell signaling is

Phosphatidylinositol is the best answer. Phosphatidylinositol derivatives generate the IP3 and DAG second-messenger pathway. Phosphatidylinositol derivatives participate in signaling; phosphatidic acid is also a signaling lipid. Both choices are valid under this broad stem.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q55

past paper asked

Dipalmitoyl lecithin acts as

Lung surfactant is the best answer. Dipalmitoyl phosphatidylcholine is the major surface-tension-lowering phospholipid of pulmonary surfactant.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q24

past paper asked

Which of the following belong to glycolipids?

Cerebrosides is the best answer. Cerebrosides are glycosphingolipids consisting of ceramide plus one sugar.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q77

past paper asked

What is the starting compound for the synthesis of sphingolipids?

Ceramide is the best answer. Ceramide is the structural core of sphingolipids. Ceramide is the common branch-point precursor for complex sphingolipids. De novo synthesis begins earlier with serine and palmitoyl-CoA; do not mistake the offered intermediate for the first substrates.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q16

past paper asked

Which of the following phospholipids is the main component of inner mitochondrial membrane?

Cardiolipin is the best answer. Cardiolipin is the signature phospholipid of the inner mitochondrial membrane. Cardiolipin is the characteristic, functionally important inner-mitochondrial phospholipid intended by the exam. It is not necessarily the most abundant phospholipid by mass; "main component" is imprecise.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q29

past paper asked

Which membrane fatty acids serve as precursors for eicosanoids?

Eicosanoids are generated from long-chain polyunsaturated fatty acids released from membrane phospholipids; arachidonic acid is the major C20 precursor. This paper uses eicosanoid loosely for lipid mediators from C20/C22 PUFAs. Strictly, eicosanoids derive from C20 precursors; C22 DHA gives docosanoids. The offered D is the intended broad PUFA choice, not the precise definition.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q36

past paper asked

The backbone of phospholipids is?

L-glycerol 3-phosphate is the best answer. Mammalian glycerophospholipids use the sn-glycerol-3-phosphate backbone. The intended answer describes glycerophospholipids: sn-glycerol-3-phosphate (historically L-glycerol-3-phosphate). Not all phospholipids have glycerol: sphingomyelin has a sphingosine backbone.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q56

past paper asked

Non-steroidal anti-inflammatory drugs (NSAIDS) like aspirin and ibuprofen act by blocking the production of

Prostaglandins is the best answer. Aspirin and ibuprofen inhibit cyclooxygenase and therefore reduce prostaglandin synthesis.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q19

past paper asked

Sphingosine is not a component of....

Cardiolipin is the best answer. Cardiolipin is a glycerophospholipid; ceramide, sphingomyelin, and gangliosides contain a sphingosine-derived backbone.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q13

past paper asked

Which of the following contains an ether-linked alkyl group?

Platelet-activating factor is the best answer. Platelet-activating factor is an ether glycerophospholipid.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q25

Test yourself

1. PIP2 cleavage products and what each does.

A. Cleavage at specific sites releases **second messengers** (DAG, IP3) or precursors like **arachidonic acid**.

C. PI → **PIP2**; cleaved by **phospholipase C** after ligand binds **GPCR** activating **Gqα** (e.g., α1-adrenergic).

B. Produces **IP3** (mobilizes intracellular **Ca²⁺**) + **DAG** (activates **PKC**); act synergistically.

D. PA = **DAG + phosphate on C3**; simplest phosphoglyceride and **precursor** of many glycerophospholipids.

2. Progressive neurodegeneration, a cherry-red macula, exaggerated startle, and no hepatosplenomegaly suggest deficiency of what?

A. Arylsulfatase A

C. Sphingomyelinase

B. Hexosaminidase A

D. Glucocerebrosidase

3. Cyclooxygenase converts arachidonic acid into precursors of which mediators?

A. Catecholamines

C. Leukotrienes only

B. Prostaglandins, prostacyclin, and thromboxanes

D. Bile acids

Answers: 1 B · 2 B · 3 B

cholesterol, lipoproteins and steroids

Lippincott 6e, Chapter 18; teacher slides cited under each idea.

Lipoprotein traffic

Particle	Main role	Recognition point
Chylomicron	Dietary TAG from intestine to tissues	ApoB-48; apoC-II activates LPL; apoE permits remnant uptake.
VLDL → IDL → LDL	Hepatic TAG export, then cholesterol delivery	ApoB-100; LDL receptor recognizes apoB-100/apoE.
HDL	Cholesterol transport and exchange	Nascent HDL is formed in liver AND intestine; apoA-I activates LCAT.
Albumin-bound fatty acids	Nonesterified fatty-acid transport	Not a lipoprotein particle; especially relevant during fasting.

cholesterol synthesis, HMG-CoA reductase, and statins

Cytosolic/ER HMG-CoA reductase converts HMG-CoA to mevalonate using NADPH and is the rate-limiting step of cholesterol synthesis. Insulin and low sterol favor activity, whereas glucagon, AMPK phosphorylation, high sterol, and enzyme degradation reduce it. Statins competitively inhibit the enzyme and increase hepatic LDL-receptor expression, lowering plasma LDL.

1. HMG-CoA reductase is rate limiting
2. Uses NADPH
3. AMPK phosphorylation inhibits it
4. Statins increase hepatic LDL clearance

→ clinic: Hypercholesterolemia

→ clinic: Statin therapy

→ clinic: Atherosclerotic cardiovascular disease

Lipid Structure Esmaili — cholesterol/lipoprotein sections (teacher-confirmed class). · Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 18 (regulatory cross-check). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Bile-acid synthesis and cholesterol elimination

Humans cannot fully oxidize the steroid nucleus, so cholesterol leaves mainly in bile as cholesterol or bile acids.

Cholesterol 7- α -hydroxylase is rate limiting for bile-acid synthesis. Conjugation with glycine or taurine makes amphipathic bile salts that emulsify dietary lipid and undergo enterohepatic recycling.

1. Biliary secretion is the net elimination route
2. 7- α -hydroxylase is rate limiting
3. Bile salts are amphipathic

4. Most bile salts are recycled

→ clinic: Cholestasis

→ clinic: Bile-acid sequestrants

→ clinic: Gallstones

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 18, bile acids (textbook-confirmed; teacher emphasis uncertain). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Lipoprotein pathways and apolipoprotein map

Chylomicrons carry dietary TAG with apoB-48; VLDL carries hepatic TAG with apoB-100 and becomes IDL then LDL. ApoC-II activates LPL, apoE mediates remnant uptake, apoB-100 binds the LDL receptor, and apoA-I activates LCAT on HDL. HDL also supplies apoC-II and apoE to nascent TAG-rich particles.

1. ApoB-48: chylomicron structure

2. ApoB-100: VLDL/LDL structure and LDL-receptor ligand

3. ApoC-II activates LPL

4. ApoE clears remnants

5. ApoA-I activates LCAT

→ clinic: Familial chylomicronemia

→ clinic: Dysbetalipoproteinemia

→ clinic: Familial hypercholesterolemia

Lipid Structure Esmaeili — 96-page deck, lipoprotein section (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

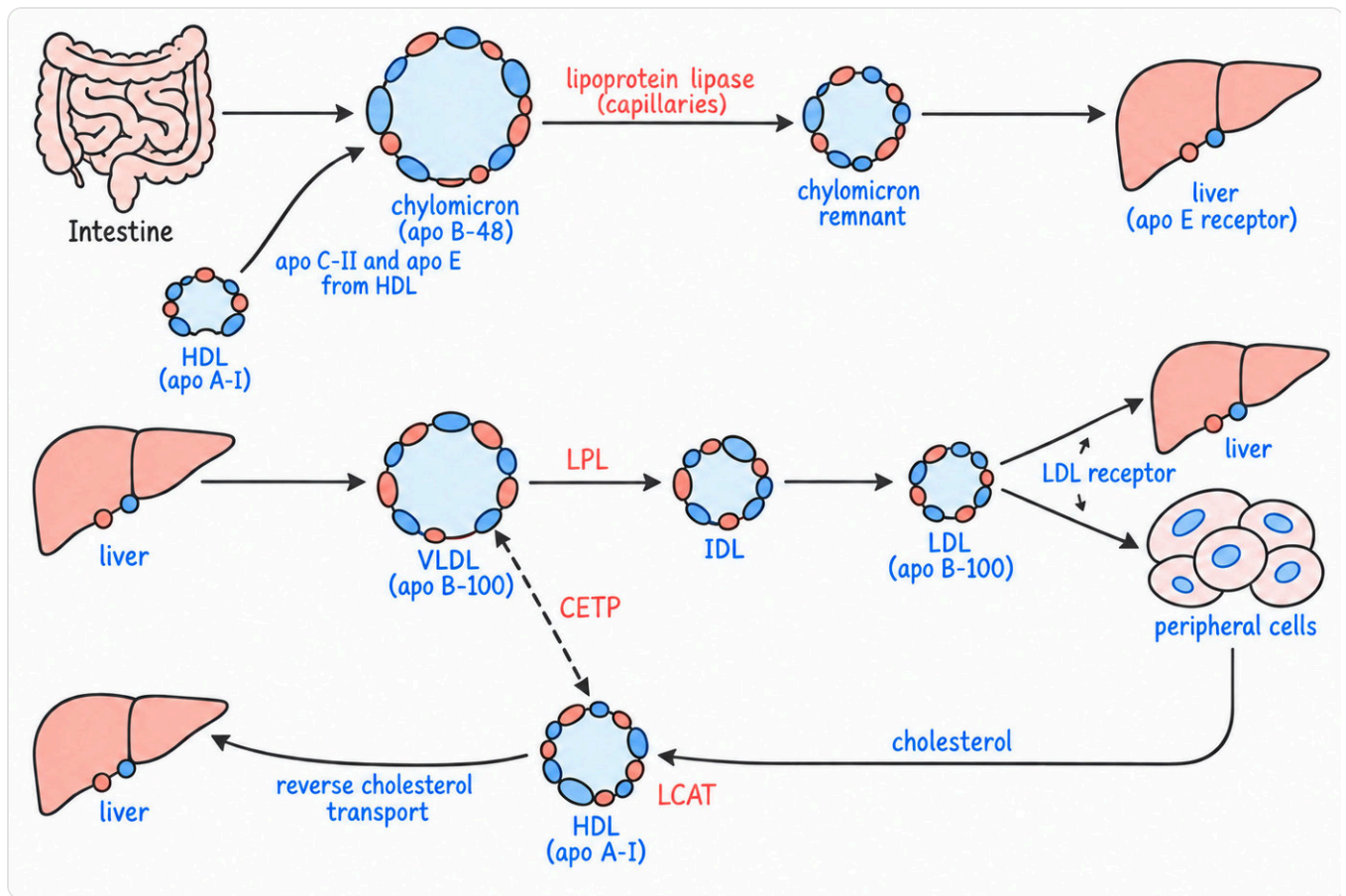


fig. 13 Exogenous and endogenous lipoprotein transport, receptor uptake, and HDL-mediated reverse cholesterol transport.

LDL-receptor uptake and atherosclerotic foam cells

LDL delivers cholesterol through apoB-100 binding to the LDL receptor and receptor-mediated endocytosis. LDL-receptor loss, defective apoB-100 binding, or gain-of-function PCSK9 variants can cause severe familial hypercholesterolemia. Macrophage scavenger receptors take up oxidized LDL without normal cholesterol feedback, creating foam cells and promoting atherosclerotic plaques.

1. LDL receptor recognizes apoB-100
2. Receptors recycle after endosomal release
3. Familial hypercholesterolemia causes early ASCVD
4. Scavenger receptors are not normally downregulated by cholesterol

→ clinic: Tendon xanthomas

→ clinic: Premature coronary disease

→ clinic: PCSK9-targeted therapy

Lipid Structure Esmaeili — lipoprotein section (teacher-confirmed). · Tracked comprehensive bank — Chapter 18 (marked confirmed August 25 scope). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

HDL maturation and reverse cholesterol transport

Lipid-poor apoA-I accepts cellular cholesterol through ABCA1 to form nascent HDL. LCAT esterifies surface cholesterol, allowing cholesteryl ester to enter the HDL core; HDL returns it directly to liver through SR-B1 or transfers it to apoB particles through CETP. ABCA1 deficiency causes Tangier disease with nearly absent HDL.

1. ABCA1 loads cholesterol onto apoA-I
2. LCAT esterifies HDL cholesterol
3. SR-B1 supports selective hepatic uptake
4. CETP exchanges HDL cholesteryl ester for TAG

→ clinic: Tangier disease

→ clinic: Reverse cholesterol transport

Lipid Structure Esmaeli — lipoprotein section (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

cholesterol as steroid precursor and acute StAR control

All steroid hormones derive from cholesterol. The acute regulated step is StAR-mediated movement of cholesterol into the inner mitochondrial membrane, where CYP11A1 converts it to pregnenolone. Tissue-specific hydroxylases then generate glucocorticoids, mineralocorticoids, and sex steroids, which act mainly through intracellular nuclear receptors.

1. Cholesterol is the common precursor
2. StAR controls mitochondrial entry
3. CYP11A1 makes pregnenolone
4. Steroid receptors regulate transcription

→ clinic: Congenital adrenal hyperplasia

→ clinic: Lipoid adrenal hyperplasia

→ clinic: Aromatase-inhibitor therapy

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 18, steroid hormones (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

LPL, hepatic lipase, and apoB-100/apoE receptors

Lipoprotein lipase hydrolyzes triacylglycerol in circulating chylomicrons and VLDL, making fatty acids available to muscle and adipose tissue. ApoC-II is its activator, and deficiency of LPL or apoC-II can cause familial chylomicronemia with severe hypertriglyceridemia. Hepatic lipase hydrolyzes remaining triacylglycerol and phospholipid as remnant particles are remodeled, including conversion of IDL toward LDL. The LDL receptor recognizes both apoB-100 and apoE, permitting recognition of LDL and certain remnant particles. ApoB-48 is required for chylomicron structure but is not an LDL-receptor ligand.

1. LPL hydrolyzes TAG in chylomicrons and VLDL.
2. LPL or apoC-II deficiency causes familial chylomicronemia.
3. Hepatic lipase helps remodel IDL into LDL.
4. LDL-receptor ligands include apoB-100 and apoE.
5. ApoB-48 does not supply the apoB-100 LDL-receptor binding site.

→ clinic: Severe chylomicronemia increases pancreatitis risk.

Lippincott 6e, Chapter 18, pp. 228–234 (LPL, remnant processing, and LDL receptor ligands). · Lipid metabolism 1.pdf, pp. 29–31 (LPL action and familial chylomicronemia). · Supplement for the specific IDL-to-LDL role of hepatic lipase: Lipoprotein metabolism in hepatic lipase deficiency (Journal of Lipid Research, 1988;29:1603–1611): <https://pubmed.ncbi.nlm.nih.gov/3244012/>

Bile-acid feedback and bile-acid sequestrants

Cholesterol 7- α -hydroxylase catalyzes the rate-limiting step of the classical bile-acid synthesis pathway. Returning bile acids suppress expression of this enzyme, limiting their own synthesis through negative feedback. Most secreted bile salts are reabsorbed in the terminal ileum and return to the liver in the enterohepatic circulation. Bile-acid sequestrants bind them in the intestinal lumen, increasing fecal loss and relieving feedback inhibition of synthesis. The liver consumes more cholesterol to replenish bile acids and increases uptake of circulating LDL cholesterol.

1. Bile acids suppress cholesterol 7- α -hydroxylase.
2. Terminal-ileal reabsorption maintains the enterohepatic bile-salt pool.
3. Sequestrants prevent intestinal bile-acid reabsorption.
4. Increased conversion of hepatic cholesterol to bile acids helps lower plasma LDL.

→ clinic: Cholestyramine is a bile-acid sequestrant.

Lippincott 6e, Chapter 18, pp. 224-227 (bile-acid synthesis, feedback, and sequestrants).

past paper asked

Dietary fats after absorption appear in the circulation as

Chylomicron is the best answer. Enterocytes package absorbed dietary lipids into chylomicrons.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q42

past paper asked

HDL is synthesized in

Liver and intestine is the best answer. Nascent HDL is produced by both liver and intestine.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q70

past paper asked

The lipoprotein richest in cholesterol is ...

LDL is the best answer. LDL is the most cholesterol-rich major plasma lipoprotein.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q30

past paper asked

Which of the following lipoproteins is produced via the exogenous lipoprotein pathway?

Chylomicrons is the best answer. Chylomicrons transport dietary lipid through the exogenous pathway.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q39

past paper asked

Which one of the following is assembled and secreted by the liver?

Very Low-Density Lipoprotein (VLDL) is the best answer. The liver packages endogenous triacylglycerol into VLDL for secretion. The liver secretes VLDL and also nascent HDL. VLDL is the intended triglyceride-export answer, but the unrestricted stem permits both HDL and VLDL.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q55

Test yourself

1. Tendon xanthomas and markedly elevated LDL from childhood most strongly suggest defective function of what pathway?
 - A. LDL-receptor-mediated clearance
 - B. Chylomicron lipolysis only
 - C. Ketone-body synthesis
 - D. HDL ApoA-I production only
2. Which apoprotein mediates hepatic uptake of both chylomicron remnants and much of IDL?
 - A. ApoC-II
 - B. ApoB-48 only
 - C. ApoE
 - D. ApoA-I
3. Which apolipoprotein ligand is recognized by the LDL receptor?
 - A. Apo C-II and apo E.
 - B. LDL-receptor complex is endocytosed.
 - C. Apo B-100 (not apo B-48).
 - D. Chylomicrons, VLDL, LDL, HDL.

Answers: 1 A · 2 C · 3 C

Amino acids: disposal of nitrogen

Lippincott 6e, Chapter 19; teacher slides cited under each idea.

practice-bank topic · outside the confirmed Term 1 syllabus, taught because the practice questions test it

urea-cycle enzymes: compartment, reaction, and defect pattern

Enzyme	compartment and reaction	Deficiency pattern
CPS I	Mitochondria: NH_4^+ + bicarbonate $\rightarrow$ carbamoyl phosphate; requires NAG	Hyperammonemia; low citrulline; no characteristic orotate rise
Ornithine transcarbamoylase	Mitochondria: ornithine + carbamoyl phosphate $\rightarrow$ citrulline	X-linked; hyperammonemia, low citrulline, high urinary orotate
Argininosuccinate synthetase	Cytosol: citrulline + aspartate $\rightarrow$ argininosuccinate	Citrullinemia
Argininosuccinate lyase	Cytosol: argininosuccinate $\rightarrow$ arginine + fumarate	Argininosuccinate accumulation
Arginase	Cytosol: arginine $\rightarrow$ ornithine + urea	Increased arginine; usually less severe hyperammonemia

Protein digestion, zymogen activation, and intestinal uptake

The amino acid pool is replenished by dietary protein digestion, body-protein turnover, and synthesis of nonessential amino acids; there is no dedicated storage protein for excess amino acids. Gastric acid denatures proteins, and pepsin begins peptide-bond hydrolysis in the stomach. Intestinal brush-border enteropeptidase activates trypsinogen, and trypsin then activates other pancreatic zymogens, including chymotrypsinogen. Most free amino acids enter enterocytes through sodium-coupled secondary active transport, whereas dipeptides and tripeptides use proton-coupled transport. Intracellular peptidases release free amino acids, which leave the basolateral membrane by facilitated transport into portal blood.

1. Hydrochloric acid denatures protein; pepsin hydrolyzes peptide bonds.
2. Enteropeptidase activates trypsinogen to trypsin.
3. Trypsin activates chymotrypsinogen and other pancreatic protease zymogens.
4. Apical free-amino-acid uptake is commonly Na^+ -coupled; small-peptide uptake is H^+ -coupled.
5. Basolateral amino-acid exit uses facilitated transport.
6. Excess amino acids are degraded after their nitrogen is removed.

→ clinic: Pancreatic insufficiency impairs both protein and fat digestion.

Lippincott 6e, Chapter 19, pp. 245–249.

Shared amino-acid transporters and cystinuria

Amino acids use carrier systems with overlapping substrate specificity in the intestine and renal proximal tubule.

Cystinuria is a defect of the transporter handling cystine and the dibasic amino acids ornithine, lysine, and arginine.

Reduced renal reabsorption increases urinary loss of all four, but poorly soluble cystine is the component that forms stones. Cystinuria is a membrane-transport defect and differs from cystinosis, in which cystine cannot leave lysosomes.

1. COLA identifies cystine, ornithine, lysine, and arginine.
2. Cystinuria affects intestinal uptake and renal proximal-tubule reabsorption.
3. Cystine precipitation causes recurrent urinary calculi.
4. Cystinosis is a separate lysosomal cystine-transport disorder.

→ **clinic:** Cystine stones can obstruct the urinary tract; hydration dilutes urinary cystine and reduces its precipitation.

Lippincott 6e, Chapter 19, pp. 249–250.

Transamination, PLP, and the ALT/AST reactions

Aminotransferases reversibly move an amino group from an amino acid to an alpha-keto acid without releasing free ammonia. Most amino groups are collected on alpha-ketoglutarate, producing glutamate, and every aminotransferase requires pyridoxal phosphate derived from vitamin B6. ALT interconverts alanine plus alpha-ketoglutarate with pyruvate plus glutamate, whereas AST interconverts aspartate plus alpha-ketoglutarate with oxaloacetate plus glutamate. The reverse AST reaction supplies aspartate nitrogen to the urea cycle. Plasma ALT is more specific for hepatocellular injury than AST, although the liver contains a larger absolute amount of AST and both enzymes also occur outside the liver.

1. Transamination exchanges amino groups; it does not directly liberate NH_4^+ .
2. PLP temporarily accepts nitrogen as pyridoxamine phosphate.
3. ALT: alanine + alpha-ketoglutarate $\rightleftharpoons$ pyruvate + glutamate.
4. AST: aspartate + alpha-ketoglutarate $\rightleftharpoons$ oxaloacetate + glutamate.
5. ALT has greater liver specificity; AST is also prominent in cardiac and skeletal muscle.

→ **clinic:** Raised serum aminotransferases indicate cell injury, not a measurement of the liver's synthetic capacity.

Lippincott 6e, Chapter 19, pp. 250–252.

Glutamate dehydrogenase releases collected amino nitrogen

Mitochondrial glutamate dehydrogenase converts glutamate to alpha-ketoglutarate while releasing ammonia.

Transamination first collects nitrogen on glutamate, so its subsequent oxidative deamination disposes of nitrogen from many different amino acids. The reversible enzyme uses NAD^+ mainly for oxidative deamination and NADPH mainly for reductive amination, with direction determined by substrates and redox state. ADP activates the enzyme and GTP inhibits it, favoring carbon-skeleton oxidation when cellular energy is low.

1. Glutamate dehydrogenase is mitochondrial and prominent in liver and kidney.
2. Oxidative deamination regenerates alpha-ketoglutarate and releases NH_4^+ .
3. Transamination followed by oxidative deamination is called transdeamination.
4. NAD^+ favors the oxidative direction; NADPH supports reductive amination.
5. ADP activates and GTP inhibits glutamate dehydrogenase.

Lippincott 6e, Chapter 19, pp. 252–253.

Glutamine and alanine safely carry peripheral nitrogen

Peripheral tissues avoid releasing large amounts of toxic free ammonia by carrying nitrogen mainly as glutamine or alanine. Glutamine synthetase consumes ATP to add ammonia to glutamate, forming glutamine; hepatic glutaminase hydrolyzes glutamine to release ammonia for urea synthesis. In muscle, ALT transfers amino nitrogen to pyruvate to make alanine, which travels to the liver. Hepatic ALT converts alanine back to pyruvate for gluconeogenesis while directing its nitrogen toward urea, completing the glucose-alanine cycle. In kidney, glutamine-derived ammonium is excreted and contributes to acid elimination.

1. Glutamine synthetase: $\text{glutamate} + \text{NH}_4^+ + \text{ATP} \rightarrow \text{glutamine} + \text{ADP} + \text{P}_i$.
2. Glutaminase releases glutamine's amide nitrogen as ammonia.
3. Alanine transports an amino group and a pyruvate carbon skeleton from muscle to liver.
4. Glutamine synthesis is a major ammonia-detoxification mechanism in the brain.
5. Renal NH_4^+ excretion supports acid-base balance.

→ **clinic:** Hyperammonemia is accompanied by increased glutamine as tissues sequester ammonia.

Lippincott 6e, Chapter 19, pp. 253, 256–257.

Urea-cycle reactions, nitrogen sources, and compartments

The liver converts nitrogen to urea, the major circulating nonprotein nitrogen end product, and sends it to the kidneys for excretion. One urea nitrogen comes from free ammonia, the other from aspartate, and its carbonyl carbon comes from bicarbonate. In the mitochondrial matrix, carbamoyl-phosphate synthetase I forms carbamoyl phosphate and ornithine transcarbamoylase transfers its carbamoyl group to ornithine, forming citrulline. Citrulline enters the cytosol, where argininosuccinate synthetase adds aspartate and argininosuccinate lyase releases fumarate and arginine. Cytosolic arginase hydrolyzes arginine to urea and ornithine, and ornithine returns to the mitochondrion. Fumarate can become malate and oxaloacetate, linking the urea cycle to the TCA cycle and regeneration of aspartate.

1. CPS I and ornithine transcarbamoylase are mitochondrial; the other three cycle enzymes are cytosolic.
2. NH_4^+ provides the first nitrogen; aspartate provides the second.
3. Citrulline exits and ornithine reenters the mitochondrion.
4. Argininosuccinate lyase produces arginine and fumarate.
5. Arginase produces urea and ornithine.
6. Urea is synthesized in liver and excreted mainly by kidney.

Lippincott 6e, Chapter 19, pp. 253–257.

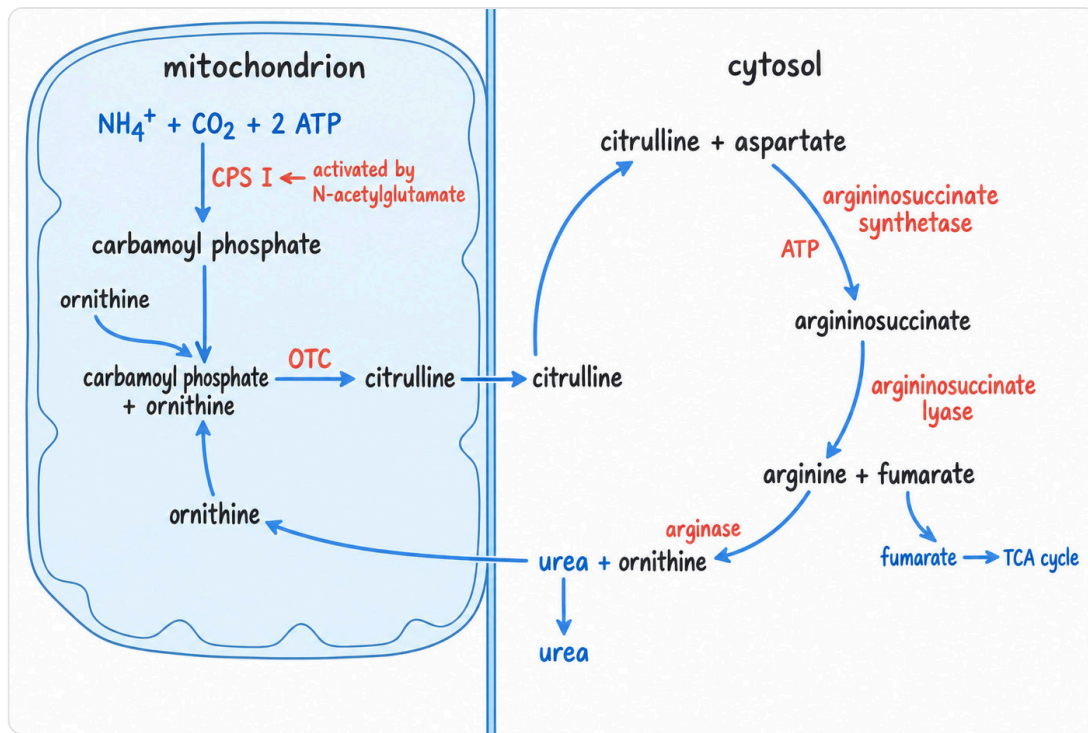


Fig. 14 The urea cycle links mitochondrial carbamoyl phosphate formation to cytosolic urea production and regeneration of ornithine.

urea-cycle energy cost and N-acetylglutamate regulation

Each turn of the urea cycle consumes three ATP molecules but four high-energy phosphate bonds. CPS I uses two ATP to form carbamoyl phosphate, whereas argininosuccinate synthetase cleaves a third ATP to AMP and pyrophosphate. N-acetylglutamate is the obligatory allosteric activator of mitochondrial CPS I. N-acetylglutamate synthase makes it from glutamate and acetyl-CoA and is stimulated by arginine, connecting increased amino-acid supply to increased nitrogen disposal. Cytosolic CPS II belongs to pyrimidine synthesis, uses glutamine nitrogen, and does not require N-acetylglutamate.

1. Urea synthesis consumes 3 ATP but 4 high-energy phosphate bonds.
2. N-acetylglutamate activates CPS I.
3. Arginine activates N-acetylglutamate synthase.
4. CPS I uses free ammonia in mitochondria; CPS II uses glutamine in cytosol.
5. Substrate supply changes flux rapidly, while enzyme induction provides longer-term adaptation.

Lippincott 6e, Chapter 19, pp. 253-255.

Hyperammonemia and recognition of urea-cycle defects

Liver failure, portosystemic shunting, or a urea-cycle defect can prevent adequate ammonia disposal and cause neurotoxicity, cerebral edema, seizures, or coma. Ornithine transcarbamoylase deficiency is the most common inherited urea-cycle disorder and is X-linked; the other classic enzyme defects are autosomal recessive. A block before citrulline formation lowers citrulline, whereas argininosuccinate synthetase deficiency raises citrulline and argininosuccinate lyase deficiency raises argininosuccinate. In OTC deficiency, excess carbamoyl phosphate can enter pyrimidine synthesis and increase urinary orotate, unlike a block at CPS I. Arginase deficiency raises arginine and generally produces less severe hyperammonemia than upstream defects. Treatment principles include limiting excess nitrogen intake while preventing catabolism and providing alternative urinary nitrogen excretion, such as phenylacetylglutamine formation from phenylbutyrate-derived phenylacetate.

1. OTC deficiency is X-linked; other classic urea-cycle enzyme deficiencies are autosomal recessive.
2. CPS I deficiency: low citrulline without the orotate increase characteristic of OTC deficiency.
3. OTC deficiency: low citrulline with increased urinary orotate.
4. Argininosuccinate synthetase deficiency causes citrullinemia.
5. Argininosuccinate lyase deficiency causes argininosuccinate accumulation.
6. Arginase deficiency raises arginine and often has milder hyperammonemia.

→ clinic: Increased plasma glutamine accompanies impaired ammonia disposal.

→ clinic: A newborn becoming lethargic after protein feeding warrants consideration of a nitrogen-disposal defect.

Lippincott 6e, Chapter 19, pp. 254–260; Chapter 22, pp. 302–303.

Amino acid degradation and synthesis

Lippincott 6e, Chapter 20; teacher slides cited under each idea.

practice-bank topic · outside the confirmed Term 1 syllabus, taught because the practice questions test it

Amino-acid disorders: blocked enzyme and distinguishing findings

Disorder	Principal defect	Distinguishing biochemical or clinical finding
Classic PKU	Phenylalanine hydroxylase	High phenylalanine; tyrosine becomes essential; musty odor
MSUD	Branched-chain alpha-ketoacid dehydrogenase	High leucine, isoleucine, valine and ketoacids; sweet urine odor
Alkaptonuria	Homogentisate dioxygenase	Urine darkens on standing; ochronosis; arthritis
Classic homocystinuria	Cystathionine beta-synthase	High homocysteine and methionine; thrombosis; ectopia lentis
Tyrosinase-negative albinism	Tyrosinase	Low melanin in skin, hair, and eyes; visual abnormalities

Glucogenic and ketogenic amino acids

After nitrogen removal, amino-acid carbon skeletons converge on pyruvate, oxaloacetate, alpha-ketoglutarate, succinyl-CoA, fumarate, acetyl-CoA, or acetoacetate. Glucogenic amino acids form pyruvate or TCA-cycle intermediates that can support net glucose production. Ketogenic amino acids form acetyl-CoA or acetoacetate, which cannot yield net glucose in humans. Leucine and lysine are exclusively ketogenic; isoleucine, phenylalanine, tyrosine, and tryptophan have both glucogenic and ketogenic products in Lippincott's human pathway classification. Valine and methionine are glucogenic because their degradation can produce succinyl-CoA.

1. Only leucine and lysine are exclusively ketogenic.
2. Phenylalanine and tyrosine yield fumarate plus acetoacetate.
3. Isoleucine yields acetyl-CoA plus a precursor of succinyl-CoA.
4. Valine, isoleucine, methionine, and threonine can form propionyl-CoA, then succinyl-CoA.
5. Propionyl-CoA conversion requires biotin and vitamin B12.
6. Alanine becomes pyruvate; aspartate becomes oxaloacetate.

Lippincott 6e, Chapter 20, pp. 261–267.

Folate one-carbon units and serine–glycine interconversion

Tetrahydrofolate carries single-carbon groups in several oxidation states on N5, N10, or both nitrogen atoms. Serine hydroxymethyltransferase uses PLP to convert serine to glycine while transferring a one-carbon unit to THF, forming N5,N10-methylene-THF. This reversible reaction links amino-acid metabolism to one-carbon requirements for nucleotide synthesis. N10-formyl-THF supplies carbons to purines, N5,N10-methylene-THF supports dTMP synthesis, and N5-methyl-THF supports homocysteine remethylation. Biotin carries CO₂ in carboxylation reactions and is distinguished from the folate one-carbon pool.

1. Serine → glycine requires PLP and THF, with THF accepting the one-carbon unit.
2. N5,N10-methylene-THF is used to convert dUMP to dTMP.
3. N10-formyl-THF supplies purine ring carbons.
4. N5-methyl-THF transfers a methyl group through B12 to homocysteine.
5. Folate deficiency can increase urinary FIGlu after histidine metabolism.

→ **clinic:** Reduced folate availability impairs nucleotide synthesis and causes megaloblastic anemia.

Lippincott 6e, Chapter 20, pp. 262–264, 267–268.

Synthesis of nonessential amino acids

Nonessential amino acids can be synthesized from common metabolic intermediates in adequate amounts under ordinary conditions. Transamination converts pyruvate, oxaloacetate, and alpha-ketoglutarate into alanine, aspartate, and glutamate, respectively. Serine is synthesized from the glycolytic intermediate 3-phosphoglycerate, and serine can supply glycine. ATP-dependent amidation makes glutamine from glutamate using free ammonia and asparagine from aspartate using glutamine as the nitrogen donor. Tyrosine depends on phenylalanine hydroxylation, and cysteine depends on sulfur from methionine, so these amino acids become conditionally essential if their precursor supply or conversion is inadequate.

1. Serine is nonessential in healthy adults and arises from 3-phosphoglycerate.
2. Alanine/pyruvate, aspartate/oxaloacetate, and glutamate/alpha-ketoglutarate are transamination pairs.
3. Glutamine synthetase uses free ammonia; asparagine synthetase uses glutamine nitrogen.
4. Proline is synthesized from glutamate via glutamate semialdehyde.
5. Phenylalanine hydroxylase requires tetrahydrobiopterin to synthesize tyrosine.
6. Cysteine synthesis depends on adequate methionine and serine.

Lippincott 6e, Chapter 20, pp. 267–269.

Methionine, S-adenosylmethionine, and transsulfuration

Methionine combines with ATP to form S-adenosylmethionine, the principal activated methyl donor in human metabolism. After methyl donation, S-adenosylhomocysteine is hydrolyzed to homocysteine and adenosine. Homocysteine can be remethylated to methionine by methionine synthase using methylcobalamin and N5-methyl-THF. Alternatively, PLP-dependent cystathionine beta-synthase joins homocysteine to serine, and cystathionine cleavage produces cysteine plus alpha-ketobutyrate. This transsulfuration route transfers methionine sulfur to cysteine while the remaining methionine carbon skeleton can become succinyl-CoA.

1. SAM is made from methionine and ATP and donates activated methyl groups.
2. SAM → S-adenosylhomocysteine → homocysteine.

3. Methionine synthase requires vitamin B12 and N5-methyl-THF.
4. Cystathionine beta-synthase and cystathionine cleavage require PLP.
5. Serine provides the cysteine carbon skeleton; methionine supplies its sulfur.

→ **clinic:** B6, B12, or folate deficiency can increase homocysteine through different metabolic blocks.

Lippincott 6e, Chapter 20, pp. 264–265.

classic phenylketonuria and tetrahydrobiopterin defects

Classic phenylketonuria results from deficient phenylalanine hydroxylase, which normally converts phenylalanine to tyrosine using tetrahydrobiopterin. Phenylalanine and alternative metabolites accumulate, while tyrosine becomes an essential dietary amino acid. Untreated disease can cause severe neurodevelopmental impairment, seizures, a musty odor, and hypopigmentation. Early, continuing restriction of phenylalanine limits toxicity, but phenylalanine cannot be eliminated because it is itself essential. Defects of BH4 synthesis or regeneration also cause hyperphenylalaninemia and impair dopamine and serotonin synthesis, so phenylalanine restriction alone does not correct their neurotransmitter deficit.

1. Classic PKU is phenylalanine hydroxylase deficiency.
2. BH4 deficiency is a distinct cause of hyperphenylalaninemia.
3. Tyrosine becomes essential when phenylalanine hydroxylation is blocked.
4. Phenylpyruvate, phenylacetate, and phenyllactate can accumulate.
5. Dietary phenylalanine is controlled, not eliminated; aspartame supplies phenylalanine.
6. Maternal hyperphenylalaninemia can harm fetal development even if the fetus does not have PKU.

→ **clinic:** Newborn screening permits treatment before irreversible neurologic injury.

Lippincott 6e, Chapter 20, pp. 270–272.

Branched-chain amino acids and maple syrup urine disease

Leucine, isoleucine, and valine are branched-chain amino acids whose initial catabolism occurs mainly in peripheral tissues, especially muscle. After PLP-dependent transamination, the branched-chain alpha-ketoacid dehydrogenase complex carries out oxidative decarboxylation. Deficiency of this complex causes maple syrup urine disease, with accumulation of the three amino acids and their corresponding alpha-keto acids. Feeding difficulty, neurologic deterioration, ketoacidosis, and a sweet urine odor are characteristic, and leucine accumulation is especially neurotoxic. Management controls branched-chain amino-acid intake while retaining enough for growth; some variants respond to thiamine.

1. MSUD is branched-chain alpha-ketoacid dehydrogenase deficiency, after the transamination step.
2. The affected amino acids are leucine, isoleucine, and valine.
3. BCKD requires TPP, lipoate, FAD, NAD⁺, and CoA.
4. Leucine is ketogenic; valine is glucogenic; isoleucine is both.
5. The shared E3 subunit also participates in PDH and alpha-ketoglutarate dehydrogenase.

→ **clinic:** A defect in shared E3 can add lactic acidosis to the MSUD phenotype.

Lippincott 6e, Chapter 20, pp. 265–267, 272–273.

Alkaptonuria and albinism affect different tyrosine pathways

Tyrosine supplies both degradative products and the pigment melanin, so defects in these separate routes have different clinical effects. Alkaptonuria is deficient homogentisate dioxygenase, also called homogentisic acid oxidase, in phenylalanine and tyrosine degradation. Homogentisate accumulates and oxidizes to dark pigment, causing urine darkening on standing, connective-tissue ochronosis, and later arthritis. Tyrosinase-negative oculocutaneous albinism instead blocks melanin synthesis because copper-dependent tyrosinase is absent or defective. Albinism causes reduced pigmentation and ocular abnormalities, without the homogentisate accumulation of alkaptonuria.

1. Alkaptonuria: homogentisate dioxygenase deficiency.
2. Alkaptonuria causes darkening urine, ochronosis, and large-joint disease.
3. Tyrosinase converts tyrosine toward DOPA and melanin.
4. Tyrosinase-negative oculocutaneous albinism causes deficient skin, hair, and eye pigment.
5. Albinism has several genetic forms; not every form is tyrosinase deficiency.

→ **clinic:** Albinism can cause photophobia, impaired vision, and increased skin-cancer risk.

Lippincott 6e, Chapter 20, pp. 263–264, 273–274.

Homocystinuria: transsulfuration versus remethylation defects

Classic homocystinuria most often reflects cystathionine beta-synthase deficiency, which blocks the PLP-dependent entry of homocysteine into transsulfuration. Homocysteine and methionine rise, while cysteine synthesis decreases. Major manifestations include thromboembolism, lens dislocation, long limbs and fingers, and variable neurodevelopmental impairment. Some patients retain enough enzyme function to respond to pyridoxine. Defects in homocysteine remethylation can also raise homocysteine but do not produce the same increased-methionine pattern, so elevated homocysteine alone does not identify the enzyme defect.

1. Classic homocystinuria is usually autosomal recessive cystathionine beta-synthase deficiency.
2. CBS deficiency raises homocysteine and methionine and lowers cysteine availability.
3. CBS requires vitamin B6-derived PLP.
4. Thrombosis is a major cause of morbidity and early death.
5. Remethylation requires folate and B12; its failure can raise homocysteine without high methionine.

→ **clinic:** The combination of ectopia lentis, skeletal findings, and thrombosis helps distinguish classic homocystinuria.

Lippincott 6e, Chapter 20, pp. 264–265, 273–274.

Heme, bilirubin and amine products

Lippincott 6e, Chapter 21; teacher slides cited under each idea.

practice-bank topic · outside the confirmed Term 1 syllabus, taught because the practice questions test it

High-yield defects in heme synthesis (Lippincott 6e, chapter 21, pp. 279–281, Figure 21.8)

Disorder	Enzyme affected	Distinguishing consequence
Acute intermittent porphyria	Hydroxymethylbilane synthase / porphobilinogen deaminase	ALA and porphobilinogen accumulate; neurovisceral attacks without photosensitivity
Porphyria cutanea tarda	Uroporphyrinogen decarboxylase	Uroporphyrin accumulation and photosensitive skin lesions
Congenital erythropoietic porphyria	Uroporphyrinogen III synthase	Type I porphyrins accumulate; photosensitivity
Erythropoietic protoporphyria	Ferrochelatase	Protoporphyrin accumulates; photosensitivity
Lead poisoning	ALA dehydratase and ferrochelatase inhibited	Reduced heme synthesis; increased ALA and anemia

Heme synthesis links glycine, succinyl-CoA, and iron

Heme consists of ferrous iron coordinated within protoporphyrin IX and is made chiefly in erythroid precursors and liver. Mitochondrial ALA synthase condenses glycine with succinyl-CoA in the PLP-dependent, committed and rate-limiting reaction. ALA then enters the cytosolic portion of the pathway, whereas the last three reactions occur in mitochondria. Ferrochelatase completes synthesis by inserting Fe^{2+} into protoporphyrin IX. Hepatic hemin suppresses ALAS1 expression and mitochondrial import, while erythroid ALAS2 production responds to iron availability.

1. ALA synthase requires pyridoxal phosphate, derived from vitamin B6.
2. The first and last three reactions are mitochondrial; intervening reactions are cytosolic.
3. ALA dehydratase joins two ALA molecules to form porphobilinogen.
4. Four porphobilinogens form hydroxymethylbilane, which becomes uroporphyrinogen III.
5. Ferrochelatase inserts ferrous iron into protoporphyrin IX.
6. Mature erythrocytes cannot synthesize heme because they lack mitochondria.

→ clinic: Loss of erythroid ALAS2 activity can cause X-linked sideroblastic anemia.

→ clinic: Drugs that induce hepatic cytochrome P450 can consume heme and increase ALAS1 expression.

Lippincott 6e, Chapter 21, pp. 277–281, porphyrin structure, heme synthesis, and Figures 21.3–21.8.

Porphyrias and lead poisoning block different heme-synthesis steps

A block in heme synthesis causes accumulation of intermediates upstream of the defective enzyme. Acute intermittent porphyria is caused by hydroxymethylbilane synthase deficiency, also called porphobilinogen deaminase deficiency, and produces neurovisceral attacks without photosensitivity. Porphyria cutanea tarda involves reduced uroporphyrinogen decarboxylase activity and photosensitive skin lesions. Lead directly inhibits both ALA dehydratase and ferrochelatase, reducing heme production and contributing to anemia. Decreased hepatic heme removes feedback suppression of ALAS1, amplifying precursor accumulation.

1. Acute intermittent porphyria: increased ALA and porphobilinogen, abdominal pain, and neurologic or psychiatric symptoms.
2. Porphyria cutanea tarda: uroporphyrinogen decarboxylase deficiency, porphyrin accumulation, and photosensitivity.
3. Ferrochelatase deficiency causes erythropoietic protoporphyria.
4. Lead inhibits ALA dehydratase and ferrochelatase; ALAS1 may increase secondarily.
5. Heme depletion by enzyme-inducing drugs can precipitate acute hepatic porphyria attacks.

→ clinic: Hemin and glucose suppress hepatic ALAS1 and reduce precursor production during acute porphyria attacks.

→ clinic: Iron overload, alcohol, estrogens, and viral infection can contribute to expression of porphyria cutanea tarda.

Lippincott 6e, Chapter 21, pp. 279–281, porphyrias and Figure 21.8; p. 290, lead-poisoning study question.

Heme degradation produces bilirubin, urobilin, and stercobilin

Most heme degradation follows removal of aging erythrocytes by macrophages in the spleen and liver. Heme oxygenase uses oxygen and NADPH to open the ring, releasing iron and carbon monoxide and forming green biliverdin; biliverdin reductase then forms bilirubin. Poorly water-soluble unconjugated bilirubin travels to the liver bound to albumin. Hepatic bilirubin UDP-glucuronosyltransferase, UGT1A1, adds glucuronic acid from UDP-glucuronate, producing water-soluble conjugated bilirubin for secretion into bile. Intestinal bacteria generate urobilinogen, most of which gives rise to brown stercobilin, while a small portion reaches urine and is oxidized to yellow urobilin.

1. Heme oxygenase opens the porphyrin ring; biliverdin reductase reduces the resulting biliverdin.
2. Heme oxygenase requires O₂ and NADPH and releases CO and Fe²⁺.
3. Unconjugated bilirubin is albumin-bound and normally absent from urine.
4. Conjugated bilirubin can enter urine when its plasma concentration rises.
5. Stercobilin colors stool brown; urobilin colors urine yellow.

→ clinic: Conjugation defects cause unconjugated hyperbilirubinemia; impaired biliary secretion can cause conjugated hyperbilirubinemia.

Lippincott 6e, Chapter 21, pp. 282–284, heme degradation, bilirubin conjugation, and Figures 21.9–21.10.

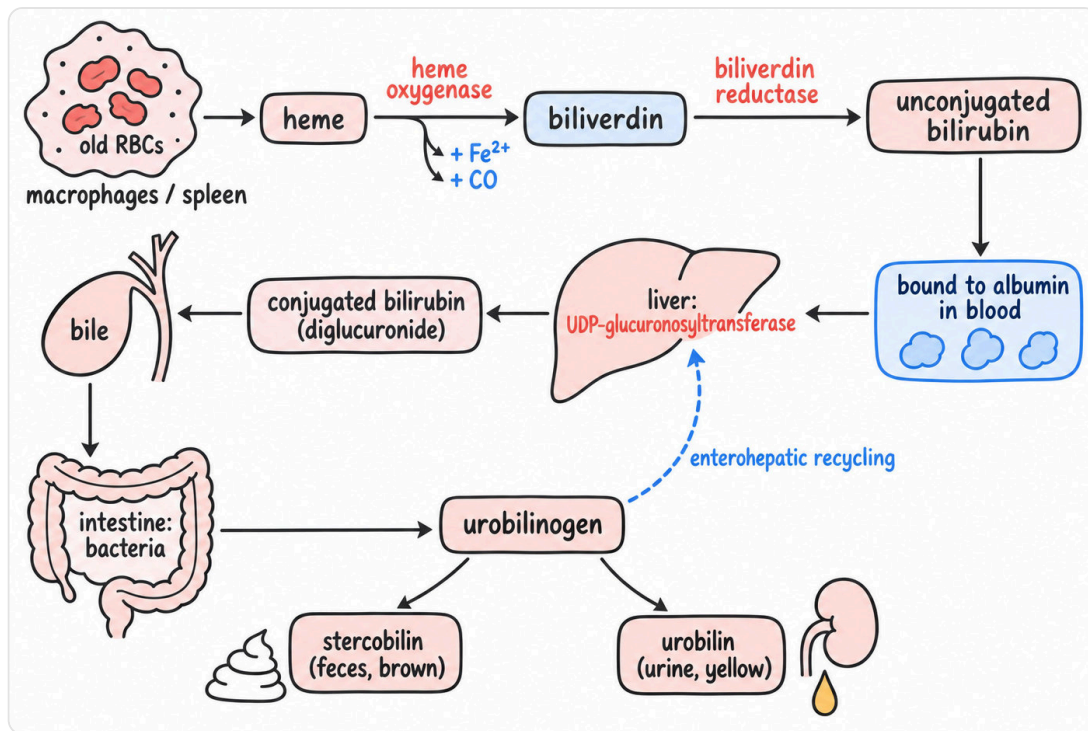


fig. 15 Heme breakdown, hepatic bilirubin conjugation, and intestinal conversion to fecal and urinary pigments.

Jaundice patterns reflect bilirubin production, conjugation, and excretion

Jaundice is yellow discoloration caused by bilirubin accumulation, so plasma bilirubin is the analyte measured directly to assess it. Hemolysis raises bilirubin production and typically increases unconjugated bilirubin and urinary urobilinogen. Hepatocellular injury can impair both conjugation and biliary secretion, creating a mixed pattern, whereas complete extrahepatic obstruction causes predominantly conjugated hyperbilirubinemia, bilirubinuria, pale stools, and absent or markedly reduced urinary urobilinogen. Gilbert syndrome causes mild, intermittent unconjugated hyperbilirubinemia through reduced UGT1A1 activity. Physiologic neonatal jaundice reflects low conjugating capacity, and indicated phototherapy uses blue light to form more water-soluble bilirubin photoisomers that can be excreted without glucuronidation.

1. Direct-reacting bilirubin approximates conjugated bilirubin; indirect bilirubin is calculated as total minus direct.
2. Hemolysis: unconjugated bilirubin rises, urine bilirubin is absent, and urinary urobilinogen increases.
3. Complete obstruction: conjugated bilirubin rises, urine bilirubin appears, and intestinal and urinary urobilinogen fall.
4. Hepatocellular jaundice can contain both conjugated and unconjugated bilirubin.
5. Gilbert syndrome is usually benign and may become apparent with fasting or illness.
6. Neonatal phototherapy changes bilirubin structure rather than activating hepatic conjugation; severe unconjugated hyperbilirubinemia can cause kernicterus.

→ **clinic:** Severe loss of bilirubin UGT activity causes Crigler-Najjar syndrome.

→ **clinic:** Unconjugated bilirubin becomes especially neurotoxic when the albumin-binding capacity is exceeded.

Lippincott 6e, Chapter 21, pp. 282–285, bilirubin UGT deficiency, jaundice types, phototherapy, and bilirubin measurement. · MedlinePlus Genetics, Gilbert syndrome, Description and Causes (supplemental support for benign intermittent disease and fasting/illness triggers): <https://medlineplus.gov/genetics/condition/gilbert-syndrome/>

Tyrosine supplies dopamine, norepinephrine, and epinephrine

Tyrosine is converted to DOPA by tetrahydrobiopterin-dependent tyrosine hydroxylase, the rate-limiting step in catecholamine synthesis. PLP-dependent decarboxylation of DOPA produces dopamine, which dopamine beta-hydroxylase converts to norepinephrine using copper and ascorbate. Phenylethanolamine N-methyltransferase (PNMT) uses S-adenosylmethionine to methylate norepinephrine into epinephrine. Monoamine oxidase and catechol-O-methyltransferase inactivate catecholamines, producing metabolites that are excreted in urine.

1. Pathway: tyrosine → DOPA → dopamine → norepinephrine → epinephrine.
2. Tyrosine hydroxylase requires BH₄; DOPA decarboxylase requires PLP.
3. Dopamine beta-hydroxylase requires copper and vitamin C.
4. SAM provides methyl groups for epinephrine synthesis and COMT reactions.
5. Vanillylmandelic acid derives from norepinephrine and epinephrine; homovanillic acid derives from dopamine.

→ **clinic:** Levodopa crosses the blood-brain barrier and supplies a dopamine precursor in Parkinson disease.

→ **clinic:** Catecholamine-producing pheochromocytomas can increase urinary catecholamine metabolites.

Lippincott 6e, Chapter 21, pp. 286–288, catecholamine synthesis and degradation, Figures 21.15–21.16.

Histidine and tryptophan supply histamine and serotonin

Histamine forms by PLP-dependent decarboxylation of histidine. It participates in allergic and inflammatory responses and stimulates gastric acid secretion. Serotonin, or 5-hydroxytryptamine, is made from tryptophan by BH₄-dependent hydroxylation followed by PLP-dependent decarboxylation. Serotonin functions in the gut and nervous system, is degraded by monoamine oxidase, and can be converted to melatonin in the pineal gland.

1. Histidine decarboxylase converts histidine to histamine.
2. Histamine synthesis removes CO₂ and requires vitamin B₆-derived PLP.
3. Tryptophan hydroxylase forms 5-hydroxytryptophan using BH₄.
4. Decarboxylation of 5-hydroxytryptophan forms serotonin.
5. Serotonin is a precursor of melatonin.

→ **clinic:** Antihistamines act at histamine receptors; blocking synthesis is not their usual mechanism.

→ **clinic:** Selective serotonin reuptake inhibitors increase serotonin availability in synapses.

Lippincott 6e, Chapter 21, pp. 286–288, histamine and serotonin, Figures 21.17–21.18. · Vitamin 2024.pdf, PDF p. 27, vitamin B₆ functions including amino-acid decarboxylation.

creatine buffers ATP and spontaneously forms creatinine

Creatine synthesis uses glycine and the guanidino group of arginine, followed by methylation of guanidinoacetate using S-adenosylmethionine derived from methionine. Liver and kidney participate in synthesis, and muscle accumulates creatine as a rapidly available energy buffer. Creatine kinase reversibly transfers phosphate between ATP and creatine, so phosphocreatine can immediately donate phosphate to ADP when ATP demand rises. Creatine and phosphocreatine undergo spontaneous nonenzymatic cyclization to creatinine, which is excreted in urine. Creatinine production therefore depends on muscle mass, while its blood concentration also depends on renal removal.

1. Creatine precursor amino acids are arginine, glycine, and methionine.
2. SAM supplies the methyl group; methionine contributes indirectly through SAM.

3. Creatine + ATP $\rightleftharpoons$ phosphocreatine + ADP is catalyzed by creatine kinase.
4. Phosphocreatine buffers ATP during abrupt increases in muscle energy demand.
5. Creatinine forms spontaneously rather than through a dedicated degradative enzyme.
6. Reduced muscle mass can lower creatinine production; reduced kidney clearance can raise plasma creatinine.

→ clinic: Creatine kinase in plasma can indicate tissue injury; creatinine is used to assess renal function in the context of muscle mass.

Lippincott 6e, Chapter 21, pp. 286–288, creatine synthesis and degradation, Figure 21.19.

Melanin synthesis uses tyrosine and tyrosinase

Melanocytes synthesize melanin from tyrosine to provide pigment in skin, hair, and eyes and to help protect tissues from sunlight. Copper-containing tyrosinase initiates this pathway by converting tyrosine toward DOPA and oxidized intermediates that form melanin. Tyrosinase is distinct from the BH₄-dependent tyrosine hydroxylase used in catecholamine synthesis. Defects in melanin production cause forms of oculocutaneous albinism, including tyrosinase-deficient disease.

1. Tyrosine is the amino-acid precursor of melanin.
2. Tyrosinase is a copper-containing enzyme in melanocytes.
3. Melanin contributes pigmentation and protection against solar radiation.
4. Albinism results from impaired melanin synthesis; it is not synonymous with loss of melanocytes.

→ clinic: Reduced pigment in albinism increases susceptibility to sunlight-related skin injury and is associated with ocular abnormalities.

Lippincott 6e, Chapter 21, pp. 286–288, melanin; Chapter 20, p. 273, albinism.

Nucleotide metabolism

Lippincott 6e, Chapter 22; teacher slides cited under each idea.

practice-bank topic · outside the confirmed Term 1 syllabus, taught because the practice questions test it

Purine versus pyrimidine metabolism (Lippincott 6e, chapter 22, pp. 293-304)

Feature	Purines	Pyrimidines
Ring assembly	Ring assembled on PRPP-derived ribose phosphate	Ring assembled first, then attached to PRPP-derived ribose phosphate
Ring donors	Glycine, glutamine, aspartate, CO ₂ , and formyl-THF	Aspartate, glutamine, and CO ₂
Regulated committed entry	Glutamine-PRPP amidotransferase; inhibited by AMP/GMP	CPS II; inhibited by UTP and activated by PRPP
Central de novo product	IMP, then AMP or GMP	OMP, then UMP, UTP, and CTP
Salvage example	HGPRT: hypoxanthine → IMP; guanine → GMP	Uridine salvage bypasses UMP synthase deficiency
Degradation endpoint	Uric acid in humans	Soluble beta-alanine or beta-aminoisobutyrate, NH ₃ , and CO ₂

De novo purine synthesis builds the ring on PRPP

Purine synthesis assembles the ring on ribose phosphate supplied by PRPP, producing inosine monophosphate as the first completed purine nucleotide. Glutamine-PRPP amidotransferase catalyzes the committed step and is inhibited by the end products AMP and GMP. Ring atoms come from glycine, glutamine, aspartate, CO₂, and two formyl groups carried by tetrahydrofolate. IMP branches to AMP and GMP, with GTP supporting AMP synthesis and ATP supporting GMP synthesis. Mycophenolic acid inhibits IMP dehydrogenase, blocking the conversion of IMP toward GMP.

1. PRPP synthetase activates ribose 5-phosphate; its reaction is not the purine-specific committed step.
2. IMP is the first fully formed purine nucleotide and contains hypoxanthine.
3. Glutamine-PRPP amidotransferase is stimulated by PRPP and inhibited by AMP and GMP.
4. Glycine donates atoms to purines, but not to the pyrimidine ring.
5. Glutamine, aspartate, and CO₂ contribute to both purine and pyrimidine rings.
6. Mycophenolate acts through mycophenolic acid to inhibit IMP dehydrogenase and guanine nucleotide synthesis.

→ clinic: Rapidly dividing cells are vulnerable to drugs that reduce nucleotide production.

Lippincott 6e, Chapter 22, pp. 293-296, de novo purine synthesis and regulation; pp. 305-306, study question 22.3 on mycophenolic acid.

Purine salvage conserves bases and restrains urate production

Purine salvage attaches preformed bases to ribose phosphate donated by PRPP. HGPRT converts hypoxanthine to IMP and guanine to GMP, whereas APRT salvages adenine to AMP. Near-complete HGPRT deficiency causes X-linked Lesch-Nyhan syndrome. Failed salvage leaves more bases for degradation and raises PRPP while lowering feedback inhibition by purine nucleotides, increasing de novo synthesis and uric acid production.

1. HGPRT salvages hypoxanthine and guanine; APRT salvages adenine.
2. Both phosphoribosyltransferases use PRPP.
3. Lesch-Nyhan syndrome results from severe HGPRT deficiency.
4. Increased PRPP and reduced IMP/GMP feedback drive additional purine synthesis.
5. Hyperuricemia, uric acid stones, motor dysfunction, and self-injurious behavior are characteristic.

→ **clinic:** Lowering urate addresses the metabolic complication of Lesch-Nyhan syndrome but does not correct its neurologic phenotype.

Lippincott 6e, Chapter 22, p. 296, purine salvage and Lesch-Nyhan syndrome, Figures 22.10–22.11.

Purine degradation and renal handling determine urate levels

Human purine degradation ends in uric acid, which is predominantly present as urate at physiologic pH. Xanthine oxidase catalyzes hypoxanthine to xanthine and xanthine to uric acid; humans do not further convert uric acid to allantoin. Hyperuricemia can reflect increased purine turnover or synthesis, but reduced renal excretion is the more common mechanism. Gout occurs when monosodium urate crystals deposit and trigger inflammation; allopurinol and its metabolite oxypurinol inhibit xanthine oxidase and lower urate production. Conversely, generalized proximal tubular dysfunction in Fanconi syndrome can increase renal urate loss and cause hypouricemia.

1. Purine breakdown ends at uric acid in humans.
2. Xanthine oxidase produces uric acid and is inhibited by allopurinol/oxypurinol.
3. Gout crystals are needle-shaped monosodium urate with negative birefringence.
4. Hyperuricemia alone does not establish gout.
5. Purine overproduction and renal underexcretion both raise urate.
6. Fanconi syndrome can cause renal urate wasting through impaired tubular reabsorption.

→ **clinic:** High cell turnover can increase uric acid production and promote gout or urate stones.

→ **clinic:** Uricosuric drugs increase renal urate excretion, whereas xanthine oxidase inhibitors reduce its production.

Lippincott 6e, Chapter 22, pp. 298–301, purine degradation, hyperuricemia, gout, and xanthine oxidase inhibitors. · Meisel AD and Diamond HS. Hyperuricosuria in the Fanconi syndrome. American Journal of the Medical Sciences. 1977;273:109–115 (supplemental primary evidence for Fanconi-related renal urate wasting): <https://pubmed.ncbi.nlm.nih.gov/848505/>

Ribonucleotide reductase supplies deoxyribonucleotides for DNA

Ribonucleotide reductase converts ribonucleoside diphosphates into their corresponding deoxyribonucleoside diphosphates by reducing the sugar 2'-hydroxyl group. It acts on both purine and pyrimidine substrates, connecting RNA-type precursors to DNA synthesis. Thioredoxin transfers reducing equivalents to the enzyme and is regenerated using NADPH. ATP activates overall enzyme activity, whereas excessive dATP inhibits it; additional specificity sites help balance individual deoxyribonucleotide pools. Hydroxyurea inhibits ribonucleotide reductase and reduces the supply of DNA precursors.

1. Substrates are ADP, GDP, CDP, and UDP; products are dADP, dGDP, dCDP, and dUDP.

2. NADPH supplies the ultimate reducing power through thioredoxin.
3. dATP inhibits overall reductase activity; ATP activates it.
4. Hydroxyurea targets ribonucleotide reductase rather than thymidylate synthase.

→ **clinic:** Excess dATP in adenosine deaminase deficiency can block deoxyribonucleotide production.

Lippincott 6e, Chapter 22, pp. 297–298, ribonucleotide reductase, allosteric regulation, and hydroxyurea.

ADA deficiency injures developing lymphocytes

Adenosine deaminase normally participates in degradation of adenosine and deoxyadenosine. Its deficiency permits these metabolites and their phosphorylated derivatives to accumulate, including dATP. High dATP inhibits ribonucleotide reductase, disrupting DNA precursor production, while accumulated purine metabolites also promote lymphocyte developmental arrest and apoptosis. Severe untreated ADA deficiency therefore causes combined immunodeficiency with profound reduction of T, B, and natural killer lymphocytes.

1. ADA deficiency causes accumulation of adenosine, deoxyadenosine, and dATP.
2. dATP inhibits ribonucleotide reductase and impairs DNA synthesis.
3. Severe ADA-deficient SCID affects T, B, and NK lineages.
4. The characteristic blood finding is severe lymphopenia rather than lymphocytosis.
5. ADA-deficient SCID is inherited as an autosomal recessive disorder.

→ **clinic:** Affected infants are vulnerable to severe infections; enzyme replacement, hematopoietic transplantation, and gene therapy address the underlying deficiency.

Lippincott 6e, Chapter 22, p. 301, adenosine deaminase deficiency and severe combined immunodeficiency.

Pyrimidines are assembled before attachment to ribose

Pyrimidine synthesis builds the ring from aspartate and carbamoyl phosphate before attaching it to the ribose phosphate of PRPP. Cytosolic carbamoyl phosphate synthetase II uses glutamine nitrogen and CO₂ and is inhibited by UTP and activated by PRPP. Orotate is attached to PRPP to form OMP and then decarboxylated to UMP by the two activities of UMP synthase. UMP supplies UDP and UTP, and glutamine-dependent amination of UTP forms CTP. Pyrimidine nucleoside kinases salvage uridine, cytidine, and thymidine; degradation of pyrimidine rings instead yields soluble beta-alanine or beta-aminoisobutyrate, with release of ammonia and CO₂.

1. Pyrimidine ring donors are aspartate, glutamine, and CO₂; glycine is not a donor.
2. CPS II is cytosolic and uses glutamine; urea-cycle CPS I is mitochondrial and uses free ammonia.
3. Pyrimidine synthesis is predominantly cytosolic, with dihydroorotate dehydrogenase associated with the inner mitochondrial membrane.
4. UMP synthase combines orotate phosphoribosyltransferase and OMP decarboxylase activities.
5. Hereditary orotic aciduria from UMP synthase deficiency can be treated with uridine to bypass the block.
6. CMP and UMP yield beta-alanine; TMP yields beta-aminoisobutyrate rather than uric acid.

→ **clinic:** UMP synthase deficiency causes orotic aciduria and megaloblastic anemia without the hyperammonemia expected in OTC deficiency.

Lippincott 6e, Chapter 22, pp. 302–304, pyrimidine synthesis, degradation, salvage, and Figure 22.21; pp. 305–306, study question 22.4.

Folate-dependent nucleotide synthesis explains key drug targets

Thymidylate synthase converts dUMP to dTMP using N⁵,N¹⁰-methylene tetrahydrofolate, which is oxidized to dihydrofolate during the reaction. Dihydrofolate reductase regenerates tetrahydrofolate, sustaining both thymidylate and purine synthesis. Methotrexate inhibits dihydrofolate reductase, whereas the active 5-fluorouracil metabolite FdUMP forms a stable inhibitory complex with thymidylate synthase. Bacteria also require de novo folate synthesis: sulfamethoxazole competes with PABA at dihydropteroate synthase, and trimethoprim inhibits bacterial dihydrofolate reductase.

1. Thymidylate synthase: dUMP → dTMP using N⁵,N¹⁰-methylene-THF.
2. Methotrexate inhibits DHFR and decreases reduced folate availability.
3. 5-FU is converted to FdUMP, which inhibits thymidylate synthase.
4. Sulfamethoxazole inhibits dihydropteroate synthase, not DHFR.
5. Trimethoprim inhibits bacterial DHFR.
6. Humans obtain folate from diet and lack the bacterial pathway that assembles folate from PABA.

→ clinic: Rapidly proliferating marrow and intestinal cells are particularly susceptible to impaired nucleotide synthesis.

Lippincott 6e, Chapter 22, pp. 293–295, Figure 22.7 on sulfonamides and antifolates; pp. 303–304, thymidylate synthase, methotrexate, and 5-fluorouracil. · Vitamin 2024.pdf, PDF pp. 34–35, folate metabolism, DHFR inhibitors, and impaired TMP/purine synthesis. · Nahimana A et al. Sulfa Resistance and Dihydropteroate Synthase Mutants in Recurrent Pneumocystis carinii Pneumonia. Emerging Infectious Diseases. 2003;9:864–867 (supplemental primary support for the distinct sulfamethoxazole and trimethoprim enzyme targets): https://wwwnc.cdc.gov/eid/article/9/7/02-0753_article

Insulin and glucagon

Lippincott 6e, Chapter 23; teacher slides cited under each idea.

Insulin biosynthesis and glucose-stimulated secretion

Preproinsulin becomes proinsulin and then insulin plus C-peptide in secretory granules. Beta-cell glucose metabolism raises ATP, closes ATP-sensitive K⁺ channels, depolarizes the membrane, and opens voltage-gated Ca²⁺ channels to trigger exocytosis. Endogenous insulin secretion releases equimolar C-peptide, unlike injected insulin.

1. Preproinsulin to proinsulin to insulin plus C-peptide
2. ATP closes KATP channels
3. Ca²⁺ triggers granule release
4. C-peptide marks endogenous secretion

→ clinic: Insulinoma

→ clinic: Sulfonylurea action

→ clinic: Distinguishing injected insulin

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 23 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Insulin receptor signaling and metabolic effects

The insulin receptor is a receptor tyrosine kinase that signals through IRS proteins, PI3K, and Akt. Insulin recruits GLUT4 to skeletal-muscle and adipose membranes and promotes glycogen, lipid, and protein synthesis while suppressing gluconeogenesis and lipolysis. Liver uses GLUT2 rather than insulin-recruited GLUT4, but insulin still changes hepatic enzyme activity and gene expression.

1. Insulin receptor is a receptor tyrosine kinase
2. GLUT4 is in muscle and adipose tissue
3. Insulin promotes storage and synthesis
4. Insulin suppresses hepatic glucose output

→ clinic: Insulin resistance

→ clinic: Type 2 diabetes

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 23 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Glucagon signaling and counterregulatory hormones

Glucagon binds a hepatic Gs-coupled receptor, raises cAMP, and activates PKA to promote glycogenolysis and gluconeogenesis while suppressing glycolysis. Skeletal muscle lacks a meaningful glucagon response and mobilizes glycogen mainly through epinephrine and contraction signals. Glucagon and epinephrine provide rapid defense against falling glucose, with cortisol and growth hormone acting more slowly.

1. Glucagon uses Gs-cAMP-PKA
2. Primary target is liver
3. No direct glucagon-driven muscle glycogenolysis
4. Counterregulation protects against hypoglycemia

→ clinic: Glucagon rescue for severe hypoglycemia

→ clinic: Fasting response

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 23 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Hypoglycemia recognition, mechanism, and biochemical diagnosis

Falling plasma glucose first triggers counterregulatory hormones and autonomic symptoms such as tremor, palpitations, anxiety, and sweating; more severe or prolonged deficiency deprives the brain of fuel and causes confusion, abnormal behavior, seizures, or coma. Ethanol can cause fasting hypoglycemia because its high NADH state diverts pyruvate to lactate and oxaloacetate to malate, limiting gluconeogenesis. During hypoglycemia, high insulin with high C-peptide supports endogenous secretion, whereas injected insulin raises insulin but suppresses C-peptide.

1. Autonomic symptoms are early warning signs
2. Neuroglycopenia causes confusion, seizures, and coma
3. Ethanol inhibits gluconeogenesis through a high-NADH redox shift
4. C-peptide distinguishes endogenous from injected insulin
5. Interpret insulin and C-peptide only in the context of documented low glucose

→ clinic: Insulinoma versus exogenous-insulin exposure

→ clinic: Alcohol-associated fasting hypoglycemia

→ clinic: Urgent recognition of neuroglycopenia

exam-scope.md (updated 2026-08-20): the active August 25 scope explicitly includes insulin, glucagon, and hypoglycemia. · Lippincott's Illustrated Reviews: Biochemistry, 6e, Chapter 23, hypoglycemia mechanisms and evaluation.

Test yourself

1. During symptomatic hypoglycemia, which laboratory pattern supports an insulin-secreting tumor rather than injected insulin?

A. Normal glucose with high ketones only	B. High insulin with high C-peptide
C. High insulin with absent C-peptide	D. Low insulin with low C-peptide
2. Which metabolic pattern is most consistent with insulin action after a meal?

A. Increased hepatic gluconeogenesis	B. Increased glycogen, fatty-acid, triacylglycerol, and protein synthesis
C. Increased hepatic ketogenesis and proteolysis	D. Increased adipose hormone-sensitive lipase
3. The insulin receptor belongs to which receptor class?

A. Receptor tyrosine kinase	B. G-protein-coupled receptor
C. Ligand-gated chloride channel	D. Intracellular steroid receptor

Answers: 1 B · 2 B · 3 A

Fed-fast integration

Lippincott 6e, Chapter 24; teacher slides cited under each idea.

Absorptive-state metabolism by organ

After a carbohydrate-containing meal, high insulin and low glucagon promote hepatic glycolysis, glycogen synthesis, fatty-acid synthesis, and TAG export. Adipose tissue increases GLUT4-mediated glucose uptake and TAG storage, while muscle takes up glucose and amino acids for glycogen and protein synthesis. Brain primarily uses glucose, and RBCs always rely on glycolysis.

1. High insulin-to-glucagon ratio
2. Liver stores glucose and makes lipid
3. Adipose stores TAG
4. Muscle makes glycogen and protein
5. RBCs remain glycolytic

→ clinic: Postprandial hyperglycemia

→ clinic: Insulin action

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 24 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Early fasting: glycogenolysis and gluconeogenesis

In early fasting, lower insulin and higher glucagon cause hepatic glycogenolysis to maintain blood glucose. As hepatic glycogen wanes, gluconeogenesis from lactate, glycerol, and glucogenic amino acids becomes dominant. Adipose lipolysis supplies fatty acids for tissue and hepatic energy plus glycerol for glucose synthesis.

1. Glycogenolysis dominates first
2. Gluconeogenesis rises with fasting duration
3. Lipolysis supplies fatty acids and glycerol
4. Liver preserves glucose for dependent tissues

→ clinic: Fasting hypoglycemia

→ clinic: Alcohol-associated impaired gluconeogenesis

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 24 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Prolonged fasting and protein-sparing ketone adaptation

With prolonged fasting, hepatic ketone production rises and the brain increasingly oxidizes ketones. This reduces the brain's glucose demand and therefore decreases the amount of amino acid that must be sacrificed for gluconeogenesis. RBCs still require glucose and produce lactate because they lack mitochondria.

1. Brain progressively uses ketones
2. Ketones spare muscle protein

3. RBCs cannot use ketones
4. Liver makes but does not consume ketones

→ clinic: Starvation

→ clinic: Ketogenic adaptation

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 24 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Tissue-specific fuel constraints

RBCs lack mitochondria and need glucose for glycolysis. Liver can oxidize fatty acids and make ketones but cannot use ketones because it lacks thiophorase; muscle uses fatty acids and ketones according to state. Brain normally relies on glucose but uses substantial ketones during prolonged fasting, while long-chain fatty acids do not cross the blood-brain barrier efficiently.

1. RBC: glucose only for ATP
2. Liver: produces ketones
3. Brain: glucose, then more ketones in fasting
4. Muscle: flexible fuel use

→ clinic: Hypoglycemic neurologic symptoms

→ clinic: Ketoacidosis

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 24 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Late starvation shifts muscle away from ketones

Resting skeletal muscle relies increasingly on fatty acids as insulin falls during fasting. It can use both fatty acids and ketone bodies early in starvation, but after prolonged starvation it reduces ketone consumption and uses fatty acids as its principal fuel. This leaves more circulating ketones available to the brain. As brain glucose demand falls, the requirement for gluconeogenesis from muscle amino acids also decreases.

1. Resting muscle favors fatty acids during prolonged starvation.
2. Late reduction in muscle ketone use helps preserve ketones for the brain.
3. Brain ketone use reduces the need for muscle-protein breakdown.
4. These resting-starvation adaptations do not describe all exercise intensities.

Lippincott 6e, Chapter 24, pp. 331–333 (resting skeletal muscle and brain in fasting).

Test yourself

1. Which tissues can oxidize liver-derived ketone bodies during fasting?

- A. Adipose lipid droplets only
- B. Extrahepatic tissues with mitochondria, including muscle and eventually brain
- C. Liver itself only
- D. Mature red blood cells

2. Which fuel does the brain normally use in the fed state?

- A. Chylomicron triacylglycerol
- B. Amino acids as its exclusive fuel
- C. Glucose
- D. Long-chain fatty acids

3. Which hormonal pattern characterizes the absorptive state after a carbohydrate-containing meal?

- A. High insulin-to-glucagon ratio
- B. Low insulin-to-glucagon ratio
- C. Absence of both hormones
- D. High glucagon with no insulin response

Answers: 1 B · 2 C · 3 A

Diabetes mellitus

Lippincott 6e, Chapter 25; teacher slides cited under each idea.

Type 1 versus type 2 diabetes pathophysiology

Type 1 diabetes results from autoimmune beta-cell destruction and absolute insulin deficiency. Early type 2 diabetes combines insulin resistance with compensatory hyperinsulinemia, followed by progressive beta-cell failure. Both increase hepatic glucose output and impair peripheral glucose handling, but severe unopposed lipolysis and ketogenesis are characteristic of untreated type 1 disease.

1. Type 1: absolute insulin deficiency
2. Type 2: insulin resistance plus beta-cell dysfunction
3. Both cause hyperglycemia
4. Type 1 has greater DKA risk

→ clinic: Autoimmune type 1 diabetes

→ clinic: Metabolic syndrome and type 2 diabetes

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 25 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Diagnosis and glycemic monitoring

Diabetes can be diagnosed with reproducibly elevated fasting plasma glucose, HbA1c, oral-glucose-tolerance values, or classic symptoms with a high random glucose. HbA1c reflects nonenzymatic glycation over the approximate lifespan of circulating RBCs and therefore represents roughly two to three months of glycemia. Altered RBC survival can make HbA1c misleading.

1. Repeat abnormal testing when symptoms are absent
2. HbA1c reflects recent months
3. RBC lifespan affects HbA1c
4. Diagnosis and treatment targets are not identical

→ clinic: Diabetes screening

→ clinic: Hemoglobinopathy or hemolysis affecting HbA1c

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 25 (textbook/tracked-bank; current thresholds require guideline verification). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

DKA versus hyperosmolar hyperglycemic state

Severe insulin deficiency activates adipose lipolysis and hepatic ketogenesis, causing DKA with high-anion-gap metabolic acidosis. HHS typically retains enough insulin to restrain major ketogenesis but not enough to prevent profound hyperglycemia and osmotic diuresis. Both cause dehydration and require urgent fluid, electrolyte, and insulin management.

1. DKA has significant ketonemia and acidosis
2. HHS has more extreme hyperosmolality

3. Osmotic diuresis drives dehydration
4. Potassium balance requires close monitoring

→ clinic: Diabetic ketoacidosis

→ clinic: Hyperosmolar hyperglycemic state

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 25 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Biochemical basis of chronic diabetic complications

Chronic hyperglycemia forms advanced glycation end products that cross-link matrix proteins, trap plasma proteins, and activate inflammatory receptors. Sorbitol accumulation and oxidative stress contribute in tissues with insulin-independent glucose uptake. These mechanisms participate in retinopathy, nephropathy, neuropathy, and accelerated atherosclerosis.

1. AGEs alter proteins and vessels
2. Sorbitol creates osmotic stress
3. Microvascular disease affects retina, kidney, and nerves
4. Cardiovascular risk is increased

→ clinic: Diabetic retinopathy

→ clinic: Nephropathy

→ clinic: Neuropathy

→ clinic: Atherosclerotic disease

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 25 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Test yourself

1. What is the fundamental defect in type 1 diabetes mellitus?

- | | |
|---|--|
| A. Defective glucagon receptors only | B. Autoimmune destruction of pancreatic beta cells causing absolute insulin deficiency |
| C. Primary insulin resistance with high insulin forever | D. Excessive beta-cell mass |

2. HbA1c most closely reflects average glycemia over approximately what interval?

- | | |
|----------------------------------|--|
| A. Only the current fasting hour | B. The preceding 2 to 3 months, weighted toward recent weeks |
| C. The preceding 10 minutes | D. Exactly one year |

3. A repeated fasting plasma glucose at or above which value meets a standard diagnostic threshold for diabetes?

- | | |
|--------------|--------------|
| A. 70 mg/dL | B. 90 mg/dL |
| C. 110 mg/dL | D. 126 mg/dL |

Answers: 1 B · 2 B · 3 D

Obesity and energy balance

Lippincott 6e, Chapter 26; teacher slides cited under each idea.

Obesity measurement and metabolic risk

BMI is weight in kilograms divided by height in meters squared and is a screening measure rather than a direct measure of adiposity. Visceral/upper-body fat is more strongly associated with insulin resistance, dyslipidemia, fatty liver, and cardiovascular disease than gluteofemoral fat. Waist measures therefore add risk information beyond BMI.

1. BMI = kg/m^2
2. BMI 30 or greater conventionally indicates obesity
3. Visceral fat carries higher metabolic risk
4. Interpret BMI in clinical context

→ clinic: Metabolic syndrome

→ clinic: NAFLD

→ clinic: Cardiovascular risk

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 26 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Leptin, ghrelin, and hypothalamic appetite circuits

Leptin from adipose tissue signals energy stores and normally suppresses orexigenic NPY/AgRP neurons while favoring anorexigenic pathways. Ghrelin rises before meals and stimulates appetite. Common obesity usually features high leptin with central leptin resistance, while weight loss lowers leptin and increases hunger signals.

1. Leptin reflects adipose stores
2. Common obesity has leptin resistance
3. Ghrelin stimulates appetite
4. NPY/AgRP neurons are orexigenic

→ clinic: Adaptive hunger after weight loss

→ clinic: Rare congenital leptin deficiency

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 26 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Adipose inflammation, adiponectin, and insulin resistance

Expanded visceral adipose tissue releases excess fatty acids and proinflammatory cytokines and recruits immune cells. Adiponectin, which normally supports insulin sensitivity and fatty-acid oxidation, generally falls in obesity. These changes promote hepatic and muscle insulin resistance and contribute to type 2 diabetes.

1. Adipose tissue secretes adipokines
2. Adiponectin falls in obesity

3. Inflammation promotes insulin resistance
4. Free-fatty-acid flux worsens ectopic lipid storage

→ clinic: Type 2 diabetes

→ clinic: NAFLD

→ clinic: Metabolic syndrome

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 26 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1-7, 14-18, and 23-33.

Test yourself

1. What leptin pattern is typical of common obesity?

- A. Absent leptin in nearly every patient
- C. Leptin unrelated to fat mass

- B. Low leptin caused by no adipose tissue
- D. Elevated leptin with central leptin resistance

2. Why is upper-body visceral obesity metabolically more hazardous than predominantly gluteofemoral fat?

- A. Only visceral fat stores vitamins
- C. Visceral fat contains no triglyceride

- B. Visceral fat has high lipolytic and inflammatory activity and drains toward the liver
- D. Gluteofemoral fat always causes diabetes

3. Which gastrointestinal hormone rises before meals and stimulates appetite?

- A. Adiponectin
- C. Leptin

- B. Ghrelin
- D. Insulin

Answers: 1 D · 2 B · 3 B

Nutrition

Lippincott 6e, Chapter 27; teacher slides cited under each idea.

EAR, RDA, AI, and tolerable upper intake level

The EAR estimates the intake meeting the needs of half a defined healthy group, while the RDA is set higher to cover nearly all. AI is used when evidence is insufficient to establish an EAR and RDA. The tolerable upper intake level estimates the highest chronic intake unlikely to create adverse effects and is not a recommended target.

1. EAR covers about half of a group
2. RDA covers nearly all healthy individuals
3. AI substitutes when evidence is insufficient
4. UL addresses toxicity risk

→ clinic: Dietary counseling

→ clinic: Supplement toxicity

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 27 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Energy density and total energy expenditure

Carbohydrate and protein provide about 4 kcal/g, fat about 9 kcal/g, and ethanol about 7 kcal/g. Resting/basal expenditure is usually the largest component of daily energy use in sedentary adults. Physical activity is the most variable component, and protein has the greatest thermic effect of feeding.

1. Fat: 9 kcal/g
2. Alcohol: 7 kcal/g
3. Carbohydrate/protein: 4 kcal/g
4. Basal metabolism is usually the largest component
5. Protein has high thermic effect

→ clinic: Weight-management counseling

→ clinic: Nutrition support calculations

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 27 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Essential fatty acids, limiting amino acids, and nitrogen balance

Linoleic acid and alpha-linolenic acid are the parent essential fatty acids because humans cannot introduce double bonds beyond carbon 9. A limiting amino acid is the essential amino acid present in the lowest amount relative to need. Positive nitrogen balance occurs in growth, pregnancy, or recovery, whereas illness, trauma, or inadequate protein can produce negative balance.

1. Linoleic acid is omega-6
2. Alpha-linolenic acid is omega-3

3. Essential amino acids must come from diet
4. Nitrogen balance reflects protein gain or loss

→ clinic: Growth and pregnancy

→ clinic: Catabolic illness

→ clinic: Essential-fatty-acid deficiency

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 27 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Kwashiorkor versus marasmus

Kwashiorkor features disproportionate protein deficiency with edema, fatty liver, skin/hair changes, and low albumin.

Marasmus reflects severe total energy deficiency with marked wasting of muscle and subcutaneous fat but usually no prominent edema. Mixed phenotypes occur, so real patients require full nutritional and medical assessment.

1. Kwashiorkor: edema and fatty liver
2. Marasmus: severe wasting
3. Low albumin contributes to edema
4. Infection commonly complicates both

→ clinic: Childhood malnutrition

→ clinic: Refeeding risk

Lippincott's Illustrated Reviews: Biochemistry, 6e — Chapter 27 (textbook concept; this chapter is confirmed for August 25 by exam-scope.md). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Test yourself

1. When is an Adequate Intake value used?

A. Only for energy intake	B. Whenever everyone has the same requirement
C. When evidence is insufficient to establish an EAR and RDA	D. Only when a nutrient is toxic
2. In a sedentary adult, which component usually accounts for the largest share of total daily energy expenditure?

A. Voluntary exercise in every person	B. Urinary glucose loss
C. Basal or resting energy expenditure	D. Thermic effect of food only
3. Which macronutrient generally has the highest thermic effect of feeding?

A. Cholesterol	B. Protein
C. Fat	D. Alcohol always

Answers: 1 C · 2 C · 3 B

Vitamins

Lippincott 6e, Chapter 28; teacher slides cited under each idea.

vitamins: function first, then deficiency

vitamin / active form	central role	Exam association
B1 / TPP	Oxidative decarboxylation; transketolase	Beriberi, Wernicke-Korsakoff; PDH, α -KGDH and branched-chain α -ketoacid DH.
B2 / FMN, FAD	Redox electron transfer	Cheilosis, glossitis; light-sensitive riboflavin can be depleted during phototherapy.
B3 / NAD ⁺ , NADP ⁺	Redox and other NAD-dependent reactions	Pellagra: dermatitis, diarrhea, dementia. Tryptophan-to-niacin synthesis requires B6.
B5 / coenzyme A, ACP	Acyl-group transfer	CoA contains pantothenate; not TPP or PLP.
B6 / PLP	Transamination, decarboxylation and other amino-acid reactions	Sideroblastic anemia, neuropathy/seizures; excess B6 can also cause neuropathy.
B7 / biotin	Carboxylation	Pyruvate, acetyl-CoA, propionyl-CoA and methylcrotonyl-CoA carboxylases; avidin binds biotin.
B9 / THF derivatives	One-carbon transfer; DNA synthesis	Megaloblastic anemia; homocysteine increases, methylmalonate does not.
B12 / methyl- and adenosylcobalamin	Methionine synthase; methylmalonyl-CoA mutase	Megaloblastic anemia plus possible neurologic disease; homocysteine and methylmalonate increase.
C / ascorbate	Collagen hydroxylation, antioxidant, iron absorption	Scurvy; deficient hydroxylation weakens collagen.
A / retinal, retinoic acid	Vision and gene regulation	Night blindness / xerophthalmia; excess preformed vitamin A is toxic and teratogenic.
D / calcitriol	Calcium/phosphate regulation	25-OH-D is the major circulating status marker; 1,25-(OH) ₂ -D is the active hormone.
E / tocopherols	Lipid-phase antioxidant	Hemolysis and neurologic dysfunction in deficiency.
K / reduced vitamin K	γ -carboxylation of glutamate residues	Clotting factors II, VII, IX, X and proteins C/S; warfarin blocks vitamin-K recycling.

vitamin solubility, coenzymes, storage, and toxicity

Most B vitamins act as coenzyme precursors, whereas vitamins A and D also have hormone-like transcriptional roles. Water-soluble vitamins generally have limited storage and faster deficiency, with B12 as a major exception. Fat-soluble vitamins require normal lipid absorption, are stored more readily, and therefore have greater potential for accumulation and toxicity.

1. B and C are water soluble
2. A, D, E, and K are fat soluble
3. Coenzymes may be loosely or tightly bound
4. Fat malabsorption threatens A/D/E/K
5. Storage increases toxicity potential

→ clinic: Cholestatic vitamin deficiency

→ clinic: Supplement toxicity

Vitamin 2024, TUMS Clinical Biochemistry — pp. 4–12 (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Thiamine (B1): TPP reactions and neurologic deficiency

Thiamine pyrophosphate supports oxidative decarboxylation by PDH, alpha-ketoglutarate dehydrogenase, and branched-chain alpha-ketoacid dehydrogenase, plus transketolase reactions. Deficiency impairs glucose oxidation and can raise lactate. Wernicke encephalopathy presents with confusion, ataxia, and ocular abnormalities and requires thiamine before glucose in an at-risk patient.

1. Active form is TPP
2. Required by three major dehydrogenase complexes
3. Required by transketolase
4. Deficiency causes Wernicke-Korsakoff and beriberi

→ clinic: Alcohol-use disorder

→ clinic: Refeeding

→ clinic: Lactic acidosis

Vitamin 2024 — water-soluble vitamin section, pp. 13–45 (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Riboflavin (B2) and niacin (B3) in redox metabolism

Riboflavin forms FMN and FAD, tightly associated redox cofactors in dehydrogenases and the respiratory chain. Niacin forms NAD⁺ and NADP⁺; NADH is used mainly for ATP-generating oxidation, while NADPH supports reductive synthesis and antioxidant defense. Niacin deficiency causes pellagra with dermatitis, diarrhea, and dementia.

1. B2 gives FMN and FAD
2. B3 gives NAD and NADP
3. NADH favors energy production
4. NADPH favors reductive work
5. Pellagra is niacin deficiency

→ clinic: Pellagra

→ clinic: Hartnup disease

→ clinic: Carcinoid-associated niacin depletion

Vitamin 2024 — water-soluble vitamin section, pp. 13–45 (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Pantothenate, pyridoxine, and biotin reaction patterns

Pantothenate is part of CoA and acyl-carrier protein, supporting acyl-group transfer. Pyridoxal phosphate from B6 supports amino-acid transamination and decarboxylation and glycogen phosphorylase; isoniazid can cause functional B6 deficiency. Biotin carries CO₂ in carboxylase reactions, and avidin in raw egg white can impair its absorption.

1. B5 supports acyl transfer
2. PLP supports amino-acid reactions
3. Isoniazid can cause B6 deficiency
4. Biotin supports carboxylation
5. Avidin binds biotin

→ clinic: Isoniazid neuropathy prevention

→ clinic: Multiple carboxylase deficiency

Vitamin 2024 — water-soluble vitamin section, pp. 13–45 (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Folate and B12 one-carbon metabolism

Tetrahydrofolate transfers one-carbon units for nucleotide synthesis, while B12 is needed for methionine synthase and methylmalonyl-CoA mutase. Both deficiencies cause megaloblastic anemia and elevated homocysteine, but only B12 deficiency raises methylmalonic acid and causes characteristic neurologic injury. B12 absorption requires gastric intrinsic factor and terminal ileum.

1. Folate supports nucleotide synthesis
2. B12 requires intrinsic factor
3. Both can cause megaloblastic anemia
4. Neurologic deficits and methylmalonate favor B12 deficiency
5. Folate prevents neural-tube defects

→ clinic: Pernicious anemia

→ clinic: Neural-tube defects

→ clinic: Methotrexate therapy

Vitamin 2024 — folate p. 34; B12 pp. 37–39 (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Vitamins A and D: vision, transcription, and calcium balance

Retinal participates in the visual cycle and retinoic acid regulates transcription through nuclear receptors; vitamin-A deficiency causes night blindness and may progress to xerophthalmia. Vitamin D3 is made in skin, 25-hydroxylated in liver, and 1-alpha-hydroxylated in kidney to calcitriol. Calcitriol supports calcium/phosphate balance and bone mineralization; deficiency causes rickets/osteomalacia and excess causes hypercalcemia.

1. 11-cis-retinal supports vision
2. Retinoic acid controls transcription
3. Vitamin D activation uses skin, liver, kidney
4. Calcitriol increases calcium/phosphate availability

5. A and D can be toxic in excess

→ clinic: Night blindness

→ clinic: Rickets and osteomalacia

→ clinic: Vitamin-D toxicity

Vitamin 2024 — vitamin A pp. 46–55; vitamin D pp. 56–63 (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Vitamins E, K, and C: antioxidant, coagulation, and collagen roles

Vitamin E is a lipid-phase antioxidant that protects membranes and RBCs from peroxidation; deficiency is most likely with fat malabsorption and can cause hemolysis or neurologic dysfunction. Vitamin K supports gamma-carboxylation of glutamate residues in clotting and bone proteins, and deficiency prolongs PT early because factor VII has a short half-life. Vitamin C supports collagen hydroxylation and iron absorption; deficiency causes scurvy with poor wound healing and bleeding.

1. Vitamin E protects membranes

2. Vitamin K enables gamma-carboxylation

3. PT rises early in vitamin-K deficiency

4. Vitamin C supports collagen hydroxylation

5. Fat malabsorption threatens E and K

→ clinic: Hemolysis in vitamin-E deficiency

→ clinic: Warfarin mechanism

→ clinic: Scurvy

Vitamin 2024 — vitamin C pp. 40–45; vitamins E/K pp. 63–70 (teacher-confirmed). · exam-scope.md (updated 2026-08-20): August 25 includes the complete confirmed Lippincott scaffold for Chapters 1–7, 14–18, and 23–33.

Calcitriol increases intestinal absorption and renal calcium conservation

Calcitriol, 1,25-dihydroxycholecalciferol, is the active hormone derived from vitamin D. Its production involves 25-hydroxylation in the liver followed by 1-alpha-hydroxylation in the kidney, with the latter stimulated by parathyroid hormone in response to low calcium. Calcitriol increases intestinal calcium absorption and helps increase renal tubular calcium reabsorption. It also works with parathyroid hormone to mobilize bone calcium when needed, while adequate vitamin D supports normal mineralization.

1. Calcitriol is active vitamin D, not a B-vitamin coenzyme.

2. The kidney performs the final activating hydroxylation.

3. Calcitriol increases intestinal calcium uptake and renal calcium reabsorption.

4. Deficiency causes rickets in children and osteomalacia in adults.

Lippincott 6e, Chapter 28, pp. 385–388 (vitamin D activation and actions). · Vitamin 2024.pdf, PDF pp. 59–61 (vitamin D metabolism and functions).

past paper asked

Which of the following vitamins provides the coenzyme for γ -carboxylation reactions?

Menaquinone is the best answer. Fat-soluble vitamins A, D, E, and K have distinct stored functions and deficiency patterns.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q7

past paper asked

How many vitamins are classified as fat-soluble?

4 is the best answer. There are four fat-soluble vitamins: A, D, E, and K.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q38

past paper asked

Which of the following is correctly matched with the vitamin it is derived from?

Pyridoxal phosphate-Vitamin B6 is the best answer. Pyridoxal phosphate is the active coenzyme form of vitamin B6.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q51

past paper asked

Which vitamin is a methyl group acceptor?

Vitamin B12 is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q52

past paper asked

Which of the following is not a clinical outcome of niacin deficiency?

Night blindness is the best answer. Niacin deficiency causes pellagra—dermatitis, diarrhea, and dementia—not night blindness.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q54

past paper asked

Which of these is a vitamin A precursor?

Beta-carotene is the best answer. β -Carotene can be cleaved to form vitamin A.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q59

past paper asked

Which of the following is the active form of pantothenic acid?

Coenzyme A is the best answer. Pantothenate is incorporated into coenzyme A.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q69

past paper asked

Compounds that are needed for enzymes to function properly are?

Vitamins is the best answer. Many vitamins are precursors of coenzymes required for enzyme function.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q75

past paper asked

A 60-year old female presents with severe back pain for the past week, she is diagnosed with osteoporosis, a condition resulting from calcium depletion in bones. As treatment which of the following vitamins can be prescribed?

Vit D is the best answer. Vitamin D supports calcium and phosphate absorption and normal bone mineralization.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q76

past paper asked

Which vitamin(s) play important roles in the metabolism of branched-chain amino acids and odd-carbon fatty acids?

Biotin supports propionyl-CoA carboxylase, shared by odd-chain fatty-acid and several amino-acid catabolic routes. B6 and B1 participate elsewhere in branched-chain amino-acid breakdown; they do not supply this shared carboxylase. B12 is also important downstream but is not offered.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q81

past paper asked

Which vitamin deficiency is seen in infants with jaundice who undergo phototherapy?

B2 is the best answer. Phototherapy can degrade riboflavin, making vitamin B2 deficiency a recognized risk.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q1

past paper asked

Which of the following vitamins is involved in the structure of coenzyme A?

Pantothenate (B 5) is the best answer. Pantothenate, vitamin B5, is a structural component of coenzyme A.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q6

past paper asked

In vitamin B6 deficiency, which other vitamin deficiency may also be observed?

B3 is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q10

past paper asked

In which reaction does biotin participate?

Carboxylation is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q17

past paper asked

Which one of the following is the main form of vitamin D in plasma?

25-Hydroxycholecalciferol (25 (OH)-D3) is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q18

past paper asked

Which of the following vitamins is directly involved in the conversion of homocysteine to methionine?

B12 (Cobalamin) is the best answer. Vitamin B12 is the cofactor for methionine synthase.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q24

past paper asked

Which vitamin deficiency causes hemolytic anemia?

E is the best answer. Vitamin E deficiency permits oxidative membrane damage and can cause hemolytic anemia.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q25

past paper asked

Which vitamin has thiazole and pyrimidine ring in its structure?

B1 (Thiamine) is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q45

past paper asked

Which vitamin is involved in the synthesis of collagen protein?

C is the best answer. Water-soluble vitamins commonly serve as coenzyme precursors and have characteristic deficiency syndromes.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q53

past paper asked

Deficiency of which one of these vitamins is the major cause of blindness?

Vitamin A is the best answer. Vitamin A is required for retinal function and epithelial integrity; deficiency can progress from night blindness to xerophthalmia.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q41

past paper asked

Which vitamin is a major lipid-soluble antioxidant in cell membrane?

Vitamin E is the best answer. Vitamin E is the principal lipid-phase antioxidant protecting membranes from peroxidation.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q17

past paper asked

All of the following vitamins provide the cofactor for pyruvate dehydrogenase, except...

Folate is the best answer. Pyruvate dehydrogenase requires TPP, lipoate, CoA, FAD, and NAD⁺; folate is not one of its cofactors.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q45

past paper asked

Which vitamin requires cobalt for its activity?

Vitamin B12 is the best answer. Cobalamin is vitamin B12; cobalt is the central metal in its corrin ring.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q11

past paper asked

Which is incorrect statement about vitamin C?

It can be synthesized in the body is the best answer. Humans lack gulonolactone oxidase and therefore cannot synthesize vitamin C.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q38

past paper asked

Deficiency of which one of these vitamins may lead to megaloblastic anemia?

Vitamin B12 is the best answer. Vitamin B12 or folate deficiency impairs DNA synthesis and causes megaloblastic anemia.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q57

past paper asked

In scurvy, which amino acid that is normally a part of collagen is not synthesized?

Hydroxyproline is the best answer. Vitamin C deficiency impairs hydroxylation of proline and lysine in collagen.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q48

past paper asked

Which of the following vitamins provides the cofactor for transamination of amino acids?

Pyridoxine is the best answer. Pyridoxal phosphate, derived from vitamin B6, is the cofactor for aminotransferases.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q6

past paper asked

Thiamine deficiency leads to_____?

Beri-beri is the best answer. Thiamine deficiency causes beriberi and may cause Wernicke-Korsakoff syndrome.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q2

Test yourself

1. Which vitamin supports collagen hydroxylation and enhances intestinal iron absorption?

- | | |
|----------------|--------------|
| A. Vitamin B12 | B. Vitamin C |
| C. Vitamin K | D. Vitamin A |

2. Which vitamin A form is the chromophore precursor required for vision?

- | | |
|---------------|-----------------------|
| A. Retinal | B. Retinoic acid only |
| C. Calcitriol | D. Tocopherol |

Answers: 1 B · 2 A

DNA structure, replication and repair

Lippincott 6e, Chapter 29; teacher slides cited under each idea.

DNA repair: do not mix organisms

Mechanism	What is repaired	Key distinction
Base-excision repair	Small base lesions, e.g. uracil in DNA	DNA glycosylase removes the base first; AP processing, resynthesis, ligation follow.
Nucleotide-excision repair	Bulky helix-distorting lesions	Human XP commonly involves XPA-XPG; bacterial UvrABC is not a human gene. XP variant: POLH.
Mismatch repair	Replication mismatches missed by proofreading	Recognizes the newly synthesized strand; human deficiency underlies Lynch syndrome.
Polymerase proofreading	Wrong newly incorporated nucleotide	3' → 5' exonuclease; DNA synthesis itself is 5' → 3'.
Homologous recombination / NHEJ	Double-strand breaks	HR uses homologous template; NHEJ directly rejoins ends and may introduce sequence changes.

Nucleosides, nucleotides, and the phosphodiester backbone

A nucleoside is a nitrogenous base joined to a pentose sugar, whereas a nucleotide adds one or more phosphates. Purines are adenine and guanine; pyrimidines are cytosine, thymine, and uracil. Within one nucleic-acid strand, the 3' hydroxyl of one sugar is joined to the 5' phosphate of the next nucleotide by a phosphodiester bond, giving the strand directionality.

1. Nucleoside = base + sugar; nucleotide = nucleoside + phosphate.
2. Ribose has a 2'-OH, whereas deoxyribose has 2'-H.
3. A and G are purines; C, T, and U are pyrimidines.
4. The backbone linkage is 3' → 5' phosphodiester, and sequences are written 5' → 3'.

→ **clinic:** Nucleoside and nucleotide analogs are used as antiviral and anticancer drugs because they can inhibit polymerases or terminate chains.

Teacher: Deoxyribonucleic Acid(DNA)Structure (2).pdf, pp. 6-14 and 25-28. · Lippincott Illustrated Reviews: Biochemistry, 6e, Chapter 29, DNA primary structure and polarity.

complementary antiparallel DNA, helix forms, and melting

Double-stranded DNA contains complementary strands running in opposite 5' → 3' directions. Adenine pairs with thymine through two hydrogen bonds and guanine pairs with cytosine through three, so Chargaff's relation is A = T and G = C. B-DNA is the usual right-handed physiologic form; heat, extreme pH, or low ionic strength denatures DNA, and greater GC content generally raises its melting temperature.

1. The strands are complementary and antiparallel.
2. A-T has two hydrogen bonds; G-C has three.

3. Base stacking also makes a major contribution to duplex stability.
4. B-DNA is physiologic and right-handed; Z-DNA is left-handed.
5. Denaturation is strand separation without backbone hydrolysis and can be reversed by controlled annealing.

→ **clinic:** Hybridization-based diagnostics and PCR primer design depend on complementarity and melting temperature.

Teacher: Deoxyribonucleic Acid(DNA)Structure (2).pdf, pp. 22-34. · Lippincott 6e, Chapter 29, DNA secondary structure, denaturation, and A/B/Z forms.

Nucleosomes, histone H1, and chromatin accessibility

A nucleosome core contains two copies each of H2A, H2B, H3, and H4 around which DNA is wrapped. Histone H1 binds linker DNA and promotes higher-order compaction but is not part of the core octamer. Acetylation of lysine residues reduces histone positive charge, loosens DNA association, and generally makes chromatin more transcriptionally accessible.

1. Core octamer = 2 × H2A, H2B, H3, and H4.
2. Histones bind negatively charged DNA because they are rich in lysine and arginine.
3. H1 is the linker histone.
4. Open euchromatin is generally more transcriptionally active than compact heterochromatin.
5. Histone-tail modifications can recruit proteins that activate or repress genes.

→ **clinic:** Histone deacetylase inhibitors are used in selected malignancies because chromatin state can alter gene expression.

Teacher: Deoxyribonucleic Acid(DNA)Structure (2).pdf, pp. 40-43. · Lippincott 6e, Chapter 29, chromatin packaging and nucleosome organization.

Semiconservative, bidirectional replication from origins

DNA replication is semiconservative because each daughter duplex retains one parental strand and contains one newly synthesized strand. Replication starts at defined origins and usually proceeds bidirectionally through replication forks. Eukaryotic chromosomes use many licensed origins so their large linear genomes can be copied completely during S phase.

1. Each parental strand is a template for a complementary daughter strand.
2. An origin opens into a replication bubble with two forks.
3. Bacterial chromosomes usually use one main origin; eukaryotic chromosomes use many.
4. Origin firing is restricted so each segment is copied once per cell cycle.

→ **clinic:** Replication stress and defective origin control contribute to genomic instability in cancer.

Teacher: Deoxyribonucleic Acid(DNA Replication (1)).pptx, slides 3-11 and 53-54. · Lippincott 6e, Chapter 29, semiconservative and bidirectional replication.

Helicase, single-strand binding proteins, and topoisomerases

Helicase separates parental strands, while single-strand DNA-binding proteins keep the templates apart and protect them. Unwinding produces positive supercoiling ahead of the fork, which topoisomerases relieve by transiently breaking and resealing DNA. Type I topoisomerases cut one strand without ATP, whereas type II enzymes cut both strands and require ATP.

1. DnaB is the principal bacterial replicative helicase.
2. SSB prevents reannealing and protects exposed single-stranded DNA.
3. Topoisomerase I changes linking number in steps of one.
4. Topoisomerase II changes linking number in steps of two and includes bacterial DNA gyrase.

→ clinic: Fluoroquinolones inhibit bacterial gyrase/topoisomerase IV; several anticancer drugs target human

topoisomerases.

Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, slides 15 onward in the fork-machinery block. · Lippincott 6e, Chapter 29, prepriming complex and topoisomerases.

5' → 3' synthesis, primers, and leading versus lagging strands

DNA polymerases add nucleotides only to a free 3'-OH, so every new strand grows 5' → 3' and requires a primer. The leading strand is synthesized continuously toward the fork, whereas the lagging strand is synthesized away from the fork as Okazaki fragments. Primase makes the RNA primers, and the repeated primers on the lagging strand must later be removed and the fragments joined.

1. Polymerase reads template 3' → 5' while synthesizing 5' → 3'.
2. A primer supplies the initial paired 3'-OH.
3. Leading synthesis is continuous; lagging synthesis is discontinuous.
4. Okazaki fragments arise from antiparallel template geometry.

Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, strand-synthesis and Okazaki-fragment slides. · Lippincott 6e, Chapter 29, DNA synthesis directionality and primer requirement.

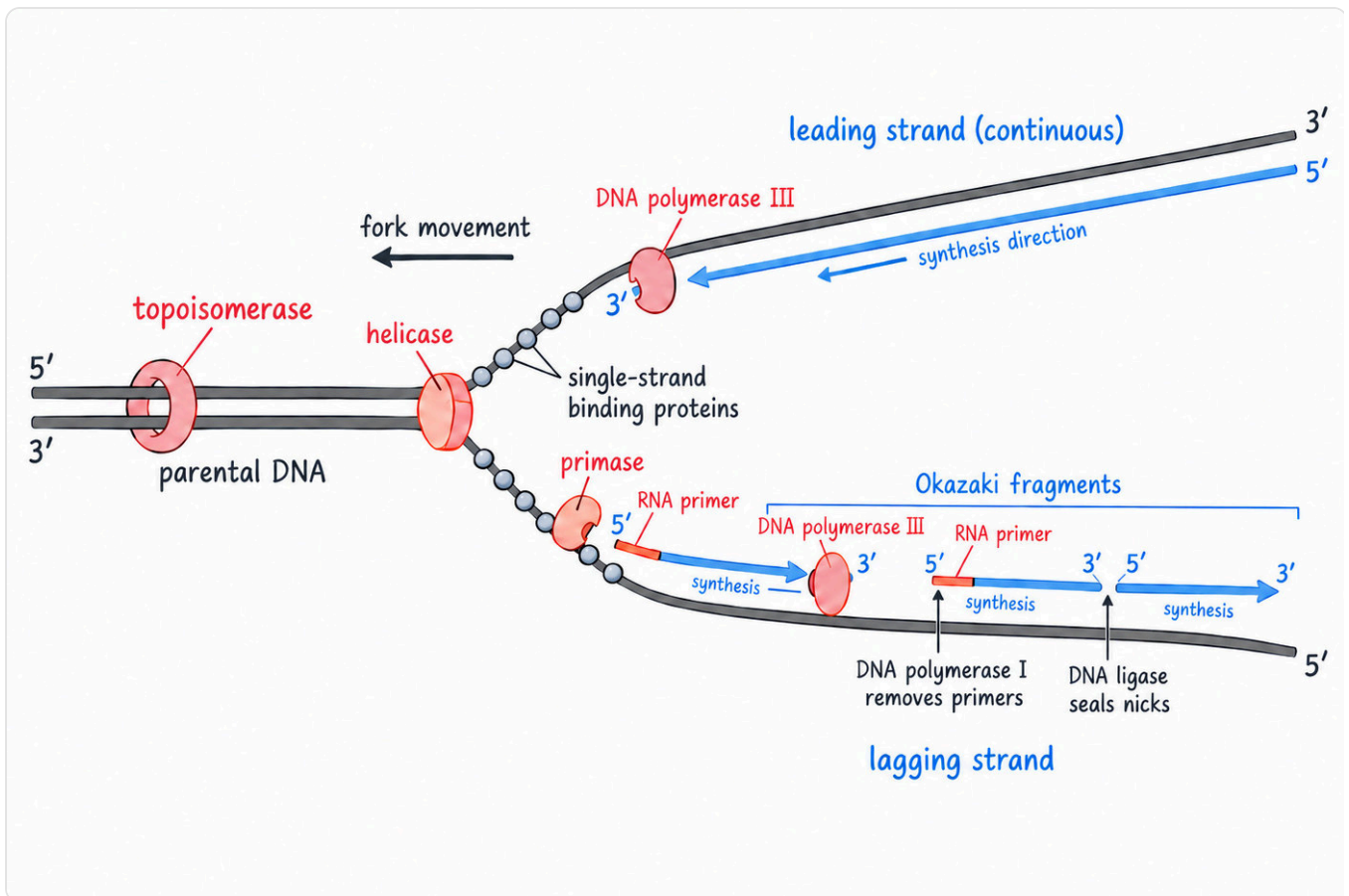


Fig. 16 A bacterial replication fork showing continuous leading-strand synthesis and discontinuous lagging-strand synthesis, both in the 5' to 3' direction.

Polymerase specialization, proofreading, primer removal, and

Ligation

Replicative polymerases extend DNA processively, and their 3'→5' exonuclease activity removes a newly misincorporated nucleotide for proofreading. In bacteria, DNA polymerase I uses 5'→3' exonuclease activity to remove RNA primers and fills the resulting gaps with DNA. DNA ligase then seals the remaining nick by creating the final phosphodiester bond.

1. Bacterial DNA polymerase III performs most fork elongation.
2. Proofreading exonuclease is 3'→5'.
3. Bacterial DNA polymerase I removes primers with 5'→3' exonuclease.
4. DNA ligase seals, but does not fill, a nick.
5. PCNA is the eukaryotic sliding clamp.

→ **clinic:** Polymerase proofreading defects produce mutator phenotypes and increase cancer susceptibility.

Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, polymerase, exonuclease, and ligase block. · Lippincott 6e, Chapter 29, polymerases I/III, proofreading, and ligation.

Eukaryotic polymerases and the telomere end-replication problem

In eukaryotes, polymerase α works with primase to initiate synthesis, while high-processivity polymerases extend the leading and lagging strands with PCNA as their clamp. Linear chromosome ends cannot be completely copied by ordinary lagging-strand synthesis, causing telomere shortening in most somatic cells. Telomerase solves this problem using an intrinsic RNA template and a reverse-transcriptase protein subunit to extend the 3' chromosome end.

1. Polymerase α /primase initiates; polymerases ϵ and δ perform most nuclear elongation.
2. Polymerase γ replicates mitochondrial DNA.
3. Telomerase contains RNA plus telomerase reverse transcriptase.
4. Telomerase is active in germ/stem cells and commonly reactivated in cancer.
5. Progressive telomere shortening contributes to replicative senescence.

→ **clinic:** Telomerase reactivation supports unlimited proliferation in many cancers.

→ **clinic:** Telomere-maintenance defects cause bone-marrow failure and other telomere biology disorders.

Teacher: Deoxyribonucleic Acid(DNA Replication (1).pptx, eukaryotic polymerase and telomere sections. · Lippincott 6e, Chapter 29, telomere synthesis and reverse transcriptase.

Mismatch, nucleotide-excision, and base-excision repair

Mismatch repair removes replication errors that escape proofreading. Nucleotide-excision repair removes a short oligonucleotide containing a bulky, helix-distorting lesion such as a UV-induced pyrimidine dimer. Base-excision repair begins with a lesion-specific glycosylase that removes a damaged base, followed by AP-site incision, gap filling, and ligation.

1. Mismatch repair corrects base mismatches and small insertion/deletion loops.
2. NER handles bulky, helix-distorting lesions.
3. BER handles small, nonbulky base damage such as deamination.
4. All three pathways use an intact complementary strand as sequence information.

→ **clinic:** Mismatch-repair defects cause Lynch syndrome and microsatellite instability.

→ **clinic:** NER defects cause xeroderma pigmentosum with extreme UV sensitivity and early skin cancers.

Lippincott 6e, Chapter 29, DNA repair section; mismatch, nucleotide-excision, and base-excision repair. · The teacher replication deck is titled DNA replication and the textbook supplies the repair mechanisms needed to complete that chapter outcome.

Homologous recombination versus nonhomologous end joining

Homologous recombination repairs a double-strand break using a homologous template, usually the sister chromatid, and is therefore relatively accurate. Nonhomologous end joining directly processes and ligates broken ends without requiring a homologous template, so sequence may be lost or altered. Failure or misuse of either pathway creates chromosome instability and can initiate malignancy.

1. HR is template directed and favored in S/G2 when a sister chromatid is available.
2. NHEJ is available more broadly but is intrinsically error prone.
3. BRCA1 and BRCA2 participate in homologous recombination.
4. End-joining errors can create deletions and translocations.

→ clinic: BRCA1/2 loss increases breast, ovarian, prostate, and pancreatic cancer risk.

→ clinic: HR-deficient tumors can be selectively vulnerable to PARP inhibition.

Lippincott 6e, Chapter 29, double-strand break repair by HR and NHEJ. · Lippincott Chapter 29 question extraction includes BRCA and clinical consequences of defective end joining.

past paper asked

Which of the following enzymes is required for end to end joining of DNA?

DNA ligase is the best answer. Replication requires directional synthesis, proofreading, and coordinated fork enzymes.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q21

past paper asked

Which statement about DNA structure and denaturation is true?

GC-rich DNA has greater thermal stability and a higher melting temperature than AT-rich DNA because GC base pairs provide three hydrogen bonds and stronger stacking interactions.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q57

past paper asked

Xeroderma pigmentosum results from a defect in?

Editorial repair: the source offered bacterial UvrABC as the human xeroderma pigmentosum defect. Human XP usually involves NER proteins XPA-XPB; the XP variant involves POLH/translesion synthesis. UvrABC is bacterial, not a human gene.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q72

past paper asked

Which one of the following nucleic acids has a left-handed helix?

Z-DNA is the best answer. Z-DNA is a left-handed helix.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q73

past paper asked

Okazaki fragments are formed during the synthesis of?

DNA is the best answer. Okazaki fragments are short DNA segments synthesized on the lagging strand.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q8

past paper asked

Ciprofloxacin inhibits.....

DNA gyrase is the best answer. Ciprofloxacin inhibits bacterial DNA gyrase/topoisomerase II, disrupting DNA topology and replication.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q26

past paper asked

Which of the following mechanisms will remove uracil and incorporate the correct base?

Base excision repair is the best answer. DNA repair pathways recognize a lesion, remove it, resynthesize DNA, and seal the strand.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q20

past paper asked

According to Chargaff's rule, in a DNA molecule the percentage of?

$T+G=A+C$ is the best answer. Because $A=T$ and $G=C$ in double-stranded DNA, $T+G$ equals $A+C$.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q23

past paper asked

The enzyme responsible for the removal of supercoiling in replicating DNA, ahead of the replication fork, is?

Topoisomerase is the best answer. Topoisomerase relieves torsional stress ahead of the replication fork.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q33

past paper asked

Which of the following reactions is required for proofreading, during DNA replication, by DNA polymerase III?

3' to 5' exonuclease activity is the best answer. Replication requires directional synthesis, proofreading, and coordinated fork enzymes.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q42

past paper asked

Which of the following leads to disruption of nucleosome structure?

Acetylation is the best answer. Histone acetylation weakens histone-DNA attraction and opens chromatin.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q46

past paper asked

Which is an example of RNA-dependent DNA polymerase?

Reverse transcriptase is the best answer. Reverse transcriptase synthesizes DNA from an RNA template.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q47

past paper asked

The proofreading function of DNA polymerase involves all of the following except

Reversal of the polymerization reaction is the best answer. Proofreading detects a mismatch and removes it with 3'→5' exonuclease activity; it is not reversal of polymerization.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q40

past paper asked

When double-stranded DNA is heated at neutral pH, which change does not occur?

The covalent N-glycosidic bond between the base and the pentose breaks is the best answer. Heat disrupts hydrogen bonding and base stacking, but does not cleave the covalent base-sugar N-glycosidic bond.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q4

past paper asked

Histones are _____ that are usually associated with _____.

Basic proteins - DNA is the best answer. Histones are basic, lysine/arginine-rich proteins that bind negatively charged DNA.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q33

past paper asked

The compound that consists of ribose linked by an N-glycosidic bond to N-9 of adenine is

Adenosine is the best answer. Adenine plus ribose is the nucleoside adenosine; a phosphate would make it a nucleotide.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q28

past paper asked

In base-excision repair, the first enzyme to act is

DNA glycosylase is the best answer. A DNA glycosylase first removes the damaged base, creating an abasic site.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q9

past paper asked

When a DNA molecule is described as replicating bidirectionally, it means that it has two ...

Replication forks is the best answer. Bidirectional replication produces two forks moving away from an origin.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q35

Test yourself

1. A nucleoside consists of which components?

A. A base plus sugar plus phosphate

C. A nitrogenous base plus a pentose sugar

B. A phosphate plus fatty acid

D. A sugar plus amino acid

2. How can histone-tail acetylation favor gene expression?

A. It increases positive charge and closes chromatin

C. It cleaves DNA into Okazaki fragments

B. It neutralizes positive charge and loosens histone-DNA interaction

D. It methylates every cytosine

3. What is nucleotide excision repair (NER) best known for fixing (in this chapter)?

A. Cells become senescent (stop dividing).

C. UV-induced pyrimidine (thymine) dimers.

B. Pol epsilon (with 3'→5' proofreading).

D. RNase H and flap endonuclease-1 (FEN1).

Answers: 1 C · 2 B · 3 C

RNA, transcription and processing

Lippincott 6e, Chapter 30; teacher slides cited under each idea.

RNA types and bacterial versus eukaryotic mRNA architecture

mRNA carries coding information, tRNA is the adaptor that matches codons to amino acids, and rRNA provides structural and catalytic functions in ribosomes. Bacterial mRNA is often polycistronic and can be translated while it is transcribed. Eukaryotic mRNA is usually monocistronic and is processed with a 5' cap, splicing, and a 3' poly(A) tail before export to the cytoplasm.

1. mRNA is the translated RNA; tRNA decodes; rRNA forms the catalytic ribosome.
2. RNA uses ribose and uracil and is usually single stranded.
3. Bacterial transcription and translation can be coupled.
4. Eukaryotic pre-mRNA is processed in the nucleus before translation.

Teacher: Transcription 2025.pptx, RNA-structure and RNA-type introduction. · Lippincott 6e, Chapter 30, major RNA types and eukaryotic mRNA architecture.

Primer-independent bacterial transcription initiation

RNA polymerase can begin RNA synthesis de novo and therefore does not require a primer. In bacteria, the sigma factor joins the core enzyme to form a holoenzyme that recognizes promoter consensus sequences. The -35 region supports recognition and the AT-rich -10 Pribnow box facilitates local DNA opening near the +1 start site.

1. RNA polymerase synthesizes RNA 5' → 3' without a primer.
2. Sigma confers promoter specificity but is not the elongating catalytic core.
3. Canonical bacterial promoter elements are near -35 and -10.
4. Promoter sequences are not themselves transcribed.

→ **clinic:** Rifampicin selectively inhibits bacterial RNA polymerase and is central to tuberculosis therapy.

Teacher: Transcription 2025.pptx, bacterial RNA polymerase and promoter-initiation block. · Lippincott 6e, Chapter 30, bacterial promoter recognition and sigma factor.

Transcript direction, coding/template strands, and bacterial termination

RNA polymerase reads the DNA template strand 3' → 5' and synthesizes RNA 5' → 3'. The RNA sequence is complementary to the template and matches the coding strand except that uracil replaces thymine. Intrinsic termination uses a GC-rich RNA hairpin followed by uridines, whereas rho-dependent termination uses an ATP-dependent helicase to release the transcript.

1. RNA grows 5' → 3' and the template is read 3' → 5'.
2. Coding-strand sequence corresponds to RNA with T replaced by U.
3. Sigma commonly dissociates after promoter clearance.
4. Intrinsic termination requires a hairpin plus a U-rich tract.
5. Rho binds RNA and catches a paused polymerase.

Division of labor among RNA polymerases I, II, and III

RNA polymerase I produces the 45S precursor that yields 18S, 5.8S, and 28S rRNAs. RNA polymerase II produces pre-mRNA and several regulatory RNAs. RNA polymerase III produces tRNAs and 5S rRNA, so the exceptional 5S product must not be assigned to polymerase I.

1. Pol I: 45S pre-rRNA → 18S, 5.8S, 28S.
2. Pol II: pre-mRNA and selected sn/miRNAs.
3. Pol III: tRNA and 5S rRNA.
4. Pol I acts in the nucleolus; Pol II and III act in the nucleoplasm.

→ **clinic:** α -Amanitin strongly inhibits RNA polymerase II and, at higher doses, polymerase III.

Teacher: Transcription 2025.pptx, eukaryotic polymerase section. · Lippincott 6e, Chapter 30, eukaryotic nuclear RNA polymerases.

RNA polymerase II promoters and general transcription factors

RNA polymerase II does not recognize its promoter alone; general transcription factors assemble a preinitiation complex. TFIID, through TBP, recognizes the TATA box when present, and other core promoter elements can substitute in TATA-less genes. TFIIH opens DNA with helicase activity and phosphorylates the polymerase II C-terminal domain to permit promoter escape.

1. TATA/Hogness box is a core promoter element near -25 in many genes.
2. TBP is part of TFIID.
3. General factors position Pol II and create an open complex.
4. TFIIH has helicase and kinase activities.

→ **clinic:** Defects in TFIIH-associated nucleotide-excision repair/transcription proteins can cause xeroderma pigmentosum or trichothiodystrophy.

Teacher: Transcription 2025.pptx, Pol II general-factor section. · Lippincott 6e, Chapter 30, TATA box, TFIID, and TFIIH.

5' capping and 3' polyadenylation of pre-mRNA

Eukaryotic pre-mRNA receives a 7-methylguanosine cap through an unusual 5'-5' triphosphate linkage early during transcription. The 3' end is cleaved downstream of the AAUAAA signal and poly(A) polymerase adds a tail that is not directly encoded by the DNA template. The cap and tail protect the transcript, support nuclear export, and cooperate in efficient translation initiation.

1. The cap is 7-methylguanosine linked 5'-5'.
2. The cap supports stability, export, and ribosome recruitment.
3. AAUAAA directs cleavage/polyadenylation.
4. Poly(A) polymerase adds the non-templated tail.
5. Histone mRNAs are an important poly(A)-tail exception.

→ **clinic:** Mutations in polyadenylation signals can reduce normal mRNA production, including in globin disorders.

Lippincott 6e, Chapter 30, pre-mRNA capping and polyadenylation. · Teacher: Transcription 2025.pptx includes eukaryotic transcription and RNA processing in its confirmed scope.

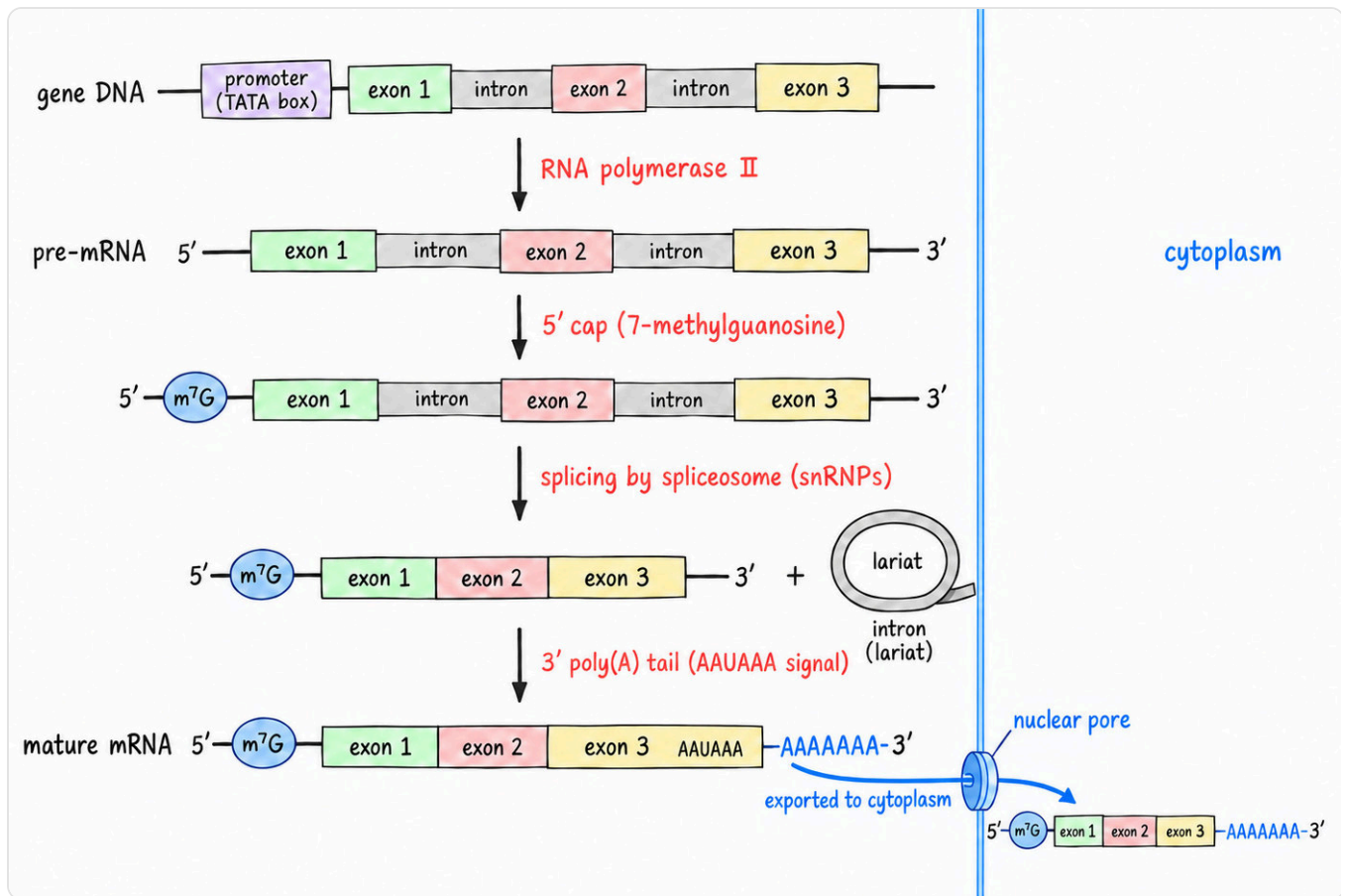


fig. 17 Eukaryotic pre-mRNA receives a 5' cap, undergoes intron splicing, and gains a 3' poly(A) tail before nuclear export.

Spliceosome mechanism, splice-site disease, and alternative splicing

The spliceosome uses snRNPs to recognize the 5' GU site, branch-point adenine, polypyrimidine tract, and 3' AG site. Two transesterification reactions form a lariat and join neighboring exons without consuming ATP for the bond-exchange chemistry itself. Alternative splice-site choice allows one gene to produce different protein isoforms, while pathogenic splice mutations can cause exon skipping, intron retention, or activation of cryptic sites.

1. Introns are removed and exons are ligated by the spliceosome.
2. The branch-point A forms a 2'-5' bond in the lariat.
3. Consensus boundaries are usually GU at the 5' end and AG at the 3' end.
4. Alternative splicing expands the proteome without adding genes.
5. Splice-site mutations can alter the reading frame or introduce premature stops.

→ **clinic:** β -thalassemia can result from abnormal β -globin pre-mRNA splicing.

→ **clinic:** Antisense oligonucleotides can therapeutically redirect splicing in selected diseases.

Lippincott 6e, Chapter 30, spliceosome, splice-site mutations, and alternative splicing. · Teacher: Transcription 2025.pptx includes eukaryotic RNA processing in the confirmed course block.

Transcription inhibitors as mechanism probes and drugs

Actinomycin D intercalates into DNA and broadly blocks RNA-chain elongation. Rifampicin binds the β subunit of bacterial RNA polymerase and prevents productive initiation. α -Amanitin from Amanita mushrooms most strongly inhibits eukaryotic RNA polymerase II, shutting down mRNA synthesis.

1. Actinomycin D targets DNA and inhibits transcription broadly.
2. Rifampicin selectively targets bacterial RNA polymerase.
3. α -Amanitin is most potent against Pol II.
4. Selectivity depends on whether the target is DNA itself or a domain-specific polymerase.

→ clinic: Rifampicin is a key antimycobacterial drug and a strong hepatic enzyme inducer.

→ clinic: Amanita poisoning can produce fulminant hepatic failure.

Teacher: Transcription 2025.pptx, transcription-inhibitor section. · Lippincott 6e, Chapter 30, rifampin, dactinomycin, and α -amanitin.

past paper asked

The initiation site for transcription is recognized by

Sigma factor is the best answer. Bacterial sigma factor gives RNA polymerase promoter specificity.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q48

past paper asked

Splice sites are present in

Eukaryotic hnRNA is the best answer. Splice sites are present in eukaryotic hnRNA/pre-mRNA before introns are removed.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q50

past paper asked

mRNA is complementary to the nucleotide sequence of

Template strand is the best answer. mRNA is complementary to the DNA template strand and matches the coding strand except U replaces T.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q78

past paper asked

The structure of tRNA appears like a

Clover leaf is the best answer. The classic two-dimensional structure of tRNA is a cloverleaf.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q82

past paper asked

Binding of the prokaryotic RNA polymerase to promoter site is inhibited by?

Rifampicin is the best answer. RNA polymerase and its factors copy a DNA template into RNA. Rifampicin is the intended drug. More precisely, it binds the bacterial RNA-polymerase beta subunit and blocks early RNA-chain extension, rather than simply preventing promoter recognition.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q3

past paper asked

Actinomycin D is an inhibitor of?

Transcription is the best answer. RNA polymerase and its factors copy a DNA template into RNA.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q15

past paper asked

What is the role of sigma factor in bacterial RNA polymerase?

Positioning RNA polymerase correctly on the DNA template is the best answer. Sigma positions bacterial RNA polymerase at the correct promoter.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q22

past paper asked

Which of the following is an incorrect statement about mRNA?

B and C are both incorrect. Replication-dependent histone mRNAs are capped but usually lack poly-A tails. Splicing removes introns and joins exons. Polyadenylation is typical, not universal.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q26

past paper asked

RNA polymerase from E. coli (core enzyme alone) has all of the following properties except that it

Recognizes specific start signals in DNA is the best answer. The bacterial core RNA polymerase can elongate RNA but needs sigma factor to recognize specific promoters.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q50

past paper asked

Which of the followings inhibit the DNA-dependent RNA polymerase in prokaryotes?

Rifampicin is the best answer. Rifampicin inhibits bacterial DNA-dependent RNA polymerase.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q42

Test yourself

1. Poly-A tail: list 3 functions.

- A. Stabilizes mRNA, facilitates nuclear export, aids translation; tail is gradually shortened in cytosol.
- C. 5' capping, 3' poly-A tail addition, intron removal (splicing) and exon joining.

- B. 3' poly-A tail and 5' cap of 7-methylguanosine linked by an unusual 5'→5' triphosphate.
- D. AAUAAA near the 3' end; pre-mRNA is cleaved downstream of it and poly-A is added to the new 3' end.

2. What converts the core E. coli RNA polymerase into a promoter-recognizing holoenzyme?

- A. Binding of DNA ligase
- C. Binding of a ribosomal subunit

- B. Addition of histone H1
- D. Binding of a sigma factor

3. Which RNA species serves as the adaptor between an mRNA codon and an amino acid?

- A. rRNA
- C. DNA

- B. snRNA
- D. tRNA

ANSWERS: 1 A · 2 D · 3 D

Translation

Lippincott 6e, Chapter 31; teacher slides cited under each idea.

Inhibitors: target and process

Agent	Target / action	Process
Rifampicin	Bacterial RNA-polymerase β subunit; early RNA extension	Transcription, not translation
Actinomycin D	DNA intercalation; blocks RNA synthesis	Transcription
Tetracyclines	Prevent aminoacyl-tRNA access to bacterial A site	Translation, 30S
Aminoglycosides	Initiation disruption / misreading	Translation, 30S
Chloramphenicol	Peptidyl-transferase inhibition	Translation, 50S
Puromycin	Aminoacyl-tRNA mimic, premature chain termination	Translation
Ricin	Depurinates 28S rRNA in the 60S subunit	Eukaryotic ribosome inactivation
Diphtheria toxin	ADP-ribosylates eEF2	Eukaryotic translocation
Cycloheximide	Inhibits eukaryotic elongation	Translation, 60S

Genetic code properties, reading frames, and mutation consequences

The genetic code is degenerate because several codons can specify the same amino acid, but it is unambiguous because each codon specifies only one amino acid. An open reading frame extends from an initiation codon through a stop codon in one nucleotide grouping. A nonsense mutation creates a premature stop, while an insertion or deletion not divisible by three shifts the downstream reading frame.

1. Degenerate does not mean ambiguous.
2. AUG usually initiates translation and also encodes methionine.
3. UAA, UAG, and UGA are stop codons.
4. Missense changes an amino acid; nonsense creates a stop; frameshift changes downstream codon grouping.
5. Silent substitutions can still affect splicing or translation efficiency.

→ **clinic:** Nonsense and frameshift variants often cause severe loss-of-function genetic disease.

Teacher: Translation 2025.pptx, slides 6 and 9-10. · Lippincott 6e, Chapter 31, genetic code and mutation effects.

Aminoacyl-tRNA synthetases, proofreading, and wobble

An aminoacyl-tRNA synthetase uses ATP to attach a specific amino acid to the 3' CCA end of its cognate tRNA. Many synthetases proofread by removing an incorrectly activated or attached amino acid before the tRNA reaches the ribosome. Flexible pairing between the third codon base and first anticodon base, called wobble, lets one tRNA decode several synonymous codons.

1. Each amino acid has a corresponding synthetase specificity system.
2. Charging consumes ATP to AMP, equivalent to two high-energy phosphate bonds.
3. The ribosome checks codon-anticodon pairing but cannot verify the attached amino acid.
4. Wobble explains why fewer tRNAs than sense codons are required.

→ **clinic:** Some antimicrobial drugs exploit differences in microbial aminoacyl-tRNA synthetases.

Teacher: Translation 2025.pptx, slides 17–25. · Lippincott 6e, Chapter 31, tRNA charging and wobble.

Ribosomal A/P/E sites, peptidyl transferase, and directionality

The ribosome reads mRNA 5' → 3' and synthesizes protein from the amino terminus toward the carboxyl terminus.

Incoming aminoacyl-tRNA enters the A site, peptidyl-tRNA occupies the P site, and deacylated tRNA exits through the E site. The large-subunit rRNA catalyzes peptide-bond formation, demonstrating that the ribosome is a ribozyme.

1. A = aminoacyl entry, P = peptidyl chain, E = exit.
2. Protein elongates N → C.
3. Peptide-bond energy comes from the high-energy aminoacyl-tRNA ester bond.
4. Bacterial 23S rRNA is the peptidyl-transferase catalyst.

→ **clinic:** Many antibiotics selectively bind bacterial ribosomal sites or the peptidyl-transferase center.

Teacher: Translation 2025.pptx, slides 26, 28, and 41–42. · Lippincott 6e, Chapter 31, ribosome structure and directionality.

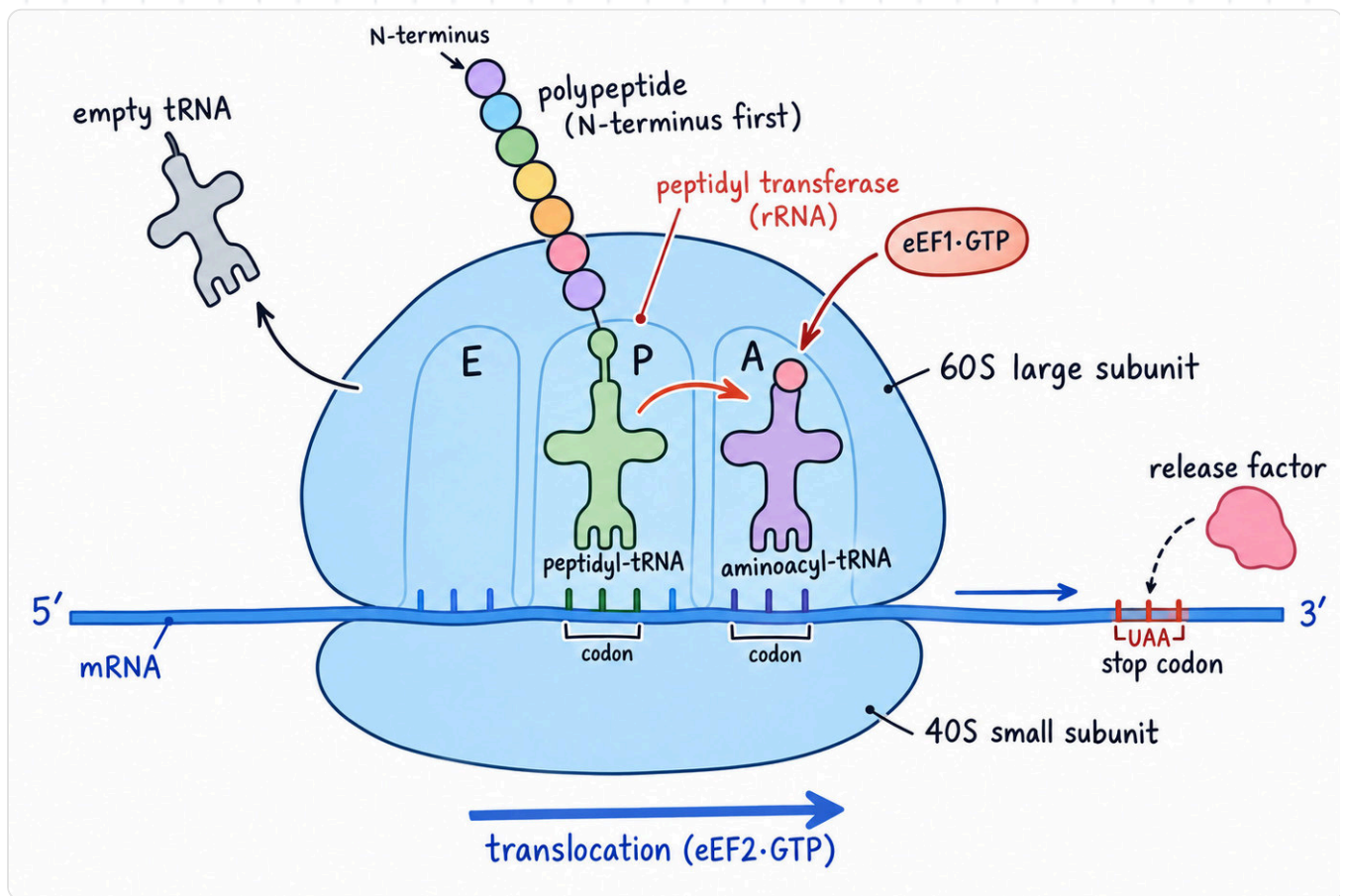


fig. 18 Eukaryotic translation uses the 40S and 60S ribosomal subunits, with tRNAs moving through A, P, and E sites as the ribosome reads mRNA 5' to 3'.

Prokaryotic Shine–Dalgarno alignment versus eukaryotic cap scanning

In bacteria, the Shine–Dalgarno sequence base-pairs with 16S rRNA to position the start codon in the P site. In eukaryotes, the small subunit is recruited near the 5' cap and scans to an AUG in a favorable Kozak context. The initiator tRNA is special because it enters the P site directly; bacteria use formylmethionine whereas cytosolic eukaryotic translation begins with methionine.

1. Shine–Dalgarno is bacterial; Kozak/cap scanning is eukaryotic.
2. Initiator tRNA starts in the P site, not the A site.
3. Bacterial initiation uses fMet–tRNA.
4. Initiation factors and GTP assemble the complete ribosome only after correct start recognition.

Teacher: Translation 2025.pptx, slides 27, 29–30, and 37–38. · Lippincott 6e, Chapter 31, prokaryotic and eukaryotic initiation.

Elongation–factor cycle and translocation

During elongation, EF-Tu–GTP delivers an aminoacyl–tRNA to the bacterial A site and GTP hydrolysis helps enforce correct codon recognition. Peptidyl transfer moves the growing chain from the P-site tRNA to the A-site amino acid. EF-G–GTP then drives translocation by one codon, shifting peptidyl–tRNA to P and deacylated tRNA toward E.

1. EF-Tu delivers charged tRNA; EF-Ts reloads EF-Tu with GTP.
2. Peptidyl transfer precedes translocation.
3. EF-G is the bacterial translocase.
4. Eukaryotic homologs include eEF1A and eEF2.
5. At least two GTP molecules are hydrolyzed per elongation cycle.

→ **clinic:** Diphtheria toxin and Pseudomonas exotoxin A inactivate eEF2 and stop translocation.

Teacher: Translation 2025.pptx, slides 39–43. · Lippincott 6e, Chapter 31, elongation cycle and energy use.

Stop-codon termination, ribosome recycling, and polysomes

No tRNA recognizes a stop codon; release factors instead enter the A site and stimulate hydrolysis of the completed peptide from the P-site tRNA. GTP-dependent factors then promote release-factor departure and ribosome recycling. Multiple ribosomes can translate one mRNA simultaneously as a polysome, increasing protein output without increasing mRNA number.

1. Stop codons are recognized by release factors, not suppressor tRNAs under normal conditions.
2. Termination hydrolyzes the peptidyl–tRNA ester bond.
3. Ribosomal subunits, mRNA, and tRNA are recycled.
4. A polysome is many ribosomes translating one mRNA.

→ **clinic:** Premature termination codons can trigger nonsense-mediated mRNA decay.

Lippincott 6e, Chapter 31, termination, recycling, and polysomes. · Teacher: Translation 2025.pptx covers the complete translation cycle.

Signal sequences direct proteins to ER, nucleus, and mitochondria

Secreted and many membrane proteins carry an N-terminal signal peptide recognized by signal-recognition particle, which pauses translation and directs the ribosome to the rough ER. Nuclear proteins use nuclear-localization signals and are generally imported after synthesis. Many mitochondrial matrix proteins carry an N-terminal amphipathic targeting sequence and are imported posttranslationally.

1. ER targeting is usually cotranslational through SRP and its receptor.
2. Free and ER-bound ribosomes are structurally identical; the signal sequence determines destination.
3. Nuclear localization signals are not usually cleaved.
4. Mitochondrial targeting sequences commonly form amphipathic helices.

→ **clinic:** Defective peroxisomal targeting causes Zellweger-spectrum disorders.

→ **clinic:** Mistargeting or failed trafficking can cause protein accumulation and organelle dysfunction.

Lippincott 6e, Chapter 31, protein targeting to ER, nucleus, and mitochondria. · Teacher translation deck establishes translation machinery; the book supplies the medically essential destination step.

Posttranslational modification, folding, and ubiquitin-proteasome quality control

Proteins can be cleaved, phosphorylated, glycosylated, hydroxylated, carboxylated, or lipidated after or during translation. N-linked glycosylation begins on asparagine in the ER, whereas O-linked sugars are added mainly to serine or threonine in the Golgi. Persistently misfolded cytosolic proteins are polyubiquitinated and degraded by the proteasome.

1. Modification can activate a precursor, target a protein, or change stability.
2. N-linked glycosylation is on Asn; O-linked is commonly on Ser/Thr.
3. Collagen proline/lysine hydroxylation requires vitamin C.
4. Vitamin K supports γ -carboxylation of glutamate in clotting proteins.
5. Polyubiquitin commonly marks proteins for proteasomal degradation.

→ **clinic:** $\Delta F508$ CFTR is often degraded by quality control before reaching the plasma membrane.

→ **clinic:** Vitamin C or K deficiency disrupts specific posttranslational modifications.

Lippincott 6e, Chapter 31, posttranslational modifications and protein degradation. · Lippincott Chapter 31 clinical examples include collagen, clotting proteins, and CFTR quality control.

Antibiotics and toxins mapped to translation steps

Tetracyclines block aminoacyl-tRNA entry into the bacterial A site, whereas macrolides inhibit translocation through the 50S subunit. Puromycin mimics aminoacyl-tRNA and causes premature chain release in both domains. Diphtheria toxin ADP-ribosylates eEF2, and ricin depurinates 28S rRNA, so both arrest eukaryotic protein synthesis by distinct mechanisms.

1. Tetracycline: blocks A-site entry on bacterial ribosomes.
2. Macrolides: 50S binding and impaired translocation.
3. Chloramphenicol: inhibits bacterial/mitochondrial peptidyl transferase.
4. Puromycin: aminoacyl-tRNA analog causing premature termination.
5. Diphtheria toxin targets eEF2; ricin targets 28S rRNA.

→ **clinic:** Mitochondrial ribosomes partly resemble bacterial ribosomes, explaining selected antimicrobial toxicities.

past paper asked

Tetracycline prevents synthesis of polypeptide by:

Preventing binding of aminoacyl tRNA is the best answer. Tetracycline binds the bacterial 30S subunit and blocks aminoacyl-tRNA entry into the A site.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q7

past paper asked

Which of the following inactivates the 60S subunit of eukaryotic ribosomes?

Ricin is the best answer. Ricin depurinates 28S rRNA in the eukaryotic 60S subunit and inactivates it. Ricin irreversibly depurinates 28S rRNA in the 60S subunit. Cycloheximide also inhibits eukaryotic elongation at the large subunit; the intended distinction is ricin-mediated ribosome damage.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q4

past paper asked

Which of the following mediates carrying genetic information from DNA to protein synthesis?

m-RNA is the best answer. mRNA carries coding information from DNA to ribosomes.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q5

past paper asked

What is the main function of t-RNA?

Identifies amino acids and transports them to ribosomes is the best answer. Ribosomes read mRNA through tRNA-mediated amino-acid delivery.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q19

past paper asked

In bacteria the elongation stage of protein synthesis does not involve

IF-2 is the best answer. IF-2 is an initiation factor; elongation uses EF-Tu, GTP, and peptidyl-transferase activity.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q16

past paper asked

Regarding protein synthesis in eukaryotes, all proteins are initially synthesized with?

Methionine at their N-terminus is the best answer. Eukaryotic translation initiates with methionine at the N-terminus.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q3

Past paper asked

Which one of the following antibiotics does not function by interfering with the translational process?

Penicillin is the best answer. Penicillin inhibits bacterial cell-wall cross-linking, not translation.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q37

Test yourself

1. Which antibiotic blocks bacterial aminoacyl-tRNA entry into the A site?

- | | |
|-------------------|------------------|
| A. Rifampicin | B. Actinomycin D |
| C. Alpha-amanitin | D. Tetracycline |

2. Termination trigger: what event starts termination?

- | | |
|---|--|
| A. A base change that creates a stop codon, causing premature termination, e.g., UCA → UAA. | B. An internal short basic nuclear localization signal. |
| C. A stop codon (UAA/UAG/UGA) enters the A site. | D. A codon is a 3-base "word" in mRNA; written 5' → 3' using A, G, C, U. |

3. In which direction does a ribosome move along mRNA, and in which direction does the polypeptide grow?

- | | |
|-------------------------------|---|
| A. 3-prime to 5-prime; C to N | B. 5-prime to 3-prime; C to N |
| C. No fixed directions | D. 5-prime to 3-prime on mRNA; N-terminus to C-terminus |

Answers: 1 D · 2 C · 3 D

Gene regulation

Lippincott 6e, Chapter 32; teacher slides cited under each idea.

constitutive versus regulated genes, cis elements, trans factors, and operons

Constitutive genes are expressed relatively continuously for basal functions, whereas regulated genes change expression with conditions, cell type, or signals. A cis-acting element is a DNA sequence that affects linked genes on the same DNA molecule; a trans-acting factor is a diffusible product, usually a protein, that binds a target sequence. A bacterial operon places multiple structural genes under shared regulatory DNA so they can be transcribed together as a polycistronic mRNA.

1. Cis = local DNA sequence; trans = diffusible regulator.
2. Promoters, operators, enhancers, and response elements are cis elements.
3. Repressors, activators, and many transcription factors act in trans.
4. Operons coordinate related bacterial genes efficiently.

→ clinic: Pathogenic variants can affect either a coding trans factor or a noncoding cis-regulatory element.

Teacher: Regulation of Gene Expression 2025.pptx, constitutive/controllable genes, cis/trans factors, and operons. · Lippincott 6e, Chapter 32, universal transcriptional-regulation logic.

Lac operon integrates lactose availability with glucose status

Without lactose, active lac repressor binds the operator and blocks transcription. Lactose-derived allolactose binds the repressor and lowers its operator affinity, providing negative-control release. When glucose is low, cAMP rises and cAMP-CAP activates RNA-polymerase recruitment, so maximal expression requires lactose present and glucose absent.

1. Allolactose is the inducer that inactivates lac repressor.
2. Low glucose → high cAMP → active CAP.
3. Lactose present removes negative control; low glucose provides positive control.
4. Maximal lacZYA transcription: lactose present, glucose absent.
5. Glucose-mediated suppression is catabolite repression.

Teacher: Regulation of Gene Expression 2025.pptx, lac-operon block. · Lippincott 6e, Chapter 32, lac negative and positive regulation.

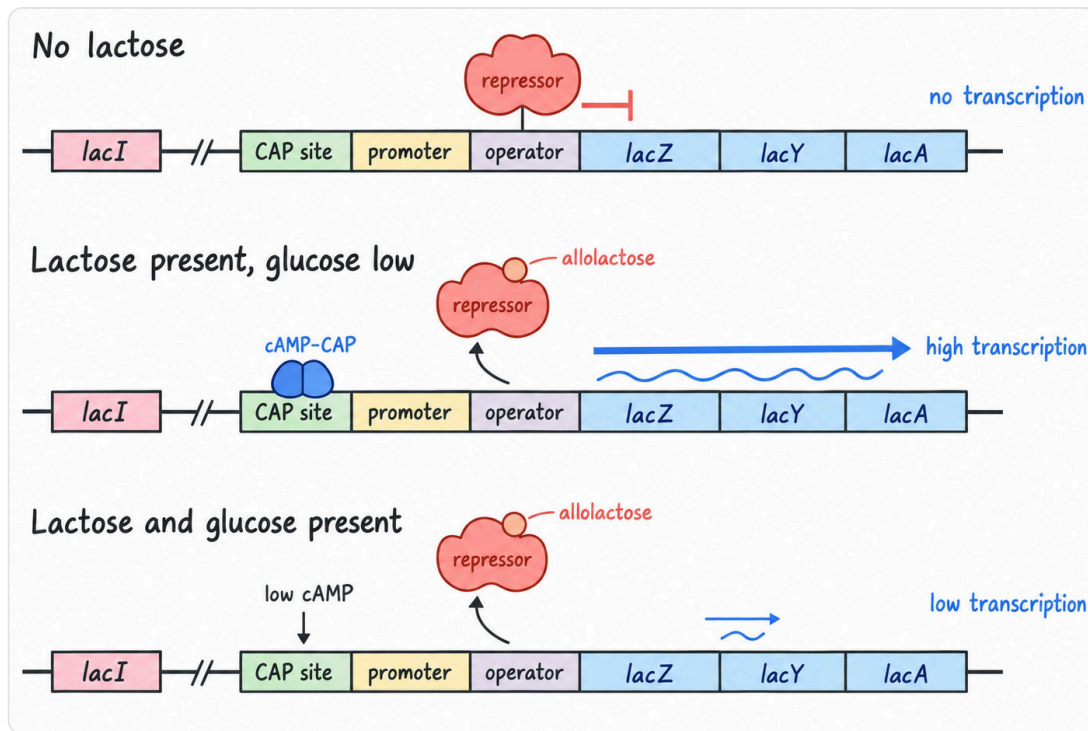


Fig. 19 The lac operon is repressed without lactose, strongly expressed when lactose is present and glucose is low, and weakly expressed when both sugars are present.

Trp operon repression and attenuation sense amino-acid supply

The trp operon is normally on when tryptophan is scarce and is repressed when tryptophan acts as a corepressor.

Attenuation adds a second layer by coupling translation of a leader peptide to formation of alternative mRNA hairpins.

Abundant charged tRNA-Trp allows the ribosome to advance and favors a terminator, whereas scarcity stalls the ribosome and favors an antiterminator so biosynthetic genes are transcribed.

1. Lac is inducible and catabolic; trp is repressible and anabolic.
2. Tryptophan activates its repressor as a corepressor.
3. Attenuation depends on coupled transcription and translation.
4. High Trp favors termination; low Trp permits transcription.

Teacher: Regulation of Gene Expression 2025.pptx, trp-operon section. · Lippincott 6e, Chapter 32, trp attenuation mechanism.

Enhancers, specific transcription factors, and combinatorial control

Specific transcription factors combine a DNA-binding domain with an activation or repression domain. Enhancers can act upstream, downstream, or at long distance and communicate with promoters through DNA looping and coactivator complexes. Different combinations of factors generate cell-specific expression, and genes on different chromosomes can be coordinated if they share response elements recognized by the same trans-acting factor.

1. Enhancers are cis DNA elements; activators are trans proteins.
2. Enhancer action is often distance and orientation independent.
3. Combinatorial control lets a limited factor set specify many expression patterns.
4. Shared consensus elements coordinate unlinked genes.

→ **clinic:** Enhancer mutations and enhancer hijacking can activate oncogenes without altering their coding sequence.

Nuclear receptors and cAMP–CREB connect hormones to transcription

Lipid-soluble hormones can enter cells, bind intracellular nuclear receptors, and direct receptor binding to hormone-response elements such as GREs. Peptide hormones instead signal through cell-surface receptors; glucagon raises cAMP, activates PKA, and phosphorylates CREB. Activated CREB binds cAMP-response elements and recruits transcriptional coactivators to change target-gene expression.

1. Nuclear receptors are ligand-activated transcription factors.
2. HRE is cis acting; hormone-receptor complex is trans acting.
3. Glucagon → GPCR → cAMP → PKA → CREB → CRE.
4. Coactivator versus corepressor recruitment determines transcriptional direction.

→ clinic: Glucocorticoid and thyroid-hormone therapies alter transcription through nuclear receptors.

→ clinic: Defects in receptor or response-element signaling can cause hormone resistance.

Lippincott 6e, Chapter 32, hormone-response elements, nuclear receptors, and CREB. · Teacher regulation deck confirms multi-level eukaryotic gene regulation as exam scope.

Histone modifications, chromatin remodeling, and DNA methylation

Gene expression can change heritably without changing DNA sequence through epigenetic mechanisms. Histone acetylation and ATP-dependent chromatin remodeling generally increase access to regulatory DNA, whereas promoter CpG methylation is usually associated with reduced transcription. The pattern is maintained through cell division yet remains more reversible than a DNA mutation.

1. Epigenetic change alters expression without changing base sequence.
2. Histone acetylation commonly opens chromatin.
3. DNA methyltransferases use S-adenosylmethionine as methyl donor.
4. Promoter CpG-island methylation usually represses transcription.
5. Chromatin remodeling repositions or ejects nucleosomes.

→ clinic: Tumor-suppressor promoters can be silenced by aberrant hypermethylation.

→ clinic: DNA-methyltransferase and histone-deacetylase inhibitors are used in selected cancers.

Teacher: Regulation of Gene Expression 2025.pptx, DNA-methylation and eukaryotic-regulation content. · Lippincott 6e, Chapter 32, DNA-level and chromatin-level regulation.

Alternative processing, RNA editing, and mRNA stability

Cells regulate gene output after transcription by selecting splice and polyadenylation sites, editing RNA sequence, controlling nuclear export, and changing mRNA half-life. ApoB editing converts a cytidine to uridine and creates a stop codon, producing shorter ApoB-48 in intestine versus full-length ApoB-100 in liver. Stability elements and RNA-binding proteins determine how long a transcript remains available for translation.

1. One pre-mRNA can yield different isoforms through alternative processing.

2. RNA editing changes transcript information without changing genomic DNA.
3. ApoB-48 is intestinal; ApoB-100 is hepatic.
4. Longer mRNA half-life generally permits more protein synthesis.
5. Iron-response elements provide a classic stability/translation control system.

→ **clinic:** ApoB-100 is required for LDL-receptor recognition, whereas ApoB-48 packages dietary lipid in chylomicrons.

Teacher: Regulation of Gene Expression 2025.pptx, posttranscriptional regulation and ApoB editing. · Lippincott 6e, Chapter 32, mRNA processing and stability.

RNA interference and regulation of translation initiation

MicroRNAs are processed into short guide strands loaded into RISC, where base pairing directs translational repression or mRNA cleavage depending on complementarity. Small interfering RNAs use the same core machinery and usually pair extensively with a target for cleavage. Cells also reduce global translation during stress by phosphorylating eIF2, which limits regeneration of the initiation ternary complex.

1. Dicer helps generate short RNA duplexes; Argonaute is the RISC effector.
2. Partial pairing usually represses translation; extensive pairing favors cleavage.
3. siRNA can be experimentally or therapeutically supplied.
4. Phosphorylated eIF2 inhibits translation initiation during stress.

→ **clinic:** Approved siRNA drugs silence pathogenic hepatic transcripts.

→ **clinic:** The integrated stress response uses eIF2 phosphorylation to rebalance protein synthesis.

Lippincott 6e, Chapter 32, RNA interference and eIF2-mediated translational control. · Teacher regulation deck confirms posttranscriptional and translational regulation in the course scope.

Gene rearrangement, amplification, and transposable elements

V(D)J recombination rearranges immunoglobulin gene segments to create enormous antibody diversity from a limited inherited segment set. Gene amplification increases copy number and can raise protein abundance, as in DHFR amplification during methotrexate resistance. Transposons and retrotransposons move through DNA- or RNA-based mechanisms and can disrupt genes or spread antimicrobial-resistance determinants.

1. Somatic V(D)J recombination permanently rearranges lymphocyte DNA.
2. Copy-number increase can drive overexpression and drug resistance.
3. Transposons use transposase; retrotransposons use an RNA intermediate.
4. Mobile elements can cause insertional mutation or move resistance genes.

→ **clinic:** Defective V(D)J machinery causes severe immunodeficiency.

→ **clinic:** Oncogene amplification and mobile resistance cassettes are clinically important forms of genome change.

Teacher: Regulation of Gene Expression 2025.pptx, immunoglobulin rearrangement in the eukaryotic-regulation section. · Lippincott 6e, Chapter 32, gene amplification and transposons.

past paper asked

The gene of lac operon which has constitutive expression is.....

i is the best answer. lacI is expressed constitutively to supply the repressor protein.

Cell & Molecules · September 2021 · Biochemistry: Biochemistry · original Q39

past paper asked

Which of these operons is anabolic?

The trp operon encodes enzymes for tryptophan biosynthesis, so it controls an anabolic pathway and is repressible by its end product. Both tryptophan and phenylalanine biosynthetic operons are anabolic. The original intended trp answer is not uniquely correct when phe is also offered.

Cell & Molecules · April 2021 report · Biochemistry; Biochemistry · original Q44

past paper asked

Where does a repressor bind to an operon?

Operator is the best answer. Operators, repressors, activators, and feedback control gene expression.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q9

past paper asked

Lactose utilization by E. coli requires which lac gene products?

LacY permease imports lactose and LacZ beta-galactosidase hydrolyzes it. Both A and B are valid for the unrestricted original question.

Cell & Molecules · February 2021 · Biochemistry; Biochemistry · original Q12

past paper asked

The operator region normally can be bound by

Repressor is the best answer. A repressor binds the operator and alters transcription of the adjacent structural genes.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q31

past paper asked

The tryptophan operon of E. coli is repressed by tryptophan added to the growth medium. The tryptophan repressor probably

Binds to the trp operator in the presence of tryptophan is the best answer. Tryptophan acts as a corepressor, enabling the repressor to bind the trp operator.

Cell & Molecules · September 2021 · Biochemistry; Biochemistry · original Q29

Test yourself

1. Which mechanism is an important posttranscriptional control of eukaryotic gene expression?

- A. Alternative mRNA splicing
- B. Changing the bacterial -10 promoter
- C. Tus-Ter binding
- D. Okazaki-fragment ligation

2. Define RNA interference (RNAi) in one sentence.

- A. Gene expression is the multistep process that produces a functional gene product: either RNA or protein.
- B. Gene silencing via decreased mRNA expression by repressing translation or increasing mRNA degradation.
- C. RNA-induced silencing complex; guide strand base-pairs with complementary target mRNA to direct silencing.
- D. RNA polymerase proceeds and transcribes the structural genes → mRNA produced → proteins made.

3. Why does low glucose favor strong lac-operon expression when lactose is present?

- A. The lac repressor binds operator irreversibly
- B. cAMP rises and the CAP-cAMP complex activates transcription
- C. cAMP destroys RNA polymerase
- D. Glucose converts lactose to tryptophan

Answers: 1 A · 2 B · 3 B

Biotechnology

Lippincott 6e, Chapter 33; teacher slides cited under each idea.

Restriction enzymes, sticky ends, and plasmid cloning vectors

Type II restriction endonucleases recognize short, often palindromic DNA sequences and cut reproducibly. Staggered cuts can create complementary sticky ends that align insert and vector before DNA ligase seals the backbone. A basic plasmid vector needs an origin of replication, a selectable marker, and a cloning site for the inserted DNA.

1. Restriction enzymes cut sequence-specific sites.
2. Compatible sticky ends improve directional assembly efficiency.
3. DNA ligase covalently seals vector and insert.
4. Origin, selectable marker, and cloning site are core vector features.

→ **clinic:** Recombinant plasmids enable production of therapeutic proteins such as human insulin.

Lippincott Illustrated Reviews: Biochemistry, 6e, Chapter 33, PDF pp. 838–877. · Tracked bank source labels Chapter 33 as confirmed August 25 Cell & Molecules scope.

cDNA and complementary nucleic-acid probes

Reverse transcriptase copies mature mRNA into complementary DNA, so cDNA represents expressed, already-spliced sequence and lacks genomic introns. That property allows a human coding sequence to be expressed in bacteria, which cannot remove most eukaryotic introns. A labeled probe detects a target by complementary base pairing under controlled stringency.

1. cDNA reflects the transcripts present in the sampled cell.
2. Mature-mRNA-derived cDNA lacks introns.
3. Reverse transcriptase makes the initial DNA copy.
4. Probe specificity depends on complementarity, temperature, salt, and mismatch tolerance.

→ **clinic:** cDNA and probes are used to measure gene expression and detect infectious or inherited sequences.

Lippincott 6e, Chapter 33, cDNA synthesis and nucleic-acid hybridization. · Confirmed August 25 Chapter 33 question bank, PDF pp. 838–877.

PCR amplification and reverse-transcription PCR

Each PCR cycle proceeds through strand denaturation, primer annealing, and thermostable-polymerase extension. The two primers bind opposite strands with their 3' ends aimed toward the target, thereby defining the amplified interval. To analyze RNA, reverse transcriptase first creates cDNA, which then serves as the PCR template.

1. Cycle order: denature → anneal → extend.
2. Primers determine specificity and amplicon boundaries.
3. Thermostable polymerase survives repeated heating.
4. RT-PCR begins with reverse transcription of RNA.
5. Quantitative PCR follows amplification using fluorescence.

→ **clinic:** PCR is used for pathogen detection, genotyping, minimal-residual-disease testing, and viral-load assays.

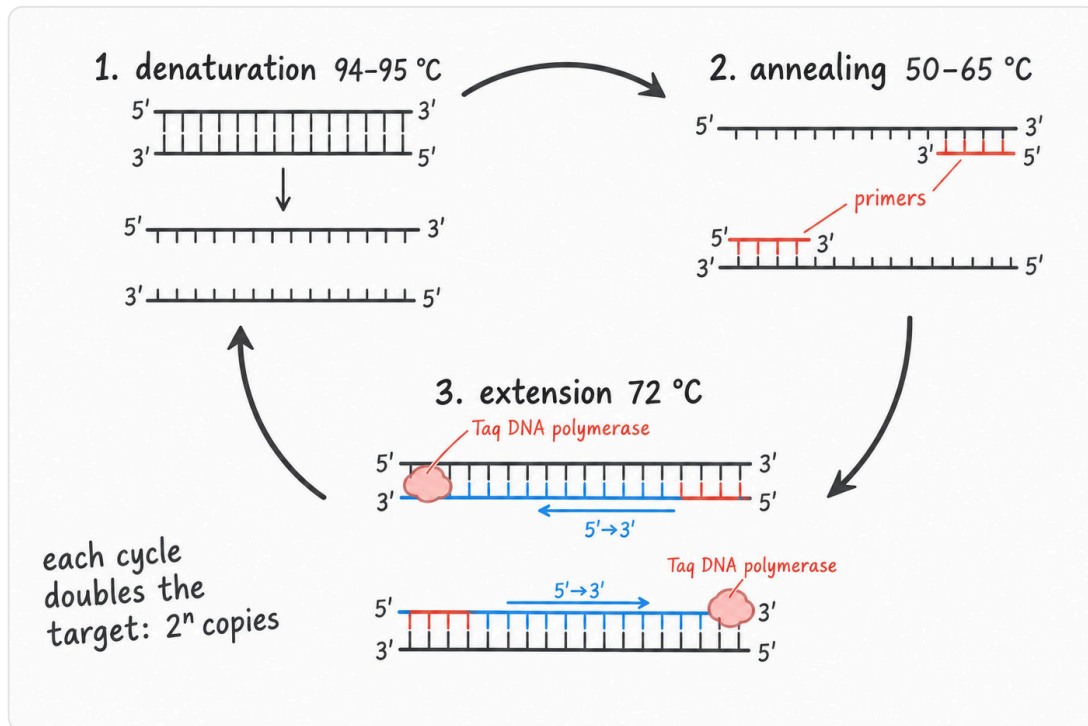


fig. 20 A PCR cycle denatures DNA, anneals two inward-facing primers, and extends new strands with Taq DNA polymerase.

Southern, Northern, and Western blot targets

Southern blotting detects a specific DNA sequence after fragment separation and transfer to a membrane. Northern blotting detects RNA transcripts and can report transcript size and relative abundance. Western blotting detects a protein using target-specific antibodies after electrophoresis and membrane transfer.

1. Southern = DNA with a nucleic-acid probe.
2. Northern = RNA with a nucleic-acid probe.
3. Western = protein with antibodies.
4. All separate a sample, transfer it, and detect a specific target.

→ clinic: Western-type immunoblots can confirm protein identity or pathogen-specific antibodies, though many clinical workflows now use other platforms.

Lippincott 6e, Chapter 33, blotting methods. · Confirmed August 25 Chapter 33 question bank, PDF pp. 838-877.

Restriction-fragment polymorphism and Sanger sequencing

A variant that creates or abolishes a restriction site changes fragment lengths and can be detected as an RFLP after digestion and suitable fragment analysis. Sanger sequencing uses occasional dideoxynucleotide incorporation to terminate extension because ddNTPs lack a 3'-OH. Sorting terminated fragments by length reveals the newly synthesized DNA sequence.

1. RFLP detects sequence-dependent fragment-length differences.
2. ddNTP incorporation prevents the next phosphodiester bond.
3. Sanger products differ by one nucleotide in length.
4. Sequence is read from the order of labeled terminated fragments.

→ clinic: Sanger sequencing remains a common confirmatory method for targeted variants.

Lippincott 6e, Chapter 33, RFLP analysis and chain-termination sequencing. · Confirmed August 25 Chapter 33 question bank, PDF pp. 838-877.

Expression microarrays and allele-specific hybridization

A gene-expression microarray hybridizes labeled cDNA from a sample to thousands of fixed probes, allowing relative transcript abundance to be compared across conditions. An allele-specific oligonucleotide can distinguish sequences differing by one nucleotide if hybridization stringency preserves the perfect match but destabilizes the mismatch. Both methods require careful controls because signal measures hybridization, not necessarily protein abundance or function.

1. Microarray signal estimates relative mRNA abundance through cDNA.
2. Allele-specific probes depend on mismatch-sensitive stringency.
3. Hybridization results require normalization and suitable controls.
4. Expression change does not prove a causal functional change.

→ clinic: Expression signatures can classify selected tumors and predict biologic behavior.

Lippincott 6e, Chapter 33, microarray and allele-specific detection methods. · Confirmed August 25 Chapter 33 question bank, PDF pp. 838-877.

Test yourself

1. Which features are essential in a basic plasmid cloning vector?

- | | |
|-----------------------------------|--|
| A. A ribosome and a mitochondrion | B. Only a poly-A tail |
| C. A complete human chromosome | D. An origin of replication, a selectable marker, and a cloning site |

2. How do dideoxynucleotides terminate DNA synthesis in Sanger sequencing?

- | | |
|---|--|
| A. They lack the 3-prime hydroxyl required for the next phosphodiester bond | B. They remove the template strand |
| C. They inhibit all base pairing | D. They contain ribose instead of deoxyribose only |

3. What type of DNA sequence is commonly recognized by a restriction endonuclease?

- | | |
|---------------------------------|---------------------------------|
| A. An RNA poly-A tail | B. A short palindromic sequence |
| C. A random amino-acid sequence | D. Only a telomere repeat |

Answers: 1 D · 2 A · 3 B

Laboratory principles and tests

Teacher slides and laboratory material; teacher slides cited under each idea.

Bench-test discrimination

Test / method	Detects or measures	Interpretation caution
Benedict / Fehling	Reducing sugars	Not specific to glucose; sucrose is negative before hydrolysis.
Barfoed	Monosaccharides reduce reagent relatively rapidly	Heating time matters; prolonged heating blurs the distinction.
Seliwanoff	Ketoses react rapidly in acid	A timed distinction; sucrose can yield fructose after hydrolysis.
Iodine	Starch-like helical polysaccharide complexes	Not a general test for all carbohydrates.
Biuret	Peptide-bond copper complex	Free amino acids do not give the protein reaction.
Ninhydrin	Free amino groups	Most amino acids purple; proline gives a different yellow response.
Beer-Lambert law	$A = \epsilon lc$; $A = -\log_{10}(T)$	Blank correctly; use linear range and matching wavelength/path length.
Chromatography	Differential partition/interaction	R_f = solute distance / solvent-front distance in planar chromatography.

General carbohydrate, starch, and reducing-sugar tests

Molisch is a broad carbohydrate test that produces a violet ring at the acid-aqueous interface. Iodine forms a blue-black complex with the helical structure of starch and is useful for polysaccharide recognition. Benedict reagent detects reducing sugars by reduction of Cu^{2+} to Cu_2O , with color progressing from green toward orange or brick red as reducing-sugar concentration increases.

1. Molisch: general carbohydrate; violet interfacial ring.
2. Iodine: starch/polysaccharide; blue-black complex.
3. Benedict: reducing sugar; colored Cu_2O precipitate after heating.
4. A positive broad test does not identify the exact carbohydrate.

→ clinic: Copper-reduction tests are historical screening methods; modern glucose assays are more specific and quantitative.

Teacher: Carbohydrate Qualification Test Esmaeili Jan 2026.pdf, pp. 7–15. · Telegram scope audit identifies this 21-page file as confirmed practical biochemistry material.

Use reaction conditions and timing to distinguish carbohydrate classes

Benedict color is semiquantitative because increasing reducing-sugar concentration produces progressively more colored Cu_2O precipitate rather than an exact calibrated concentration. Barfoed uses acidic copper reagent, in which monosaccharides reduce copper rapidly while disaccharides react more slowly. Seliwanoff uses acid and resorcinol; ketoses dehydrate rapidly and form a red product sooner than aldoses.

1. Benedict estimates reducing-sugar amount only roughly by color.
2. Barfoed separates rapid monosaccharide reduction from slower disaccharide reduction.
3. Seliwanoff favors rapid ketose detection.
4. Timing matters: prolonged heating can create misleading late positives.

Teacher: Carbohydrate Qualification Test Esmaili Jan 2026.pdf, pp. 14–19. · Teacher-derived image spotters confirm the expected Benedict, Barfoed, and Seliwanoff appearances.

Ninhydrin, Biuret, and xanthoproteic reaction targets

Ninhydrin reacts with free amino groups and gives Ruhemann's purple for most amino acids, while proline characteristically gives yellow. Biuret detects multiple peptide bonds through a violet Cu^{2+} complex in alkaline solution and is therefore a protein/peptide test rather than a free-amino-acid test. The xanthoproteic reaction nitrates aromatic residues, producing yellow that deepens toward orange after alkali.

1. Ninhydrin: free amino groups; purple for most amino acids.
2. Proline is the yellow ninhydrin exception.
3. Biuret: Cu^{2+} complex with multiple peptide bonds in alkali.
4. Xanthoproteic: aromatic amino-acid residues; yellow-to-orange product.

→ **clinic:** Modern clinical protein quantification uses standardized colorimetric or immunochemical assays, but Biuret chemistry remains widely used for total protein.

Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. · Tracked curriculum-gap bank: Practical amino acid and protein tests. No separate local teacher deck is recorded in the current Telegram scope audit.

Precision glassware, meniscus reading, and burette delivery

A volumetric pipette is calibrated to deliver one specified volume with high precision, whereas a graduated pipette can deliver variable volumes with lower precision. A typical aqueous meniscus should be read at eye level from its bottom to avoid parallax. A burette delivers controlled variable volumes of titrant through a stopcock and is read from the change between initial and final graduations.

1. Volumetric pipette: one precise delivered volume.
2. Volumetric flask: one precise contained final volume.
3. Read the lower aqueous meniscus at eye level.
4. Burette: variable controlled delivery for titration.
5. Correct pipette fillers are used; mouth pipetting is never acceptable.

→ **clinic:** Preanalytical volume errors can bias every downstream calculated concentration.

Teacher: Laboratory Equipment Esmaili Jan 2026.pdf, pp. 18–27. · Telegram scope audit confirms the 46-page laboratory-equipment deck.

Recognize core equipment and match it to safety or separation function

A fume hood captures hazardous vapors and exhausts them away from the worker; it is not a sterile cabinet. An analytical balance uses a draft shield to support high-precision mass measurement. A centrifuge accelerates sedimentation so components separate according to a combination of size, shape, and density under the applied relative centrifugal force.

1. Fume hood protects from chemical vapors; it does not guarantee product sterility.
2. Analytical balance doors reduce air-current error.
3. Centrifugation separates by sedimentation behavior, not pH alone.
4. Rotors must be balanced by mass and tube position before spinning.
5. Use relative centrifugal force when protocols require cross-instrument reproducibility.

→ **clinic:** Improper centrifugation can hemolyze samples or leave cells in plasma/serum and distort results.

Teacher: Laboratory Equipment Esmaeili Jan 2026.pdf, pp. 36 and 43-44. · Teacher-derived image spotters identify analytical balance, fume hood, and centrifuges.

Acid-base titration endpoint and normality calculation

In the teacher experiment, standardized 0.1 N NaOH is delivered from the burette to determine an unknown acid concentration. Phenolphthalein marks the endpoint as the first faint pink color persisting for roughly 30 seconds; a strong pink indicates overshooting. At the stated equivalence relationship, unknown normality follows from $N_1V_1 = N_2V_2$ using consistent volume units.

1. Titrant is known; analyte is unknown.
2. Teacher standard: 0.1 N NaOH.
3. Endpoint is persistent pale pink, not deep pink.
4. Normality counts reactive equivalents per liter.
5. Record initial and final burette readings and use their difference.

→ **clinic:** Titrimetric principles underlie several classical laboratory standardizations and acid-base measurements.

Teacher: Titration Esmaeili Jan 2026.pdf, pp. 6-9. · Telegram scope audit confirms this 10-page practical titration file.

Initial velocity, reagent blanks, and enzyme units

Enzyme activity is measured in the early linear phase so substrate is not yet substantially depleted and product accumulation has not driven the reverse reaction. A reagent blank contains the nonenzymatic assay components but lacks active enzyme or otherwise prevents reaction, allowing background signal to be subtracted. One conventional enzyme unit is the amount catalyzing conversion of one micromole of substrate per minute under defined conditions.

1. Use initial velocity in the linear range.
2. A blank measures non-analyte/reagent background.
3. Activity depends on temperature, pH, substrate concentration, and assay time.
4. An enzyme unit is a rate under specified conditions, not a mass.
5. Boiling usually destroys activity by denaturing protein structure.

→ **clinic:** Serum ALT, AST, ALP, CK, and lipase are commonly reported as catalytic activity rather than enzyme concentration.

Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. · Tracked curriculum-gap bank: Practical enzyme assays. No separate local teacher deck is recorded in the current Telegram scope audit.

Beer-Lambert law, blanking, and wavelength selection

Within its linear range, Beer-Lambert law states $A = \epsilon bc$, so absorbance is proportional to molar absorptivity, path length, and concentration. A reagent blank removes absorbance caused by solvent, reagents, and cuvette so the remaining signal is attributed to analyte or reaction product. Measuring near the wavelength of maximum absorbance improves sensitivity and reduces the effect of small wavelength-setting errors.

1. $A = \epsilon bc$ within the validated linear range.
2. Absorbance is $-\log_{10}$ of fractional transmittance.
3. Blank composition should match the sample except for the measured analyte/reaction.
4. Use $\lambda_{\max}$ for sensitivity.
5. Dilute and apply the dilution factor when absorbance exceeds the calibration range.

→ clinic: Many glucose, bilirubin, protein, and enzyme assays convert concentration or reaction rate into absorbance.

Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md · Tracked curriculum-gap bank: Spectrophotometry. No separate local teacher deck is recorded in the current Telegram scope audit.

DNA extraction by lysis, nuclease inhibition, and alcohol precipitation

Detergent disrupts lipid membranes and helps release DNA from cells and nuclei. EDTA chelates divalent cations required by many nucleases, while protease removes protein contaminants including histones and nucleases. Salt shields phosphate charges and cold ethanol lowers DNA solubility, allowing DNA to aggregate and precipitate from the aqueous extract.

1. Detergent lyses membranes.
2. EDTA chelates Mg^{2+}/Ca^{2+} and suppresses nucleases.
3. Protease removes DNA-bound and contaminating proteins.
4. Salt plus cold alcohol precipitates DNA.
5. Avoid vigorous shearing when intact high-molecular-weight DNA is needed.

→ clinic: Poor extraction quality can cause false-negative PCR or failed sequencing.

Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md · Tracked curriculum-gap bank: DNA extraction. No separate local teacher deck is recorded in the current Telegram scope audit.

Chromatographic separation by mobility, charge, size, or affinity

In paper or thin-layer chromatography, R_f is the distance traveled by solute divided by the solvent-front distance under the same run conditions. Ion-exchange chromatography separates by net charge through reversible binding to oppositely charged groups. Affinity chromatography uses a specific immobilized ligand to retain a target, while size-exclusion chromatography makes larger molecules elute first because they enter fewer bead pores.

1. R_f is unitless and lies between zero and one in a valid planar run.
2. Ion exchange separates principally by net charge.
3. Affinity uses specific target-ligand recognition.
4. Size exclusion: large molecules elute before small molecules.
5. Changing pH or ionic strength can alter binding and elution.

→ clinic: Chromatographic methods separate hemoglobin variants, drugs, metabolites, and proteins in clinical laboratories.

Atomic emission, calibration, and clinical ions in flame photometry

A flame excites atoms, and light emitted as electrons return to lower energy levels is measured at element-specific wavelengths. Sodium, potassium, and lithium are common analytes because their atomic emissions are strong and characteristic. An unknown concentration is obtained from standards and a calibration curve; out-of-range samples must be diluted into the validated range and corrected by the dilution factor.

1. Flame photometry measures emitted light, not absorbance.
2. Emission wavelength identifies the element; intensity relates to concentration.
3. Na^+ and K^+ are classic analytes.
4. Standards and blanks establish a valid calibration.
5. Never extrapolate far beyond the highest standard.

→ clinic: Flame emission was widely used for serum sodium and potassium and remains conceptually important in electrolyte analysis.

Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. · Tracked curriculum-gap bank: Flame photometry. No separate local teacher deck is recorded in the current Telegram scope audit.

Osmolarity, tonicity, RBC behavior, and dialysis

Hypotonic extracellular fluid causes red cells to gain water, swell, and potentially hemolyze, whereas hypertonic fluid causes water loss and crenation. Tonicity depends on effective nonpenetrating solutes rather than total osmolarity alone, so an isosmotic solution of a readily permeant solute such as urea can still be hypotonic. Across a dialysis membrane, water initially moves toward the compartment with the higher concentration of retained solute.

1. Hypotonic → cell swelling/hemolysis.
2. Hypertonic → cell shrinkage/crenation.
3. Isosmotic does not always mean isotonic.
4. Penetrating solutes contribute little to sustained tonicity after equilibration.
5. Water moves toward higher effective osmolarity across a water-permeable membrane.

→ clinic: Incorrect IV-fluid tonicity can cause dangerous cellular swelling or dehydration, especially in the brain.

Confirmed TUMS Cell & Molecules practical curriculum and data/teacher-materials/exam-scope.md. · Tracked curriculum-gap bank: Osmosis practical. No separate local teacher deck is recorded in the current Telegram scope audit.

Test yourself

1. Barfoed's test is primarily used to distinguish which carbohydrates?
 - A. Ketoses from aldoses only
 - B. Monosaccharides from disaccharides by rapid reduction in acidic medium
 - C. Starch from glycogen
 - D. Proteins from lipids
2. The violet color in the Biuret test results from Cu^{2+} complexing with what?
 - A. Free glucose carbonyls
 - B. Aromatic rings after nitration
 - C. Multiple peptide bonds in alkaline solution
 - D. DNA phosphate groups in acid
3. What happens to red blood cells placed in a hypertonic solution?
 - A. They take up water and lyse
 - B. They lose water and crenate
 - C. They synthesize more membrane immediately
 - D. They become isotonic by importing the nonpenetrating solute

Answers: 1 B · 2 C · 3 B

Sources and limits

1. Latest original course-scope reconciliation, 20 August 2026; original syllabus and teacher-material archive.
2. Ferrier, D. R. Lippincott Illustrated Reviews: Biochemistry, 6th edition (2014): confirmed chapters listed above.
3. Teacher foundations: INTRODUCTION.ppt; Water and Buffer 1404; Marks Essentials water/buffers reference.
4. Teacher biomolecules: Amino acids and Proteins Parts 1-2; Lipid structure (Esmaili); Vitamin 2024.
5. Teacher enzymes and genetics: Enzyme Kinetics and Regulation; DNA Structure; DNA Replication; Transcription 2025; Translation 2025; Regulation of Gene Expression 2025.
6. Teacher laboratories: Laboratory Equipment, Titration and Carbohydrate Qualification Test (Esmaili, January 2026), plus the original laboratory scope.
7. Four source-paper biochemistry sections: April 2021 Cell & Molecules report, February 2021, September 2021, Biochemistry 1 Finals 2022. Filename dates are retained; a report date is not proof of the exam sitting date.
8. Source caveats: human DNA repair — <https://www.ncbi.nlm.nih.gov/books/NBK1397/> ; enzyme kinetics — <https://www.ncbi.nlm.nih.gov/books/NBK92007/> ; hepatic HDL/VLDL — <https://www.ncbi.nlm.nih.gov/books/NBK351/> .

Original confirmed Term 1 syllabus: Lippincott Chapters 1-7, 14-18 and 23-33, plus foundations, water/buffers and laboratory-derived theory. Retake-specific exclusions have not been announced here. Chapters 8-13 and 19-22 are outside that confirmed syllabus but are taught here as clearly marked practice-bank sections, because the reused Cells & Molecules practice bank tests them. Physiology, histology and the Term 2 metabolism course are excluded. Study keys are not certified university keys.

This review covers the confirmed original biochemistry syllabus and the four imported biochemistry past papers, plus clearly marked practice-bank topics. It does not certify a new retake announcement or predict future questions. Figures marked as diagrams are teaching illustrations; check details against Lippincott.